Finite subgroup atlases: proofs, certified tables, and source qualifications

L054 synthesis of acquired components

This manuscript concerns Connor and Leemans' seven-page author edition of *An atlas of subgroup lattices of finite almost simple groups* (arXiv version 2, 29 November 2013; PDF dated 2 December 2013), and collates its eight-page 2015 journal version, pages 259–266. The maximal descent and recursive naming constructions are the authors' ideas. The arguments below and the acquired complete proofs make their interfaces explicit and correct several ancillary formulations. No novelty or new historical measurement is asserted.

Main results and the atlas boundary. We recover the maximal-descent and structural-certificate constructions under explicit complete local subgroup and exact-operation interfaces. The acquired classified roster contains all 127 almost simple isomorphism classes of order at most one million. Complete numerical tables and structural certificates have been checked for every one of these groups, including 47 groups absent from the current small companion roster. Seven larger applications are also complete. Their sizes are recorded below.

Domain	Groups	Subgroup-class rows
All almost simple groups of order at most 10^6	127	10,694
Seven specified larger groups	7	5,294
Complete reconstruction	134	15,988

These are row occurrences in actual tables, not pairwise different abstract subgroups. Of these rows, 13,520 belong to 87 current companion tables; one printed label in the A_{11} table is refuted and replaced by its verified structure. The remaining 2,468 rows are additional complete applications with generated structural descriptions.

This proves the small-order claim for a domain containing the source's 2012 online-ATLAS domain and supplies more than one hundred complete tables. It is *not* a reproduction or verification of the entire named historical or current atlas. The original articles give no exact filename roster. The current inventory of 116 companion tables, specifically attributed outputs such as the O'Nan table, and unlinked numerical claims remain in a separate ledger. No unpaid current table acquires its structural labels through this main-result reconstruction. Historical file identity and the reported historical timings are not asserted.

1 The maximal-descent and weight argument

All groups here are finite. For $H \leq G$, write $[H]_G$ for its conjugacy class, ordered by containment of representatives. Suppose each processed H comes with an actual isomorphism $\phi_H : H \to \widetilde{H}$, a complete cover of its maximal-subgroup conjugacy classes, and exact ambient conjugacy tests. Lists may repeat a local class; duplicates must first be identified by actual local conjugacy. Completeness is an explicit mathematical/software input. It does not follow from a plausible list length or matching orders.

Theorem 1. *Maintain a permanent registry of both processed and pending classes, initially containing G. Process each registered parent once; pull back its complete local maximal cover, fuse the resulting subgroups under conjugacy in G, and register each unseen class. This terminates with exactly all subgroup classes of G. Reachability among its actual maximal-inclusion edges is the entire class containment relation.*

Proof. An isomorphism and its inverse preserve subgroup containment, maximality, conjugacy and normalizers. Hence every new class and edge is genuine. There are finitely many classes, and no registered class is processed twice. For completeness, refine any $K \leq H$ to a maximal subgroup chain. At each step a member of the complete local cover is conjugate inside the parent to the next chain term. Its ambient fusion only changes the representative by conjugacy. Induction down this chain, starting with G, discovers $[K]_G$. The same argument inside any larger representative proves that containment implies reachability. The reverse implication follows from transitivity of containment on classes. □

The permanent registry repairs the source prose's test against only the current queue unless an additional scheduling invariant is supplied. Already completed parents can be reused: their complete transported local lists and exact fusion remain valid, while every unprocessed parent is still an obligation. The acquired literal-state and continuation interfaces implement this invariant. Named display strings alone are not embedded subgroup data; a failed symmetric-group checkpoint demonstrated why actual generators must be serialized.

For representatives H, K, let $M_1, \dots, M_t$ be the distinct local H-classes of maximal subgroups that fuse to $[K]_G$. Put

$$c_H = [G : \mathrm{N}_G(H)], \qquad a_{HK} = \sum_{j=1}^{t} [H : \mathrm{N}_H(M_j)], \qquad b_{KH} = c_H a_{HK}/c_K.$$

Orbit–stabilizer proves that a_{HK} counts the actual maximal embeddings inside H. Counting the pairs of conjugates (H', K') with K' maximal in H' in both orders proves the expression for b_{KH} and its integrality. Local classes that fuse must be added, rather than counted once. Duplicate local representatives must not be added twice. The exact bounds $a_{HK} \leq c_K$ and $b_{KH} \leq c_H$ follow immediately.

The class-poset covers are obtained by transitive reduction of the complete maximal-edge graph. On a cover, every embedding is maximal, so these weights are exactly the source's n_{HK}, n_{KH}. Outside covers, the displayed maximal-edge weights do not automatically count every embedding. The acquired robust local-list proof and retained-inventory reweighting pay these distinctions. In particular, an early duplicate local-class output for the $\mathrm{P\Gamma U}_3(5)$ model inflated one computed weight from $(3, 1)$ to $(6, 2)$; the corrected complete replay matches the source. This was our execution issue, not a source-table error.

2 Recursive names as exact existence assertions

A structural expression is interpreted recursively. Parentheses in the native source are retained; otherwise products associate to the left. A cross denotes a direct product, a colon a split extension, and the centered dot a nonsplit extension. The source also uses a different full stop for an extension whose splitting has not yet been specified. Names without the action or factor data need not determine a unique group.

For $N \lhd H$, $Q = H/N$, choose a section $s : Q \to H$ with $s(1) = 1$. Set $\alpha_q(n) = s(q)ns(q)^{-1}$ and $f(q, r) = s(q)s(r)s(qr)^{-1} \in N$. Then $N \times Q \to H$, $(n, q) \mapsto ns(q)$, is bijective, and multiplication is

$$(n, q)(m, r) = (n\alpha_q(m)f(q, r), qr).$$

The acquired complete extension proof verifies associativity exactly from $\alpha_q\alpha_r = \mathrm{Inn}(f(q,r))\alpha_{qr}$ and $f(q,r)f(qr,t) = \alpha_q(f(r,t))f(q,rt)$, with the identity normalizations. An explicit complement proves splitting; a commuting complement proves a direct product. Failure of a search to find a complement proves neither nonsplitting nor absence of a specified structure.

At each nonterminal naming step choose a proper nontrivial normal subgroup. Both child orders are smaller, so recursion terminates. Exact leaf identifications and the verified extension data reconstruct the parent by induction. Without those data the expression records only an existence assertion. The source's preferences for short or recognizable factors may select one such description, but are not a canonical-isomorphism theorem.

Every accepted row uses an actual embedded subgroup. An earlier named model is transported by an explicit bijection fixing identity and respecting right multiplication by every ordered generator. Induction on words proves that this is a homomorphism; bijectivity proves it is an isomorphism. Complete Cayley-word maps, faithful smaller actions, elementary normal forms, actual normal factors and complements, and exhaustive quotient-lift obstructions supply the concrete certificates. Each criterion has its acquired complete proof. Source typography is checked separately: a flattened superscript cannot silently become a cyclic order, and a dot cannot silently be replaced by a colon. Repeated descriptions may use different concrete models, each with its own exact transport.

3 Finite application and retained inputs

The attached current small-table roster has 80 tables and 8,226 subgroup-class rows. Each numerical route fixes an actual named group, a complete subgroup inventory, actual embedded generators, local class fusion, both incidence weights, and a bijection to the native source rows. Each structural route checks the source expression and every required normal factor, quotient, complement or nonsplitting obstruction. Their complete proof texts and separate author and reviewer records are acquired without another census.

Theorem 2. *Subject to the acquired classification of finite simple groups, their exact orders and outer automorphism groups, there are 127 almost simple isomorphism classes of order at most one million. For each class, the acquired applications give its full subgroup-class inventory, maximal incidences and a certified structural description of every subgroup class. Together these tables contain 10,694 subgroup-class rows.*

Proof. The acquired small-order roster proof first bounds the simple socle using the exact family orders and the retained classified exceptional list. There are 56 socles in this range. For a fixed socle S, an almost simple group is the full inverse image of a subgroup of $\mathrm{Out}(S) = \mathrm{Aut}(S)/S$. An isomorphism preserves the characteristic simple socle and therefore identifies such groups exactly by the corresponding outer-subgroup conjugacy relation. The complete small outer-group calculations give 127 cases. The complete roster proof, classification locators and arithmetic checks are included in the proof collection; no new proof of the classification is claimed here.

The 80 current small-table applications cover their exact named targets. The remaining 47 cases have 28 complete parent descents and 19 exact index-two restrictions from those parent inventories. The descent theorem above and the acquired restriction theorem prove completeness. Explicit named-model identities join each group to its roster position. Complete generator-edge transports, normal factors, complements and nonsplitting obstructions prove the 2,468 additional structural rows. Thus all 127 roster positions have complete numerical and structural applications. □

Proposition 3. *The following seven larger groups have complete numerical tables and structural certificates, with the stated correction for A_{11}.*

Group	Subgroup-class rows
A_{10}	430
A_{11}	788
M_{23}	204
$\mathrm{PSp}(4,5)$	307
$\mathrm{PSp}(4,7)$	603
$\mathrm{Sp}(6,2)$	1,369
S_{10}	1,593

Proof. Each application acquires a complete numerical inventory and an exact bijection to the companion rows, then verifies every displayed structural assertion with the literal subgroup generators. The acquired mathematical proofs justify the reused transport, support, extension and finite normal-form tests. The seven independent table reviews bind all 5,294 rows and their native expressions. The A_{11} review recovers 787 printed labels and proves the remaining printed label false, supplying its actual structure instead. The exact correction is stated below. No inference from an unverified group name or from matching orders substitutes for these certificates. □

Only M_{23} is a sporadic group among these seven larger applications. We do not infer coverage of the source's plural historical larger-sporadic examples from that one case. The quantified main-result reconstruction and the correctness of every individually named historical output are separate claims. The latter is not made here.

The source's Section 4 reports all groups in its ATLAS-restricted small-order domain, some larger groups, and more than one hundred complete outputs. The 127-group theorem proves a stronger small-order mathematical statement; the seven further applications supply explicit larger cases. No historical membership assumption is needed for that quantified conclusion. Conversely, these conclusions do not validate every current companion PDF. The retained index has 133 group rows, 130 linked occurrences and 116 distinct primary links, with a printed update date of 4 August 2013. Its documentary date does not establish historical byte identity. The O'Nan landing page specifically attributes its current linked table to this article's programs; that particular companion obligation remains recorded and unpaid. The source-claim map preserves all such boundaries. The companion index bounds the simple socle order $|S|$, whereas the article bounds $|G|$; our small-order theorem does not establish that stronger companion domain. The unlinked Suzuki and Fischer numerical assertions are retained separately. The completed S_{11} numerical inventory remains additional evidence with its structural labels unproved.

Named constructors, complete external local covers or previously accepted ambient inventories, and the exact group-operation interfaces remain explicit premises. Source counts are not used as completeness filters. All earlier mathematical authorship and distinct reviews are retained; this synthesis neither repeats the prior proofs nor represents itself as an independent review of its own earlier components.

4 Exact source corrections and attribution

Class posets and joins. The subgroup join is $\langle H, K\rangle$, not ordinary set union. The conjugacy-class poset need not be a lattice, contrary to the literal sentence on author page 2 and journal page 261. For an elementary counterexample take $G = A_5$. The classes of subgroups C_2 and C_3 have both S_3 and A_4 as upper bounds: embed S_3 on three letters and compensate every odd permutation by swapping the other two; embed A_4 on four letters. The relevant C_2 subgroups are conjugate in A_5, as are the C_3 subgroups. Indeed an odd conjugator may be made even using an odd permutation centralizing the relevant element. Any least upper bound would lie below both S_3 and A_4, and its order would be divisible by six and divide six. It would therefore

have order six. But A_4 has no subgroup of order six: such a subgroup would have index two, whereas every homomorphism $A_4 \to C_2$ kills the generating 3-cycles. Thus no such least upper bound exists. The accepted literal A_5 catalogue provides the same incidence obstruction.

Composition and deterministic names. Strict normal inclusions alone do not define a composition series: the factors must be simple, equivalently the lower term is maximal normal in the next one. Jordan–Hölder gives the multiset of simple factors up to isomorphism, not uniqueness of the intermediate subgroups. For example, C_2^2 has three different order-two intermediate terms with the same two composition factors. These are qualifications to author page 4 and journal page 263, not new classification results.

The impossibility sentence about deterministic names on those pages is also too strong literally. Number the identity by zero, enumerate all remaining labelings of a finite group's complete multiplication table, and take the lexicographically least resulting table as its name. Isomorphic groups yield the same collection of relabeled tables; equality of the minima supplies an isomorphism. This finite deterministic construction is not an efficient or useful short notation. It shows that ambiguity of factor descriptions is not undecidability of finite group identification. The source's example of nonisomorphic groups with the same normal-section information remains attributed to its earlier cited paper.

The group-name chain. Both native author page 3 and journal page 262 print a chain beginning $S_4(3) \cong U_4(2)$ and including $O_6^-(3)$ and $W(E_6)$. As printed the chain is false. The independently acquired named $\mathrm{PSU}(4,2)$ certificate has order 25920, whereas the acquired E_6 Weyl calculation has order $27 \cdot 1920 = 51840$. These orders preclude the asserted isomorphism with $W(E_6)$. We do not silently replace the field parameter or a full Weyl group by an index-two subgroup. The valid general point—that one group may have several geometric descriptions—does not require that faulty chain.

The A_{11} companion correction. Row 186 prints the left-associated expression $(2.A_5) : 2$. Its order-120 normal section would be the nonsplit double cover of A_5. The independently reviewed application proves that this section cannot embed in A_{11} under the stated convention, and checks the actual order-240 subgroup by a complete literal certificate. Its structure is $(A_5 \times C_2) \cdot C_2$, equivalently $C_2 \cdot (A_5 : C_2)$. The printed expression is retained as a refuted source assertion; it is not silently reassociated. This correction concerns a companion label, not a newly solved open conjecture.

Source examples and historical reports. The examples $S_5 = A_5 : C_2$ and $A_4 = C_2^2 : C_3$ have their standard actual normal factors and complements, also present in the acquired registry. The groups S_5 and $A_5 \times C_2$ have the same two composition factors but centers of orders one and two, respectively, so are not isomorphic. A coordinate flag of C_2^5 gives the displayed product of five order-two factors, each split and commuting. The label $5 \times 2 \cdot 2 : 2$ can describe both D_{40} and $C_5 \times D_8$: use their cyclic order-20 normal subgroup, its order-10 subgroup, and an order-two complement. The intermediate index-two extension is nonsplit. The final action in the first case inverts the order-five factor and in the second case centralizes it. The centers have orders two and ten, respectively, so the groups are not isomorphic. This pays the source's concrete illustration of nonunique notation.

The introduction's earlier sporadic subgroup computations, the small-group count near order 1024, software-database descriptions and historical normal-section example remain attributed antecedents. The concrete $L_3(7) : 2$ degree-reduction and machine-time/memory report, and the journal's qualitative performance comparison, are empirical source reports. Correctness of the algorithm does not prove a universal speed or memory advantage, and no historical timing is reproduced or credited here. The source-clause register lists every such boundary separately.

Exact current small-table roster

Current file locator	Group order	Rows
alt5	60	9
sym5	120	19
psl27	168	15
pgl27	336	23
alt6	360	22
psl28	504	12
psl211	660	16
m10	720	25
pgl29	720	26
psigmal29	720	56
psl213	1092	16
pgl211	1320	29
pgammal29	1440	65
psigmal28	1512	25
pgl213	2184	30
psl217	2448	22
alt7	2520	40
psl219	3420	19
psl216	4080	21
pgl217	4896	32
sym7	5040	96
psl33	5616	51
psu33	6048	36
psl223	6072	23
pgl219	6840	36
psl225	7800	37
m11	7920	39
psl216d2	8160	47
psl227	9828	16
psl33d2	11232	78
pgammau33	12096	100
pgl223	12144	34
psl229	12180	22
psl231	14880	29
pgl225	15600	43
psigmal225	15600	103
psl225d2	15600	39
pgammal216	16320	69
pgl227	19656	32
alt8	20160	137
psl34	20160	95
psu42	25920	116
suz8	29120	22
psigmal227	29484	38
pgl231	29760	44
pgammal225	31200	115
psl232	32736	24
psigmal34	40320	152
psl34d21	40320	151
psl34d23	40320	102
sym8	40320	296
pgammau42	51840	350
pgammal227	58968	67
pgl34	60480	100

Current file locator	Group order	Rows
psu34	62400	34
psl34d2e2	80640	288
suz8d3	87360	39
m12	95040	147
pgammal34	120960	226
psl34d6	120960	143
psl34ds3	120960	142
psu34d2	124800	80
psu35	126000	80
pgammal232	163680	30
j1	175560	40
alt9	181440	223
m12d2	190080	213
psl34dD12	241920	409
psigmau34	249600	120
psigmau35	252000	148
psl264	262080	76
sym9	362880	554
psl35	372000	140
pgu35	378000	93
m22	443520	156
j2	604800	146
psl35d2	744000	208
pgammau35	756000	244
m22d2	887040	490
sp44	979200	496
Total: 80 tables		8226

Transport, incidence weights, and certificates for finite subgroup data

A component for the subgroup atlas of Connor and Leemans

We prove the correctness of the two algorithmic constructions in Sections 2 and 3 of [1], using its seven-page author edition dated 2 December 2013. The maximal-subgroup descent and the recursive choice of normal subgroups are the authors' ideas. Our contribution here is an explicit certificate interface: changes of permutation action, fusion of conjugacy classes, incidence weights, and extension data have separate hypotheses and conclusions. We do not claim that these elementary arguments are new group-theoretic results. In particular, this component does not reproduce the more than one hundred numerical lattices announced in Section 4.

Conventions and inputs. All groups are finite. Write $H^g = g^{-1}Hg$ and $[H]_G$ for the conjugacy class of a subgroup. Order the classes by $[K]_G \leq [H]_G$ when some conjugate of K is contained in H. This is a partial order: a relation in both directions forces equal orders, and containment between finite subgroups of equal order is equality. We use the term *poset*; none of the arguments assumes that this poset is a lattice. In the actual subgroup lattice the join is $\langle H, K\rangle$, not set union.

For each subgroup H selected for processing, the input consists of an isomorphism $\phi_H : H \longrightarrow \widetilde{H}$ and a complete list of representatives of the $\widetilde{H}$-conjugacy classes of maximal proper subgroups. Completeness of this list is an explicit input, not a consequence of the orders or number of the returned subgroups. Conjugacy tests in the ambient group G must be exact. A positive conjugacy test can be checked by its transporter; a negative test requires a complete test or certificate.

Lemma 1. *An isomorphism $\phi : H \to \widetilde{H}$ transports subgroup containment, maximality, conjugacy within H, and normalizers. In particular, if $\widetilde{M}$ runs through a complete set of maximal-subgroup class representatives, then $\phi^{-1}(\widetilde{M})$ does so in H, with*

$$[H : \mathrm{N}_H(\phi^{-1}(\widetilde{M}))] = [\widetilde{H} : \mathrm{N}_{\widetilde{H}}(\widetilde{M})].$$

Proof. The mutually inverse maps on subgroups are image and preimage. They preserve containment in both directions, hence preserve the absence of a strictly intermediate subgroup. The identity $\phi(M^h) = \phi(M)^{\phi(h)}$ transports conjugacy. Applying it with $M^h = M$ identifies the two normalizers; their indices agree because both group orders agree. □

For a reduced permutation action, it is therefore necessary to establish faithfulness, rather than merely the existence of an action homomorphism. An explicit inverse is sufficient. So is a presentation of its image on the actual images of the generators, provided that all defining relators hold on the original generators: the resulting reverse homomorphism is inverse on generators and consequently on the groups. This last criterion is proved and checked in the separate faithful-action component of this project; it is an optional way of supplying the isomorphism required here.

Theorem 2. *Start with a registry containing $[G]_G$ and a queue containing G. Process each registered class once. For a representative H, transport its complete maximal-subgroup list by Lemma 1, identify each resulting subgroup up to G-conjugacy, and register and queue every class not already registered. Record an edge from $[H]_G$ to each resulting class, retaining separately all local H-conjugacy classes that fuse there. The process terminates and returns every subgroup class of G exactly once. Reachability in the resulting graph is precisely containment in the class poset.*

Proof. There are finitely many subgroups, and a registered class is never queued again. Thus termination holds even if the queue is processed in arbitrary order. Every inserted vertex and edge is an actual subgroup class and an actual proper containment.

Given $K \leq H$, refine the interval from K to H to a chain of maximal proper inclusions; this is possible by finiteness. Induction down such a chain proves that a conjugate of K is discovered whenever the class of H is processed. At each step the complete local maximal list contains a conjugate within the current parent, and fusion changes that representative only by conjugacy in G. Starting from G proves completeness.

Edges imply containment and containment is transitive on classes. Conversely, apply the same maximal-chain argument inside a representative of the larger class to obtain a path to the smaller class. Exact ambient conjugacy tests and the registry ensure that vertices are neither duplicated nor incorrectly identified. □

The registry in this formulation contains both queued and processed classes. Testing only the current queue, as the brief prose algorithm in [1] suggests literally, needs an additional scheduling invariant to avoid rediscovering a processed class. The permanent registry makes that invariant unnecessary. A computation begun from partial data must still certify that every registered parent has a complete processed maximal list; unprocessed vertices remain obligations.

Weights after fusion. Fix representatives H, K. Let $M_1, \ldots, M_t$ be the representatives of the H-conjugacy classes of maximal subgroups of H which are G-conjugate to K. Put

$$a_{H,K}^{\max} = \sum_{j=1}^{t} [H : \mathrm{N}_H(M_j)], \qquad c_H = [G : \mathrm{N}_G(H)]. \tag{1}$$

An empty sum is zero. Several local classes may fuse into one ambient class; replacing this sum by one summand is not valid.

Theorem 3. *The number $a_{H,K}^{\max}$ counts the maximal subgroups of the actual H belonging to $[K]_G$. The number of conjugates of H in which an actual K is maximal is*

$$b_{K,H}^{\max} = \frac{c_H a_{H,K}^{\max}}{c_K}. \tag{2}$$

The cover relations of the class poset are the edges left after transitive reduction of the graph in Theorem 2. For such a cover, $a_{H,K}^{\max}$ and $b_{K,H}^{\max}$ are the full incidence numbers n_{HK} and n_{KH} defined in Section 2 of [1].

Proof. Orbit–stabilizer for the conjugation action of H gives exactly $[H : \mathrm{N}_H(M_j)]$ distinct subgroups in the jth local class. These local classes are disjoint and complete, proving (1). Count the pairs (H', K') with $H' \in [H]_G$, $K' \in [K]_G$, and K' maximal in H'. Counting by the first coordinate gives $c_H a_{H,K}^{\max}$; counting by the second gives $c_K b_{K,H}^{\max}$. Conjugacy makes each fibre size constant, proving (2), including its integrality.

The graph is acyclic because orders strictly decrease along edges. Its reachability is the class order, so its transitive reduction gives exactly the covers. If $[H]_G$ covers $[K]_G$ and $K' \leq H'$ are representatives, any strict intermediate subgroup would give a strict intermediate class. Thus every such containment is maximal and the maximal-incidence counts are the required full incidence counts. □

For a pair that is not a cover, (1) only counts maximal embeddings. It need not count every embedding of the smaller class into the larger. Likewise, normalizer orders alone do not replace the local fusion data. The preceding results recover the weighted class poset without asserting that it determines G up to isomorphism.

For example, take $G = S_4$, $H = \langle (1\,2), (3\,4) \rangle$, and $K = \langle (1\,2) \rangle$. The two subgroups generated by the individual transpositions are distinct local classes in the abelian group H, but they are conjugate in G. Hence $a^{\max}_{H,K} = 2$, $c_H = 3$, $c_K = 6$, and $b^{\max}_{K,H} = 1$. Keeping only one of the two local classes would give the wrong weight. This example occurs in the literal S_4 check.

What a recursive group name certifies. Let $N \lhd H$, let $Q = H/N$, and choose a section $s : Q \to H$ with $s(1) = 1$. Define

$$\alpha_q(n) = s(q)ns(q)^{-1}, \qquad f(q,r) = s(q)s(r)s(qr)^{-1} \in N. \tag{3}$$

Here $\mathrm{Inn}(u)(n) = unu^{-1}$, and the multiplication on N and Q is part of the data. A structural expression involving only their names and a dot omits the maps in (3).

Theorem 4. *The map $(n,q) \mapsto ns(q)$ is a bijection $N \times Q \to H$, and under this bijection the multiplication is*

$$(n,q)(m,r) = (n\alpha_q(m)f(q,r), qr). \tag{4}$$

Conversely, automorphisms α_q and elements $f(q,r)$ define a group by (4), with kernel N and quotient Q, if

$$\begin{aligned} \alpha_1 &= \mathrm{id}, & f(1,q) &= f(q,1) = 1, \\ \alpha_q\alpha_r &= \mathrm{Inn}(f(q,r))\alpha_{qr}, & f(q,r)f(qr,t) &= \alpha_q(f(r,t))f(q,rt). \end{aligned}$$

Thus the displayed data, with exact identifications of kernel and quotient, certify the extension, not merely its order or composition factors.

Proof. Every $h \in H$ lies in exactly one quotient coset q and has a unique expression $ns(q)$; this proves the bijection. Inserting $s(q)^{-1}s(q)$ in a product gives (4). Conjugation by $s(q)s(r) = f(q,r)s(qr)$ proves the first identity involving α. Associativity of $s(q)s(r)s(t)$ proves the last identity.

Conversely, compute the two associations of three pairs in (4). The identity involving α moves the conjugating factor $f(q,r)$ through α_{qr}; the last identity then makes the two products equal. The normalization gives identity $(1,1)$. Given (n,q), its product with (m,q^{-1}) is the identity for the unique $m = \alpha_q^{-1}(n^{-1}f(q,q^{-1})^{-1})$. An associative finite monoid in which each element has a right inverse is a group: if $xy = 1$ and $yz = 1$, then $x = x(yz) = (xy)z = z$, hence $yx = 1$. Finally the projection to Q is a surjective homomorphism and its kernel is $N \times \{1\}$. □

An explicit complement C with $N \cap C = 1$ and $NC = H$ certifies a split extension; its inverse quotient map supplies a section with $f = 1$. A complement centralizing N certifies a direct product. Conversely, a homomorphic section supplies a complement, and a homomorphic section whose image centralizes N supplies a direct decomposition. Failure to find a complement does not certify a nonsplit extension: that conclusion needs an exhaustive complement test or a separate obstruction proof.

For instance, let $H = \langle x \rangle \cong C_4$ and $N = \langle x^2 \rangle$. The normalized section taking the nonidentity quotient element to x centralizes N, but its factor at that pair is $x^2 \neq 1$. It is not a homomorphic section. Replacing this factor by 1 gives $C_2 \times C_2$; the kernel and quotient have not changed, but the multiplication has. The finite check rejects that replacement as a certificate for the original C_4, even though it is valid data for a different extension.

Corollary 5. *The recursive naming procedure of Section 3 of [1] terminates if, at each nonterminal step, it chooses a proper nontrivial normal subgroup and recursively treats its kernel and quotient. Exact identifications at the leaves and exact extension data at the internal vertices determine the original group. Omitting the action and factor data gives only a structural description, and distinct groups can have the same resulting description.*

Proof. Both child orders are smaller than the parent order, so induction on order proves termination. At a terminal simple group or a selected recognizable group the supplied identification is checked directly. At each internal vertex Theorem 4 reconstructs the parent from the already identified children and the recorded multiplication. Induction proves the claim for the root. No uniqueness claim is made for descriptions that discard this multiplication data. □

Evidence and limits. The accompanying finite checks compare maximal descent with an independent enumeration obtained by adjoining elements to subgroups. They check local fusion, both incidence counts, transitive reduction, and extension multiplication on small explicit groups. These checks validate the stated interfaces and supply examples. They do not establish completeness of the external maximal-subgroup algorithms on every atlas group, reproduce the atlas's numerical files, or reproduce its historical timing comparison. The full L054 manuscript therefore remains pending. No conjecture, compression ratio, or Lean verification is claimed by this component.

References

[1] T. Connor and D. Leemans, *An Atlas of subgroup lattices of finite almost simple groups*, arXiv:1306.4820v2, 29 November 2013; retained author PDF dated 2 December 2013, Sections 2–4.

Incidence weights from a list with repeated representatives

The descent and transport results in the accompanying manuscript assume a complete list of distinct local conjugacy classes. A software interface may return a complete list with repetitions. The following elementary observation allows us to use that weaker input. It changes neither the descent theorem nor the classification being computed. We claim no new group-theoretic result.

Lemma 1. *Let H be a finite group. Suppose $M_1, \dots, M_s$ are maximal proper subgroups of H, and every maximal subgroup of H is H-conjugate to at least one M_i. Process the list in order, retaining M_i precisely when it is not H-conjugate to any previously retained subgroup. The retained list contains exactly one representative of each H-conjugacy class of maximal subgroups. Its completeness does not require the original list to be irredundant or to carry correct multiplicities.*

Proof. Two retained subgroups cannot be conjugate: the later one would have been discarded. For completeness, let M be any maximal subgroup and choose an index i for which M_i is conjugate to M. If M_i is retained, its class is represented. Otherwise the test that discards it supplies a previously retained representative of the same class. Thus every class is represented and no class is represented twice. All retained subgroups are maximal because they belong to the original list. □

If H is replaced by an isomorphic permutation group $\widetilde{H}$, this operation is performed using conjugacy in $\widetilde{H}$. The transport lemma carries the resulting distinct classes back to H. Conjugacy in a larger ambient group must not be used for this step: distinct local classes may fuse in that group, and each then contributes to the incidence number.

Lemma 2. *Let $H, K \leq G$ be finite groups. Among the retained representatives above, let $M_{i_1}, \dots, M_{i_t}$ be precisely those that are G-conjugate to K. Then the number of actual maximal subgroups of H in the ambient class $[K]_G$ is*

$$a_{H,K} = \sum_{j=1}^{t} [H : \mathrm{N}_H(M_{i_j})].$$

Writing $c_H = [G : \mathrm{N}_G(H)]$ and $c_K = [G : \mathrm{N}_G(K)]$, the reverse incidence is $b_{K,H} = c_H a_{H,K}/c_K$. Necessarily

$$0 \leq a_{H,K} \leq c_K, \qquad 0 \leq b_{K,H} \leq c_H.$$

In particular, if K is maximal in G, then $a_{G,K} = c_K$ and $b_{K,G} = 1$.

Proof. Conjugation by H has stabilizer $\mathrm{N}_H(M_{i_j})$ at M_{i_j}, so its orbit has the displayed index. The first lemma makes these orbits pairwise disjoint, and its completeness shows that their union is precisely the set being counted. Counting incident pairs of ambient conjugates first by their larger and then by their smaller subgroup gives the formula for $b_{K,H}$. Each fibre is a subset of the appropriate ambient conjugacy class, which proves both bounds. When the larger subgroup is G, all conjugates of a maximal K are maximal in it, and there is just one ambient subgroup conjugate to G. □

The implementation therefore discards repeated local conjugacy classes before ambient fusion and obtains each weight from a normalizer index. It does not use a multiplicity attached to the original list. The two bounds provide useful necessary checks; satisfying them alone does not establish correctness or completeness. Completeness of the original maximal-subgroup list and exactness of the group operations remain explicit computational premises.

Normal-subgroup restriction of an acquired subgroup inventory

Human Compression project: L054 component

This is an elementary reuse method, not a new subgroup classification. The input is an already acquired complete list of actual embedded subgroup representatives, with its original completeness and group-computation premises. Printed names or published row counts are not substitutes for that input.

Theorem 1. *Let G be finite, $K \lhd G$, and $\mathcal{R}$ a complete covering list of G-conjugacy classes of subgroups. Let T be a transversal for G/K. For each $H \in \mathcal{R}$ with $H \leq K$, form H^t for every $t \in T$. Discard candidates only after actual K-conjugacy to an earlier retained candidate is established. The resulting list is a complete irredundant K-subgroup inventory. Repetitions in $\mathcal{R}$ cause no problem.*

Proof. If $L \leq K$, choose $H \in \mathcal{R}$ and $g \in G$ with $L = H^g$. Normality gives $H \leq K$. Write $g = tk$ with $t \in T$, $k \in K$. Then $L = (H^t)^k$, so its K-class is represented among the candidates. A right transversal works too, since left and right cosets coincide. Every candidate lies in K. Removing it only after a positive conjugacy test preserves its class, and comparison with all earlier survivors makes the final list irredundant. □

Normality is essential: for nonnormal K, an ambient representative outside K can have a conjugate inside K. The executable checks actual normality, input containment and transversal cardinality/coset distinctness. It retains the actual generating permutations and an origin (H, t) for each result. These checks do not replace the acquired completeness of $\mathcal{R}$. Class lengths in K are $[K : N_K(H)]$.

Lemma 2. *Suppose $[G : K] = 2$. An ambient class of subgroups lying in K consists of one or two K-classes. In the latter case every element outside K interchanges the two. The class of H does not split if and only if $G = KN_G(H)$.*

Proof. Normality makes conjugation by G act on K-conjugacy classes, and this action factors through G/K. The stabilizer of the class of H is $KN_G(H)$: if $H^g = H^k$, then $gk^{-1} \in N_G(H)$, and the converse is immediate. The assertions follow by the orbit–stabilizer formula. □

Theorem 3. *Assume $[G : K] = 2$, the ambient inventory is irredundant, and its weighted edges count all actual maximal subgroups in each fixed parent. For $H \leq K$ and an ambient child class C, let a be its maximal-subgroup multiplicity inside H. If C does not split, its multiplicity in the restricted inventory is a. If C splits but the parent class does not, each child class has multiplicity $a/2$. If both split, their separate multiplicities can instead be computed by the acquired robust local maximal-subgroup method in the actual parent. These rules give the complete restricted weighted maximal-incidence graph.*

Proof. All maximal subgroups of H lie in K, and maximality in H is independent of the ambient group. For a nonsplitting child class there is nothing to distribute. If the child splits and the parent does not, the lemma supplies $n \in N_G(H) \setminus K$. Conjugation by n permutes the actual maximal subgroups of H and exchanges the two child classes. It is a bijection between the two incidence fibres, proving equality and the evenness of a.

When both classes split, these arguments determine only their sum a. For any parent with such an ambiguous entry, the executable obtains a complete local maximal cover after checked

faithful degree reduction, deduplicates it by actual local conjugacy, and transports representatives back to H. Its direct local normalizer indices are added under actual K-conjugacy. This is exactly the acquired robust weight method. Every result must aggregate under its origin map to the acquired ambient edge row. Other parents use just the two inheritance rules already proved. Thus every actual maximal incidence is counted once. If c_H, c_L are class lengths in K and the forward weight is a_{HL}, double counting gives reverse weight $c_H a_{HL}/c_L$. The executable checks integrality and the two ambient class-size bounds. □

These edges are actual maximal inclusions, not covers of a conjugacy-class poset. No transitive reduction of the latter is used. Complete local covers, genuine maximality, exact membership/conjugacy/normalizers and faithful degree reduction remain explicit GAP interfaces as in the accepted atlas method. The first theorem needs a covering list; the weight-inheritance routine requires irredundancy and checks it in G.

Finite controls. Six tiny normal inclusions, with indices $1, 2, 4$, are checked both with the original complete ambient lists and with repetitions. These twelve controls produce 115 candidates and 46 retained classes in total and exercise actual splitting. Five index-two weight controls compare every entry against all actual subgroup inclusions, testing maximality by the absence of an actual intermediate subgroup. All 122 ordered pairs and 24 edges agree. The D_{16} to D_8 control uses two ambiguous-parent local computations; the other parents use inheritance. These are small controls, not atlas targets.

Planned application and limitations. The retained parent inventories for the current companion tables `pgammal216`, `psigmau34`, `pgammau35` and `pgammal225` contain possible normal index-two targets. The actual normal subgroup must still be identified with the intended named group by a defining action or exact transport. A source description, group order or matching numerical graph alone does not establish that identification. In particular the three index-two subgroups of the last parent must be distinguished without target row counts as a filter.

This component runs no atlas target and grants no new numerical or structural label credit. Current companion files are not asserted identical to historical online editions. The source attribution, whole-original admission, questions, formalization and compression measurements remain separate.

Restricting an acquired subgroup inventory to an arbitrary subgroup

L054 reuse component

This elementary extension of the accepted normal-subgroup restriction method uses actual embedded subgroups. It is not a new finite-group classification. Conjugation is on the right: $H^g = g^{-1}Hg$, hence $(H^g)^k = H^{gk}$.

Theorem 1. *Let G be finite, $K \leq G$, and $\mathcal{R}$ a complete covering list of subgroup G-conjugacy classes. Let T represent the left cosets tK. For every $H \in \mathcal{R}$ and every $t \in T$, form H^t and keep it if and only if $H^t \leq K$. Fuse these candidates by actual K-conjugacy. The surviving list represents every subgroup K-conjugacy class exactly once. Input repetitions are harmless.*

Proof. Given $L \leq K$, completeness supplies $H \in \mathcal{R}$ and $g \in G$ with $L = H^g$. Write $g = tk$, where $t \in T$ and $k \in K$. Then $L = (H^t)^k$, and $H^t = L^{k^{-1}} \leq K$. Thus a candidate in the K-class of L survives the containment test. Conversely every retained candidate is a subgroup of K. Removing a candidate only after positive K-conjugacy to an earlier survivor preserves coverage. Testing all earlier survivors makes the final list irredundant. Input repetitions only repeat candidate classes and do not change this argument. □

Neither normality nor an initial test $H \leq K$ is permissible as an unstated hypothesis. A chosen ambient representative may lie outside K even though its class meets K. The code therefore tests containment only *after* conjugating each ambient representative. It records the exact origin (H, t) and the final class of every admitted candidate.

GAP's `RightTransversal(G,K)` represents right cosets Kt. Inverting these representatives gives the needed left transversal: inversion bijects Kt with $t^{-1}K$. The lower-level interface checks $|T| = [G : K]$, each $t \in G$, and $t_i^{-1}t_j \notin K$ for distinct entries. These checks establish left-coset coverage. They do not establish completeness of $\mathcal{R}$, which must be acquired from its original certificate. Input containment and the presence of G and of the final top K are additional sanity checks, not substitutes for that premise.

Corollary 2. *The acquired robust local-weight method can compute the complete weighted maximal-incidence table from this restricted inventory without enumerating all subgroups of K again.*

Proof. For each retained parent H, acquire a complete cover of its maximal-subgroup classes, using checked faithful degree reduction if desired. Remove local conjugacy repetitions, and transport the actual subgroups back into H. The inventory theorem places each in exactly one K-class. A local class represented by M contains $[H : N_H(M)]$ actual maximal subgroups. Sum these indices whenever several local classes fuse to the same ambient child class. This counts every actual maximal inclusion exactly once. Writing $c_H = [K : N_K(H)]$, double counting incidences gives reverse weight $c_H a_{HL}/c_L$. No cover relation in the conjugacy-class poset is used. This is the already accepted robust weight argument; the present new kernel supplies only its complete inventory input. Ambient G-edge weights cannot in general be reused unchanged after restricting to a nonnormal K. □

Literal controls and strict countercontrols. The finite controls construct all actual subgroups of six tiny permutation groups, through order24, by repeated literal multiplication closure. They then restrict independently conjugated complete ambient class lists to *every* actual subgroup, both with and without input repetitions. Expected K-classes are formed by literal conjugation over all elements of K, separately from the kernel's subgroup-conjugacy decision. The tests check both directions, exact origins, candidate fusion and class lengths.

A nonnormal S_3 fixture chooses $K = \langle (1\,2) \rangle$ and the ambient representative $H = \langle (1\,3) \rangle$. It constructs a genuine right transversal with $H^t \neq K$ for every chosen representative. Using these representatives without inversion misses the target top; inverting them recovers it. Filtering the chosen ambient list by $H \leq K$ before transport also misses the top. Thus both tempting incorrect variants are rejected by actual group examples, not merely by differently written code.

Acquired inputs and scope. Complete ambient inventories and local maximal covers, faithful transports, and exact subgroup/conjugacy/normalizer operations retain their original GAP software and classification premises. The primitive operations are not reproved here. Intended applications are the index3 nonnormal subgroup in the already acquired $P\Gamma U_3(5)$ inventory and small-index subgroups of the acquired full automorphism model of $\mathrm{PSL}_3(4)$, once the required complete ambient inventory is available. Each target still needs its actual named-model identification and source join. This method packet runs no atlas target and grants no numerical, structural-label, whole-source, question, formalization or measurement credit.

Literal certificates for twenty-four atlas structure labels

L054 bounded recovery component

Source and scope. Connor and Leemans, *An atlas of subgroup lattices of finite almost simple groups*, Ars Math. Contemp. 8 (2015), 259–266, Section 3, distinguish an unspecified extension $N.Q$, a nonsplit extension $N \cdot Q$, a semidirect product $N : Q$ and a direct product $N \times Q$. Unparenthesized products associate to the left. The authors explicitly permit nonunique descriptions; their Figure 1 also uses an identification database and choices of desirable normal subgroups. We verify the displayed existence assertions, without claiming to reproduce those historical choices or to give canonical names. This is an elementary certificate method, not a novelty claim.

The inputs are the retained *current downloaded* companion PDFs for A_5 and M_{11}, both internally dated 25 November 2012, and their exact row maps to previously saved literal subgroup representatives. Their identity with every file available at the fixed 2013 edition is not assumed. All three table pages and both pages of *How to read the atlas* were directly inspected. In particular, $2^2, 3^2$ are native superscripts; they were not inferred from flattened strings "22" or "32". The finite roster below is the entire claim of this component.

Proposition 1 (Elementary verification rules)**.** *Let H be a finite group whose complete elements and multiplication are given literally. The following data certify the indicated assertions.*

1. *A generating element of order m certifies C_m. For $p = 2, 3$, commuting elements of order p whose exponent vectors give a bijection $\{0, \dots, p-1\}^a \to H$ certify C_p^a.*

2. *A bijective multiplication-preserving map from the standard finite permutation model certifies A_4, A_5 or S_3. Generators r, s of a group of order $2m$, with $|r| = m$, $|s| = 2$ and $srs = r^{-1}$, certify D_{2m}. Generators i, j of a group of order eight, both of order four, satisfying $i^2 = j^2$ and $j^{-1}ij = i^{-1}$ certify Q_8.*

3. *Actual subgroups $N, K \leq H$ with $N \lhd H$, $N \cap K = 1$ and $\langle N, K \rangle = H$ certify $H = N : K$. If they commute, they certify $H = N \times K$. Their two factor labels can be certified recursively.*

4. *If $N \lhd H$ has index two, its quotient is C_2. The extension is nonsplit if and only if every element of $H \setminus N$ has order different from two.*

Proof. The first assertions give explicit cyclic or elementary abelian coordinate isomorphisms. In the second, multiplication preservation and bijectivity give an isomorphism. The dihedral relations reduce every word to r^a or $r^a s$, so there are at most $2m$ elements; generation and the stated order give the dihedral presentation exactly. The quaternion relations similarly reduce words to i^a or $i^a j$ for $0 \leq a < 4$, and give the usual order-eight quaternion presentation. Normality in the third assertion gives $NK = H$, and trivial intersection makes $(n, k) \mapsto nk$ bijective. Multiplication is the semidirect multiplication for the actual conjugation action; when the factors commute it is direct multiplication. Finally, a section of $H \to H/N \cong \mathrm{C}_2$ exists exactly when the nonidentity quotient element has an outside lift of order two. This requires checking all outside elements: a single badly chosen section proves nothing. □

Proposition 2 (Reuse by explicit transport)**.** *A literal bijective homomorphism from one certified concrete finite model to an actual subgroup transports its certified structure. A shared registry may therefore store one such model and its certificate, with a separately checked isomorphism for each occurrence. A registry key need not, and must not, assert that the displayed label determines a unique isomorphism type.*

Proof. An isomorphism preserves products, orders, normality, intersections and generation. It transports complements and the entire outside-element list. Thus every hypothesis of the preceding proposition is preserved, inductively through the label tree. In the implementation each registry record has a concrete literal origin and generators. Every transport is a full bijection checked on every multiplication pair. Distinct concrete models with the same tree may have separate registry records. □

Theorem 1 (The bounded atlas roster)**.** *Under the acquired literal row maps, all nine A_5 labels and the following fifteen M_{11} labels are true of their actual saved subgroup representatives:*

row	*structure*	*row*	*structure*
12	$11:5$	13	$Q_8:S_3$
14	$(3:S_3)\cdot 2$	16	$S_3\times S_3$
17	$2^2:S_3$	18	$Q_8:3$
19	$D_{10}\cdot 2$	20	$3:S_3$
21	$S_3\times 3$	22	$Q_8:2$
27	3^2	28	Q_8
29	8	30	D_8
36	2^2		

The A_5 rows, in order, are $A_5, A_4, D_{10}, S_3, 5, 2^2, 3, 2, 1$. Here a bare integer denotes a cyclic group, and the dihedral subscript denotes its order.

Proof. The certificate files give complete actual permutation elements and original subgroup generators for each of the twenty-four rows. An independent token parser reads all thirty-nine retained M_{11} subgroup constructors and joins the chosen fifteen exactly to those generators. For A_5, the lexicographically sorted list of all sixty even permutations gives the indexing of the acquired literal subgroup array; all nine element lists are joined exactly. The accepted numerical row maps, the native label transcriptions and these actual-generator joins are checked separately. No source label is used to change a row map or to filter an ambient census.

The verifier closes each displayed generating list and compares it to the complete listed elements. It computes products by literal permutation composition. Each certificate is then checked by the preceding rules, with its exact native label tree. This pays forty-four factor nodes, 4,032 standard-model multiplication pairs and all twenty-eight outside elements of the two index-two nonsplit assertions. The outside lists are exhaustive, and the actual two quotient cosets are retained. Each colon node also records its actual nontrivial-action Boolean; each direct node checks commutation. The shared registry has twenty-four concrete models for forty-four factor occurrences, giving twenty reuses; all 15,671 transport multiplication pairs are explicitly checked. These finite certificates supply every hypothesis of the two propositions. □

Finite evidence and limits. The bounded witness discovery used only these already retained small subgroups (order at most sixty), finished in less than one second, and performed no ambient subgroup census. Its exact inputs, code and terminal receipt are preserved. Discovery is not a verification premise: the final checker uses only the literal certificates, original subgroup arrays, source joins and multiplication rules. Eighteen damaged examples are rejected, including the nonnormal diagonal S_3 in $S_3\times S_3$, noncommuting proposed direct factors, a nonhomomorphic bijection, and an index-two subgroup in $\mathrm{C}_4\times\mathrm{C}_2$ with a bad chosen section but an outside involution. The same checks pass with Python optimization enabled.

The numerical class catalogue and its completeness retain their separately accepted interfaces; this proof adds the stated labels only. It does not certify any other row, the atoms M_9, M_{10}, larger classical aliases, additive-exponent conventions, or the historical naming software. There is no whole-atlas, whole-original, conjecture, Lean or measurement credit. The current author supplied this proof and implementation; root proposed the shared concrete-model registry and authorized the narrow pilot. Distinct review is required before this component is acquired.

Seventeen further literal structure labels in the M_{11} table

L054 bounded recovery component

Acquired interface and source. We acquire the accepted twenty-four-label pilot's elementary witness rules, explicit transport proposition, actual subgroup arrays, numerical row maps and native interpretation of Connor–Leemans' extension notation. The source is the current retained two-page companion M_{11} table; its historical identity with every fixed-2013 file remains unasserted. Section 3 of the final 2015 journal article supplies the notation: a colon is a semidirect product, a centered dot is a nonsplit extension, and unparenthesized products associate to the left. Native pages 1 and 2, including their continued-table boundary, were inspected directly. No flattened numeral or punctuation heuristics determine the following label trees.

Theorem 1. *The actual saved subgroup representatives joined to the following seventeen source rows have the displayed structures:*

row	*structure*	*row*	*structure*
8	$((3 : S_3) \cdot 2) \cdot 2$	9	$((3 : S_3) \cdot 2) : 2$
10	A_5	11	A_5
15	$(3 : S_3) \cdot 2$	23	D_{12}
24	A_4	25	11
26	D_{10}	31	S_3
32	S_3	33	6
34	5	35	4
37	3	38	2
39	1		

Bare integers denote cyclic groups, and a dihedral subscript denotes order.

Proof. The exact original M11 generator list is parsed by the acquired literal token parser. Each new row has its previously accepted numerical row index and an exact generator-array join; neither an order coincidence nor a structural name is used to substitute a subgroup. The verifier computes each finite group's complete literal elements from those generators and checks the certificate against the manually transcribed native tree.

All atom, normality, intersection, complement, generation and multiplication tests use the unchanged accepted pilot verifier. In row 8 the first normal factor of order 36 is itself certified as $(3 : S_3) \cdot 2$, and its embedding is normal of index two in the actual order-72 group. Both index-two quotient maps are nonsplit: every outside element is checked and none has order two. Row 9 has an actual outer complement, while the inner order-36 factor is a nonsplit index-two extension: all eighteen outside elements have order different from two. These are distinct positive assertions about the displayed left-associated trees; no uniqueness of descriptions is inferred. The remaining rows use exactly the acquired cyclic, alternating, symmetric, dihedral and index-two rules. Thus the new witnesses pay twenty-nine factor nodes, 7,524 standard-model multiplication checks and ninety exhaustive outside-element checks.

The registry is extended by retaining all twenty-four old concrete models and all forty-four old occurrence maps as literal prefixes. Five new concrete models suffice for the twenty-nine new factor occurrences. Every new map is an explicit bijective homomorphism, checked on every product using the acquired transport verifier. The resulting twenty-nine concrete models serve seventy-three factor occurrences, with 38,910 checked transport products. In particular, the two displayed A_5 rows may share a model without identifying their ambient conjugacy classes. All hypotheses of the acquired certificate and transport propositions are thereby met. □

Execution and scope. The discovery was capped at 120 seconds and finished in less than one second, using only the seventeen retained subgroups of order at most 72. No ambient census was run. The final checker performs no discovery or isomorphism search; it verifies literal witnesses and rejects ten damaged joins, nested extension witnesses and transport maps, also with Python optimization enabled. Together with the accepted pilot this accounts for the nine A_5 and thirty-two M_{11} labels. M11 rows 1–7, involving $M_{11}, M_{10}, L_2(11), A_6, M_9 : 2, S_5, M_9$, remain outside this component. No canonical name, historical naming-software reproduction, whole original, conjecture, Lean or measurement claim is made. The current author supplied the extension and its code; root proposed registry reuse and authorized the fixed roster. The predecessor's row 9 AST wrongly transcribed the first centered dot as a colon; root review caught this source error. This successor certifies the actual printed nonsplit-inner, split-outer label and checks AST punctuation against the literal source. The other sixteen witnesses remain byte-identical. Distinct review remains required.

Four standard atoms in the current M_{11} subgroup table

L054 bounded recovery component

Source and acquired theorem. This component addresses only rows 1, 3, 4 and 6 of the retained current Connor–Leemans companion M_{11} table. Their native labels are respectively M_{11}, $L_2(11)$, A_6 and S_5. The numeric row bijection and actual saved subgroup generators are acquired from the accepted atlas computation. The printed structures are checked separately here; an order coincidence does not establish a group name.

We acquire Theorem MC.3 (Cayley transport) from the accepted project volume *Marked coordinates and exact orbit enumeration*, together with its distinct review and public admission. Its exact interface is this: for finite groups generated by ordered tuples (a_i) and (b_i), a bijection f with $f(1) = 1$ and

$$f(xa_i) = f(x)b_i \qquad \text{for every } x, i$$

is the unique isomorphism with those generator images. Every finite group is generated as a monoid by its group generators, so the positive-word induction in that acquired proof supplies full multiplicativity. We use this previously proved elementary result, with no novelty claim or new formalization. Checking all labelled edges costs $|G|$ times the number of generators, rather than checking all $|G|^2$ products.

Exact standard models. The A_6 model is the set of all even permutations of six points; its four listed 3-cycles generate exactly those 360 permutations by literal closure. The S_5 model consists of all 120 permutations on five points, with its listed 5-cycle and transposition. Thus these model identities use the definitions, not a small-group identifier or a structural-name oracle.

For $L_2(11) = \mathrm{PSL}_2(11)$, form all 2×2 matrices of determinant one over $\mathbb{F}_{11}$, and their fractional-linear actions on $\mathbb{F}_{11} \cup \{\infty\}$. A matrix fixing $0, 1, \infty$ is scalar; among determinant-one scalars it is $\pm I$. Consequently this literal image is the defining projective special linear group. The verifier forms all 1,320 matrices and all 660 distinct actions. It checks that the two specified elementary unipotent actions generate exactly that set. No classification of groups of order 660 is used.

The M_{11} model retains the already accepted named interface `MathieuGroup(11)` and its exact exported two permutations. The standard-name identification of this constructor is an explicit retained premise; it is not deduced merely from the computed order 7,920. The new verifier checks the literal constructor output against the acquired file, reconstructs its whole permutation closure and uses no GAP routine.

Theorem 1. *The actual retained subgroup representatives corresponding to the four source rows have the printed isomorphism types:*

source row	*actual record*	*order*	*generators*	*checked edges*
1 (M_{11})	39	7920	2	15840
3 ($L_2(11)$)	37	660	2	1320
4 (A_6)	36	360	4	1440
6 (S_5)	34	120	2	240

Proof. For each row the independent literal verifier parses the original saved generator array, checks its exact source-row join, and reconstructs the complete subgroup by finite positive-word closure. The shared registry contains the full literal standard model and its ordered generators. Its record is compared with the defining construction above. The supplied image vector is a bijection onto the whole actual subgroup, sends identity to identity, and satisfies every labelled right-generator edge. The total is 18,840 such checks. All hypotheses of acquired MC.3 hold, giving the four isomorphisms. The arguments apply to the actual representatives, not merely to groups of the same order. The four new registry models are indexed after the 29 acquired small models; the old registry is retained under its exact hash rather than altered. □

Evidence and limitations. Four isomorphism queries supplied the candidate maps in 2.38 seconds under a 120-second cap. GAP's `IsomorphismGroups` routine served only to find witnesses. The final checker uses neither GAP nor an isomorphism search, and independently checks every finite image. Ten damaged maps, generator arrays and source joins are rejected, also under Python optimization. The source native first page was directly inspected, including its subscripts and punctuation. Both the literal generators and their acquired numerical provenance are bound; the earlier local-class duplicate issue is handled by the separately accepted robust-weight interface and does not change these saved representatives.

Together with the accepted 24-label pilot and corrected 17-label extension, these four assertions would account for all nine A_5 labels and 36 of the 39 M_{11} labels after distinct acceptance. The remaining rows 2, 5 and 7 contain M_{10}, $M_9 : 2$ and M_9, whose precise naming conventions remain separate. No historical download identity, canonical name uniqueness, whole atlas, original-paper completion, conjecture, Lean or measurement credit is claimed. Root suggested registry reuse and the labelled-edge method; the current author found and acquires the existing complete MC.3 proof, and authored these literal models, joins and verifiers. Distinct review remains required.

The three remaining Mathieu stabilizer labels

Human Compression project: L054 finite component

This note certifies rows 2, 5 and 7 of the retained current M_{11} companion table: M_{10}, $M_9 : 2$ and M_9. The printed table is dated 25 November 2012; identity with every historical web edition is not assumed. The accepted numerical row bijection and literal subgroup arrays are inputs. The notation is not inferred from subgroup orders.

Exact naming convention. GAP's reference manual, Section 50.1–13, provides the named Mathieu constructors in degrees nine and ten. The actual implementation in `grp/basicprm.gi` explicitly identifies M_{10} as the point-11 stabilizer in its standard M_{11}, and M_9 as the point-10 stabilizer in that M_{10}. We retain this named-constructor convention, including its literal generators; we do not claim a new characterization of Mathieu groups. The standard M_{11} is exactly the already acquired model

$$G = \langle (1\,2\,\cdots\,11),\ (3\,7\,11\,8)(4\,10\,5\,6) \rangle.$$

Its complete 7,920-element array is acquired from the accepted shared registry. Embedded on these eleven points, the retained generators are

$$\begin{aligned} M_{10}: &\quad (1\,9\,6\,7\,5)(2\,10\,3\,8\,4), \quad (1\,10\,7\,8)(2\,9\,4\,6), \\ M_9: &\quad (1\,4\,9\,8)(2\,5\,3\,6), \quad (1\,6\,5\,2)(3\,7\,9\,8). \end{aligned}$$

The verifier computes their closures from literal permutations, without using the constructor's stored orders. These closures have 720 and 72 elements and equal, element for element, G_{11} and $G_{10,11}$ obtained by filtering the acquired whole array. Thus the stabilizer joins are checked directly, in addition to retaining the names.

Theorem 1. *The actual subgroups matched to source rows 2, 7 and 5 are, respectively, a conjugate of this M_{10}, a conjugate of this M_9, and a split extension of this M_9 by a cyclic group of order two, with nontrivial complement action.*

Proof. All permutations use the GAP right-action convention. For each occurrence, the certificate supplies a literal $p \in G$. The verifier checks every element of $p^{-1}M_{10}p$ or $p^{-1}M_9p$ against the full closure of the corresponding saved subgroup generators. Conjugation is an isomorphism, so equality of these actual element sets proves the named identification. It additionally checks that row 2 is exactly G_9 and row 7 is exactly $G_{7,11}$.

For row 5 the full actual subgroup H equals $G_{\{7,11\}}$, the setwise pair stabilizer. Its pointwise kernel $N = G_{7,11}$ is the transported M_9 and has index two. The explicit element

$$t = (3\,10)(5\,9)(6\,8)(7\,11)$$

belongs to $H \setminus N$ and has order two. The verifier checks normalization, $H = N \sqcup Nt$, and a noncommuting pair from N and t. Consequently $H = N \rtimes \langle t \rangle$ with nontrivial action, exactly the displayed colon assertion. Neither a nonsplit extension nor a direct product is silently substituted for it. □

Certificate boundary. The row joins compare the original generator arrays and the exact native labels and punctuation, via the accepted numerical bijection. Three source joins require 864 full conjugacy-element checks and 72 additional normalization checks for the complement. Nine damaged certificates are rejected in ordinary and optimized Python. No GAP call, isomorphism search, ambient subgroup census or source-count filter occurs. The two new shared models have registry identifiers 34 and 35; every earlier registry remains unchanged. The author's root colleague suggested the stabilizer route and the reuse of saved elements, as disclosed in the packet.

Combined with the separately accepted 24-row pilot, corrected 17-row extension and four standard atoms, this accounts for all nine A_5 labels and all 39 M_{11} labels in these retained current tables. It does not certify structural labels in other tables, impose uniqueness on extension notation, or complete the whole source atlas.

Literal structural descriptions in fourteen retained $L_2(q)$ tables

Private recovery component

Theorem 1. *For the exact current companion tables and the accepted numerical row bijections listed below, all 348 displayed structural descriptions hold for their actual retained subgroup representatives. This asserts the indicated decompositions, not that an extension label determines a unique group.*

q	7	8	11	13	17	19
rows	15	12	16	16	22	19
q	16	23	25	27	29	31
rows	21	23	37	16	22	29
q	32	64				
rows	24	76				

The tables are attributed to Thomas Connor and Dimitri Leemans. All nineteen native pages were inspected. Their dates range from November 2012 to June 2013, but identity with every historical file available to the 2013 source article is not claimed. The retained numerical classifications and weighted incidence checks are separate acquired components. The term "lattice" in the source headings does not supply a lattice property here.

The token trees in `native_assertions.json` preserve the literal punctuation and left association: a colon denotes the displayed split extension, a centered dot the displayed nonsplit extension, and p^e an elementary abelian group. Bare integers denote cyclic groups. In particular, row 5 of $L_2(32)$ is cyclic of order 33, whereas row 4 of $L_2(27)$ is 3^3. The dihedral subscript denotes group order. No exponent is recovered by guessing how to split a flattened decimal string.

Retained inputs. The acquired complete inventories supply actual permutation generators for all subgroup representatives and the exact weighted row bijections. The robust cached-weight calculation supplies the final orders and incidences; no previous local-class multiplicity assumption is reintroduced. The fourteen top rows use the original executed $\mathrm{PSL}(2,q)$ constructors, their literal saved ambient models and the full-order subgroup in each accepted inventory. A subgroup of a finite group having the full group order equals that group. Its selected generators need not be the constructor's generators. This route retains the named-constructor identification premise; order alone is never used to identify an arbitrary group with $L_2(q)$.

The registry acquires the 29 exact small models and decomposition certificates from the accepted A5/M11 structural components. It also uses the literal retained $L_2(8)$ model for the proper occurrence in $L_2(64)$. The latter occurrence is paid by an explicit isomorphism, not by an ambient name. The generator-edge transport theorem MC.3 in the accepted marked-coordinate umbrella is acquired in full. No ambient subgroup census is run here.

Efficient literal transport. All permutations act on the right: ab sends i first to i^a and then to $(i^a)^b$. Each concrete registry model has ordered generators and a complete positive-word closure. A stored vector f is checked to map identity to identity, to be a bijection onto the closure of the actual target generators, and to satisfy $f(xa) = f(x)f(a)$ on every right-generator edge. Induction on words gives the unique homomorphism; bijectivity makes it an isomorphism. This is precisely acquired MC.3. The cost is proportional to the number of group elements times the number of generators, not to a full multiplication table. New models are indexed by their literal data, never by a supposedly canonical extension name. Factor references point only to previously proved models, so the certificate dependency order is acyclic.

Atoms and split extensions. A cyclic atom has an element of the asserted order in a literal group of that order. An elementary abelian atom has the asserted number of commuting order-p generators and their complete closure has order p^e. A dihedral atom has r, s with orders $m, 2$, $srs = r^{-1}$, and their closure has order $2m$. These conditions prove the displayed atom. The A_4, A_5, S_3 atoms and the already certified small split types use the acquired registry.

At a new split node $H = N : K$, the verifier constructs N and K from their actual generators, checks membership in H, and transports their recursively certified descriptions. Conjugation of each generator of N by each generator of H remains in N. Finiteness makes these inclusions equalities and proves $N \trianglelefteq H$. It checks $N \cap K = 1$ and $|N||K| = |H|$; hence $H = NK$ and $H \cong N \rtimes K$. A noncommuting generator pair additionally pays the displayed nontrivial action. This does not assert that all actions with the same factors are isomorphic. Every nested node follows the exact source tree.

Lemma 1. *Let $N \trianglelefteq H$ have prime index p. The quotient map onto H/N splits if and only if some element outside N has order p.*

Proof. A complement has order p, so any of its nonidentity elements gives such an element. Conversely, an outside element of order p generates a subgroup meeting N trivially and mapping onto the quotient. It is a complement. □

For each centered-dot node in this cluster, the quotient has order 2 or 3. The actual normal subgroup and its complete factor certificate are checked as above. The verifier then checks $x^p \neq 1$ for *every* element of $H \setminus N$. This proves nonsplitting by the lemma. Failure to find a complement is not used as evidence. The prime-three case is the elementary extension of the previously accepted prime-two criterion.

Complete certificate and controls. The deterministic literal verifier pays all 334 proper-subgroup occurrences, with 70 new concrete registry models in addition to the 30 acquired model inputs. There are 387 occurrence/factor transports, 69,818 right-generator edge checks and 3,488 outside-element checks for nonsplitting. Together with the fourteen exact top-model routes, these account for all 348 rows. The source join checks every original generator array and numerical row map; none of the source names filtered an ambient group search.

Candidate generation only sought witnesses inside the retained proper subgroups. It completed in 3.7 seconds under a fixed total 120-second budget. GAP normal-subgroup and isomorphism routines suggested these witnesses; their search completeness is not a premise of the literal certificate. A startup failure caused by an empty generator list was preserved before the successful pass. One successful GAP output split an integer by its standard backslash-newline continuation. The raw file is preserved and the decoder permits removal only when both adjacent characters are decimal digits. No mathematical search was repeated for that formatting issue.

The controls damage identity images, a bijective nonhomomorphism, element coverage, source-row coverage, actual generator and row-map joins, source superscripts and punctuation, quotient orders, elementary bases, dihedral relations and the factor dependency order. All must be rejected. Separate literal controls distinguish the nonsplit cyclic groups of orders 4 and 9 from split elementary abelian groups of those orders. Ordinary and optimized Python execute the same explicit checks; assertions are not used as gates.

Scope and attribution. This is a finite structural-label recovery on the fourteen specified current tables, conditional on the already acquired complete inventories and explicit named-constructor inputs. It proves neither a new classification of subgroups of $L_2(q)$ nor the identities of unexamined historical editions. It does not complete the entire subgroup atlas or add a new original-source, conjecture, formalization or measurement credit. The normal-series, semidirect-product and generator-edge arguments are standard, with their existing project interfaces retained. Root suggested the cluster/registry strategy and reuse of MC.3; the author prepared the native ledger, witnesses and literal checker.

Literal structural descriptions in ten retained PGL_2 tables

Private recovery component

Theorem 1. *For the ten current companion tables and the accepted numerical row bijections below, all 329 displayed structural descriptions hold for the actual retained subgroup representatives. This includes the ten top rows. Each extension description asserts the displayed decomposition, rather than uniqueness of an isomorphism type from its factor names.*

q	7	9	11	13	17
rows	23	26	29	30	32
q	19	23	25	27	31
rows	36	34	43	32	44

The current PDFs are attributed to Thomas Connor and Dimitri Leemans. All eighteen native pages, including the empty continued table on the second page for $q = 11$, have been inspected. We do not identify these bytes with every historical file accompanying the 2013 article. The numerical classification, class lengths and weighted incidences are acquired from the previously accepted literal inventories and robust weight calculations. This component concerns their structural labels. The source's word "lattice" supplies no lattice-property premise.

Exact native assertions. The row ledger retains parentheses, subscripts, superscripts, centered dots and colons. Bare integers denote cyclic groups, p^e denotes an elementary abelian group, and the dihedral subscript denotes group order. Extension strings are associated from the left, as in the acquired naming interface. A colon is certified as a nontrivial split extension; a centered dot is certified as a nonsplit extension. In particular, row 14 for $q = 23$ is cyclic of order 22, while rows 28 and 29 print 2^2. Row 15 for $q = 31$ is cyclic of order 32, while the 3^2 entries for $q = 9, 27$ are elementary abelian. For $q = 9$, row 3 is $(3 : S_3) \cdot 2 \cdot 2$, with two nonsplit steps. None of these distinctions is inferred from a flattened decimal string or an expected search result.

Acquired actual models. Every source row is joined by its accepted row bijection to its actual saved permutation generators. A strict parser requires literal embedded generators for every subgroup, including proper projective subgroups; display names are not evaluated as constructors. The ambient models are the exact originally executed $\mathrm{PGL}(2, q)$ models. Their named-constructor identification and complete subgroup inventories are explicit acquired inputs. Order alone does not identify a group. The new top certificates prove the split descriptions, including $A_6 : 2$ at $q = 9$.

Shared registry and literal transports. The accepted PSL_2 structural component and its predecessor A5/M11 components supply the actual small models and their decompositions. The accepted literal alternating-group model supplies A_6; nine accepted PSL_2 ambient models supply the other simple factors. The fixed roster uses 76 such model inputs, retaining multiple models of a description when available. Each input is checked against its exact acquired bytes. The present certificate adds 43 concrete models, whose factor references only point backwards to acquired or earlier verified models.

We use the acquired generator-edge theorem MC.3. With right permutation composition, close the ordered generators under positive words. A stored map sends identity to identity, is bijective onto the closure of the target generators, and satisfies $f(xa) = f(x)f(a)$ for every generator a and every model element x. Induction on words proves that f is the unique homomorphism with these generator images; bijectivity makes it an isomorphism. This checks a number of products proportional to the order times the generator count, not a full quadratic multiplication table. The literal verifier checks all maps, including the simple factors inside the actual top groups. No transient GAP isomorphism assertion is acquired without these explicit checks.

The new decomposition certificates. The atom and extension criteria are exactly those proved in the accepted PSL_2 component. A cyclic certificate is an element of full order. A dihedral certificate consists of r, s of orders $m, 2$, with $srs = r^{-1}$ and complete generated group of order $2m$. Previously proved elementary and alternating atoms are transported from the registry.

For a split node $H = N : K$, the verifier constructs N, K from their literal generators, checks membership, and verifies their previously certified descriptions by generator-edge maps. Conjugating each generator of N by each generator of H stays in N; finiteness proves $N \trianglelefteq H$. The checks $N \cap K = 1$ and $|N||K| = |H|$ give $H = NK$ and the claimed semidirect product. A noncommuting generator pair verifies nontrivial action. The one new complement of composite order is the literal S_3 complement at row 6 for $q = 27$, in $3^2 : S_3$; it receives the same full checks.

Every new centered-dot step has index two. For a normal subgroup N of prime index p, a complement exists exactly when an outside element has order p: such an element generates a complement, and a complement provides such an element. The complete check $x^p \neq 1$ on every $x \in H \setminus N$ therefore proves nonsplitting. This is the acquired prime-index criterion, not a conclusion from a failed complement search. All 640 outside elements in the new dotted-node certificates are tested.

Finite coverage and development record. The final literal verifier checks all 329 row occurrences and the factors of all 43 new models: 374 transports and 424,622 right-generator edges. All ten ambient rows receive actual split-extension certificates, rather than merely retaining a printed top name. The original embedded arrays, the numerical row maps, the native assertion trees and their exact orders are checked before these certificates are used.

Witness generation was confined to the saved groups. The first bounded pass took 5.79 seconds and found 327 row witnesses. Its source-preparer exception for cyclic 22 in the $q = 23$ table had been applied too broadly to the two elementary abelian rows. The native reading and original outputs are preserved; the corrected exception is explicitly row 14. A separate 1.57-second pass supplied only the two missing witnesses. The other 327 witnesses and all 43 new model certificates were reused. The incomplete first build failed its full-roster guard, and no partial result was admitted. There was no ambient subgroup census or repeated classification. Candidate normal-subgroup, complement and isomorphism routines only suggest data that the literal verifier independently checks.

The damaged controls change identity images, bijections, element and row coverage, actual row and generator joins, the cyclic/power and dot/colon distinctions, an acquired seed, a quotient, dihedral relations and the factor dependency order. Each damaged input must be rejected. Separate cyclic versus elementary controls exercise the prime-two and prime-three non-splitting criterion inherited from the accepted verifier. Ordinary and optimized Python agree, with explicit checks rather than removable assertions.

Scope and attribution. The finite claim is confined to the displayed structural descriptions in these ten exact current tables, conditional on the accepted complete numerical inventories and named model inputs. It is not a new subgroup classification or a unique group-naming theorem, does not identify unseen historical editions, and does not complete the full atlas. It supplies no new original-source, question, formalization or measurement credit. The registry, normal-series and generator-edge arguments are acquired standard methods. Root proposed continued registry reuse; this component's author prepared the native ledger, new witnesses and literal source joins.

Literal descriptions in two semilinear projective tables

Private recovery component

Theorem 1. *In the exact current companion tables for* $\mathrm{P\Sigma L}_2(8)$ *and* $\mathrm{P\Gamma L}_2(27)$*, all 25 and 67 displayed subgroup structure descriptions hold for the actual representatives specified by the accepted numerical row bijections. In particular, the top groups have the displayed decompositions* $\mathrm{L}_2(8) : 3$ *and* $(\mathrm{L}_2(27) : 3) : 2$*. Each description asserts a decomposition, without asserting that factor names alone determine an isomorphism type.*

Source and exact notation. The tables are attributed to Thomas Connor and Dimitri Leemans. All four native pages were directly inspected. Their current bytes are retained; identity with unseen historical companion files is not asserted. The numerical subgroup inventories, class lengths, weighted incidences and row bijections are acquired inputs. The present result concerns the structural descriptions in these two tables only. No lattice-property premise is taken from the headings.

Section 3 of the acquired 2015 article specifies direct product $\times$, semidirect product :, nonsplit extension $\cdot$, and left association of products. Thus row 19 in the second table means $(2 \times 6) : S_3$, and row 13 means $(7 : 3 \times 2) \cdot 2$. The native ledger distinguishes 3^3, 3^2, 2^3 and 2^2 from cyclic orders. The standard notation 3^{1+2} is given its extraspecial meaning: a group of order 27 whose center, derived subgroup and Frattini subgroup coincide and have order 3. No exponent or extraspecial sign is specified by the source. Both exponent-three and exponent-nine groups occur, and the registry keeps their models distinct.

Actual groups and acquired identification. Every row is joined to literal embedded permutation generators in a saved complete inventory. The strict parser accepts only permutation cycles inside `Group`; no displayed subgroup name is executed as a constructor. The original top models were the recorded permutation images of $\mathrm{Aut}(\mathrm{PSL}(2, q))$, for $q = 8, 27$. Their exact ordered generators, executed drivers and named-model identification are retained from the accepted numerical component. The classical identification of these automorphism groups with the stated semilinear groups is an explicit inherited premise, not inferred from orders. The new certificates also verify the printed top decompositions.

Registry and transport. The accepted PSL_2 and PGL_2 components supply 36 actual models and already proved descriptions needed here. Multiple models of a description are retained whenever available. This packet adds 26 concrete models, with references only to acquired or earlier verified factors. The acquired marked generator-edge criterion MC.3 verifies each transport: the stored bijection sends identity to identity and satisfies $f(xa) = f(x)f(a)$ on every right-generator edge. Induction on words makes it a homomorphism, and bijectivity makes it an isomorphism. All maps are checked on literal arrays; no transient GAP isomorphism result is accepted without these checks. This uses order times generator count, rather than quadratic multiplication tables.

Lemma 1. *Let $H = \langle x, y\rangle$ have order 27, and put $z = x^{-1}y^{-1}xy$. Suppose z is central of order 3, $x^3, y^3 \in \langle z\rangle$, and $Z(H) = \langle z\rangle$. Then H is extraspecial.*

Proof. The quotient by $Z = \langle z\rangle$ is generated by two commuting order-dividing-three elements and has order nine, so it is C_3^2. Since z is a commutator and this quotient is abelian, $H' = Z$. Every maximal subgroup contains Z. Indeed, a subgroup of order nine not containing Z would have product with Z equal to H; that subgroup is abelian, and Z is central, contradicting $H' \neq 1$. A subgroup of order three not containing Z extends to its proper order-nine product with Z, so cannot be maximal. Consequently the maximal subgroups are exactly the inverse images of the four lines in H/Z. Their intersection is Z, proving $\Phi(H) = Z$. The elementary fact that groups of order p^2 are abelian follows from the class equation and the fact that a group with cyclic central quotient is abelian. □

The literal verifier checks every hypothesis, including the exact center by commuting with the two generators. Two new order-27 models have respective element-order spectra $\{1, 3\}$ and $\{1, 3, 9\}$. They account for the source's unqualified extraspecial descriptions without identifying two nonisomorphic groups.

Products and dotted extensions. For $H = N : K$, the acquired criterion checks literal membership, normality of N on generators, both factor descriptions, $N \cap K = 1$ and $|N||K| = |H|$. These imply the split decomposition; a noncommuting generator pair pays nontrivial action. For $H = N \times K$, the same product and intersection checks, together with commutation of every pair of factor generators, make $(n, k) \mapsto nk$ an isomorphism $N \times K \to H$. There are nine new direct-product models. Each centered-dot step has index two: the complete check $x^2 \neq 1$ for all outside elements proves that there is no complement. The sole new dotted model has 42 outside elements.

Finite certificate and controls. The witness-only GAP pass returned in 2.97 seconds, with all 92 row witnesses; it did not enumerate ambient subgroups. Normal-subgroup, complement and isomorphism routines only propose certificates. The literal verifier independently checks all 92 source joins, 26 new models and 137 transports, comprising 465,483 right-generator edges. Both top rows receive full certificates. Every proper subgroup generator is parsed in its actual ambient action; no smaller canonical action is substituted for an embedded subgroup.

The controls damage identity images, a bijective nonhomomorphism, element and row coverage, source row maps, actual generators, superscripts, the extraspecial atom, dot versus colon, product association, acquired models, quotient index, elementary basis orders, acyclic references, the commutator and a direct factor. Each must be rejected. Independent all-element product controls accept $C_3 \times C_2$ and reject the noncommuting factors in S_3. Both actual extraspecial spectra are checked, while the abelian groups C_{27} and $C_9 \times C_3$ are rejected. The inherited prime-index controls distinguish C_{p^2} from C_p^2 for $p = 2, 3$. Ordinary and optimized Python use the same explicit guards.

Coverage and attribution boundary. This is a proof of the displayed existence descriptions in 92 exact current rows, conditional on the acquired complete inventories, numerical bijections and named-model interfaces. It neither specifies a unique extension from its factors nor gives a new subgroup census. The extra registry models and their actual transports are the new finite evidence; standard naming conventions, prior atomic descriptions and the marked-edge lemma remain attributed inputs. Root suggested continued registry reuse; the component author prepared the native ledger, certificates, new literal tests and source joins. The whole atlas and unseen editions remain outside this component. No original-source, question, formalization or measurement count changes here.

Two further field-extension structural tables

Private recovery component

Theorem 1. *All 47 displayed structural descriptions in the current companion table for* $\mathrm{L}_2(16) : 2$*, and all 30 in the table for* $\mathrm{P\Gamma L}_2(32)$*, hold for the actual subgroup representatives specified by the accepted numerical row bijections. The top rows receive actual split-extension certificates. An extension description asserts the indicated decomposition, not unique determination of an isomorphism type by its factor names.*

Exact source and model interfaces. These two tables are attributed to Thomas Connor and Dimitri Leemans. All four native pages were inspected, including the empty continued table on the second $q = 32$ page. Current bytes are retained, without asserting identity with all historical companion editions. The complete numerical inventories and source-row bijections are acquired results. No lattice-property premise is inferred from the headings.

The native ledger distinguishes cyclic order 33 at $q = 32$, row 15, from the elementary abelian powers 2^e, $e = 2, 3, 4, 5$. For $q = 16$, row 14 is $(2^3 \cdot 2) : 2$, row 23 is $(2 \times 4) : 2$, and row 9 is $D_{34} \cdot 2$. For $q = 32$, row 5 is $(11 : 5) : S_3$ and row 8 is $(2^4 : 5) \times 2$. All products associate from the left, as specified in Section 3 of the acquired 2015 naming article. A dot denotes nonsplitting, a cross direct product, and a colon a split extension. The source targets do not select between alternative actual models of the same description.

The $q = 16$ representatives are extracted with the exact acquired strict literal-state parser from the completed normal-restriction state. Its named-model transport identifies the target with the projective $\mathrm{SL}_2(16)$ action extended by the square of the field Frobenius; the full order is 8160. The already accepted transport and its full verification are acquired here. For $q = 32$, every subgroup and ambient group is parsed from literal embedded permutation cycles in the saved complete inventory. The originally executed ambient model was the permutation image of $\mathrm{Aut}(\mathrm{PSL}(2, 32))$. The named-constructor and classical semilinear identification remain explicit inherited premises. No proper subgroup is reconstructed from a display name, and group order alone is never used as its identification.

Shared certificates. The accepted PSL_2, PGL_2 and preceding semilinear components supply 41 required concrete models and descriptions. Two previously accepted PSL_2 ambient actions supply the simple factors $\mathrm{L}_2(16)$ and $\mathrm{L}_2(32)$. The present evidence adds 18 models; every factor reference points to an acquired or earlier verified model. The acquired MC.3 generator-edge criterion checks each actual transport: a bijection fixes identity and respects right multiplication by every ordered generator, hence is a homomorphism by induction on words. No full quadratic multiplication table is required.

Actual decomposition tests. All criteria are those already proved in the acquired components. The literal verifier checks the order and complete closure of each group. For a split node $H = N : K$, it checks the two factor descriptions by actual generator-edge bijections, normality of N, $N \cap K = 1$ and $|N||K| = |H|$. A noncommuting generator pair checks nontrivial action. Composite complements, including the S_3 in $(11 : 5) : S_3$, undergo these same tests. Prime-five complements in the field-extension top group and its solvable subgroups are also tested literally.

For a direct node, all factor generators commute; together with the intersection and product-size checks, this proves the displayed direct product. There are seven new direct-product models. Each dotted step has index two and is checked by the complete outside-involution obstruction: a complement exists exactly when an outside element has order two. The new model $2^3 \cdot 2$ has eight outside elements, all with nonidentity square. Other dotted descriptions transport already accepted models. No failure of a search routine is used as a nonsplitting proof.

Finite evidence and checks. The single bounded witness pass returned all 77 row certificates in 2.37 seconds, without an ambient subgroup census. Normal-subgroup, complement and isomorphism routines only suggest data. The independent literal verifier checks all 77 source joins, all 18 new models, 112 transports and 679,553 right-generator edges. The two top rows have full certificates, and every source row is tied to its exact saved embedded generators before a structural assertion is checked.

The damaged controls alter identity images, a bijective nonhomomorphism, element and row coverage, source row maps, actual generators, the cyclic 33 and power 2^5 distinction, a centered dot, product association, quotient index, acquired models, complement generators, dependency order and a direct factor. Each fails an explicit guard. Independent all-element controls distinguish $C_3 \times C_2$ from the noncommuting factors in S_3; prime-index controls distinguish C_{p^2} and C_p^2 for $p = 2, 3$. Ordinary and optimized Python must agree. The complete literal state parser is acquired unchanged, including its rejection of nonliteral constructors and invalid embedded cycles.

Scope and attribution. The finite result concerns 77 exact current structural descriptions, conditional on accepted complete subgroup inventories, numerical source bijections and named-model interfaces. It does not assert unique extension naming, a new subgroup classification, completeness of the whole atlas, or identity of unseen editions. Standard product and transport arguments are acquired; the new evidence consists of the fixed native ledger, decomposition witnesses and literal joins. Root authorized continued registry reuse; the component author prepared these actual certificates and checks. No original-source, question, formalization or measurement count changes here.

Five structural tables related to A_6

Private L054 recovery component

Theorem 1. *Every displayed structural description in the five current companion tables S_5, A_6, M_{10}, $P\Sigma L_2(9) \cong S_6$, and $P\Gamma L_2(9)$ holds for the actual subgroup class specified by the retained numerical row map. The tables have respectively $19, 22, 25, 56, 65$ rows, hence 187 descriptions in total. An extension label asserts the indicated decomposition, without claiming that its factor names determine a unique isomorphism type.*

Source and acquired interfaces. The tables are attributed to Thomas Connor and Dimitri Leemans. All eight native source pages were directly read. Current PDF bytes, not an assumed historical edition identity, define these assertions. Their complete numerical classifications and source row maps are acquired from the accepted robust-weight, M_{10}, and final-six components. The filenames `psigmal29` and `pgammal29` are locators; the source titles and exact native row syntax are recorded separately.

Products associate from the left, following the acquired naming article. A dot means non-splitting, a colon a split extension, and a cross a direct product. For example the $P\Gamma L_2(9)$ rows 6, 9 and 13 are $(((3 : S_3) \cdot 2) \cdot 2) : 2$, $((3 : S_3) \cdot 2) \cdot 2$, and $(D_{10} \cdot 2) \times 2$. Row 17 is $Q_8 : 2^2$, not $Q_8 : 22$ with cyclic complement of order 22. The native elementary abelian powers $2^2, 2^3, 3^2$ are explicit syntax nodes. The dihedral subscript denotes the total group order.

Embedded representative reconstruction. The M_{10} saved array and the $P\Gamma L_2(9)$ completed state contain literal embedded generators. Some earlier S_5, A_6, S_6 arrays instead print `SymmetricGroup` or `AlternatingGroup` on an explicit point set. We do not assume that a display name preserves the producer's hidden embedding. Those expressions supply candidate reconstructions: the full symmetric or alternating group on exactly that point set, fixing its complement. All other function expressions are rejected.

Here is the independent check that pays this boundary. Close each candidate subgroup literally and check membership in the actual ambient permutation group. Enumerate its conjugacy orbit under the ambient generators, as sets of literal permutations. These orbits are pairwise disjoint, have the acquired class lengths, and their number equals the acquired complete number of subgroup classes. Consequently their union is the complete actual subgroup set. For each retained representative, select its maximal proper members in this finite set by descending order and subset tests. Count them by ambient class. If the downward count is a and the two ambient class lengths are c_H, c_K, the upward count is ac_H/c_K by counting incidence pairs. The checker verifies integrality and both counts against every source pair, including nonedges. This proves the actual source join without an assumption about the old printed embeddings. All 8,831 ordered pairs pass; no new ambient subgroup search is used.

Reusable structural certificates. Forty concrete certified models are acquired from the preceding PSL_2, PGL_2, semilinear, field-extension and Mathieu/alternating atom packets. In particular M_{10} and M_9 retain their exact point-stabilizer interfaces in the standard M_{11} action; they are not identified by orders 720 and 72 alone. Two elementary additional models are S_4, S_6: the literal groups on four and six letters have orders $4!, 6!$, so equal the full symmetric groups on those sets. These are not classifications inferred from the orders of arbitrary groups.

Ten further models carry actual decompositions. Every factor is an acquired or earlier verified model. A source occurrence is identified with its model by a literal bijection fixing identity and preserving all right-generator edges. The acquired MC.3 argument proves that such a bijection is a homomorphism, by induction on words. The same argument transports every factor certificate. Identical printed labels can index several different models; a successful actual bijection is always required.

The split tests check a normal subgroup N, a complement K, their verified factor types, $N \cap K = 1$ and $|N||K| = |H|$. A noncommuting pair of factor generators verifies nontrivial action. Direct nodes additionally require all cross-generator pairs to commute. These conditions prove the claimed products. A dotted prime-index node is checked by listing every outside element and requiring that its pth power is nonidentity: a complement at index p exists exactly when an outside element has order p. The only new dotted model has index two and eight outside elements. Other dotted descriptions use acquired model transports. No failed complement search is a nonsplitting proof.

Finite evidence and validation. The literal reader using the acquired criteria checks all 187 row joins, ten new models, 206 transports and 31,324 right-generator edges. Candidate generation uses only the retained actual groups. Its changed, bounded pass returned in 3.45 seconds. An earlier unused recursive candidate had a trivial split action; the unchanged literal reader rejected it. That failed check and its exact code are preserved. Candidate selection was strengthened to require nontrivial action, with no relaxation of the final criterion.

Damaged certificates test identity, bijective nonhomomorphisms, missing or duplicate rows and elements, altered actual generators/maps, powers, dots, association, factors and cyclic dependencies. Additional controls reject duplicate reconstructed classes, a changed source weight, a missing ambient generator list and unsafe named expressions. Independent direct product and prime-index examples distinguish the positive and negative cases. Ordinary and optimized Python must agree. No discovery routine is rerun by the frozen checker.

Scope. The proof acquires the prior complete numerical classifications and named model definitions explicitly. Its new result is the 187 exact current structural descriptions and their embedded source joins. It neither asserts unique extension names nor completes all structural atlas rows. No original-source, question, formalization or measurement count changes here. The author prepared the finite application under root's registry reuse instruction; the standard structural and transport arguments are acquired, not claimed as new mathematics.

Five remaining PSL_2 extension tables

Private L054 structural recovery component

Theorem 1. *The five current companion tables specified below contain a total of 364 rows. Each structural description has an actual decomposition certificate for its subgroup in the accepted numerical row map, with the explicit Q_{12} convention below. A displayed extension asserts existence of that decomposition; factor names are not asserted to determine a unique group.*

`psigmal227`	38 rows
`psl225d2`	39 rows
`pgammal216`	69 rows
`psigmal225`	103 rows
`pgammal225`	115 rows

Source and acquired inputs. The current tables are attributed to Thomas Connor and Dimitri Leemans. All fifteen native pages were directly inspected, and the first `psl225d2` page was also inspected at higher resolution. We acquire the accepted complete numerical classifications, their exact row bijections, and the named ambient-model transports. We do not assume that all current PDF bytes are identical to a historical 2013 edition. No lattice-property claim follows from the tables' headings.

Every imported proper subgroup and ambient group is given by literal embedded permutations, using either the acquired strict completed-state parser or the literal saved-array parser. There is no reconstruction from a printed group name. The three restriction inputs retain their actual projective defining-model transports. The two full semilinear ambient inventories retain the accepted automorphism-model identification; these are acquired classification interfaces, not conclusions from their orders.

Products associate from the left, as explicitly prescribed in Section 3 of the 2015 naming article. A dot denotes a nonsplit extension, a colon a split extension, and a cross a direct product. Native exponents remain explicit syntax nodes. In `pgammal225`, row 47 is cyclic 26 and row 52 is cyclic 24; neither is an elementary abelian power. In `pgammal216`, row 38 is 2^4. In `psigmal227`, 3^{1+2} is extraspecial of order 27. In `psl225d2`, row 21 is $D_8 \cdot 2$, with a centered dot. For example the top `pgammal216` row means $(\mathrm{L}_2(16) : 2) \cdot 2$. The caption is not substituted for the displayed row.

Reusable literal certificates. Ninety-seven concrete models and their descriptions are acquired from the accepted PSL_2, PGL_2, semilinear, field-extension and A_6-related components. Different concrete models may have the same description. Each use requires an actual isomorphism, checked by the acquired MC.3 criterion: a bijection fixing identity and preserving every right-generator edge is a homomorphism, by induction on words. Factor references point only to acquired or previously verified models. The present application adds 42 models and uses 430 such transports.

The existing split and direct-product criteria apply unchanged. They check a normal subgroup N, a verified complement K, trivial intersection and $|N||K| = |H|$. A colon additionally has a noncommuting pair of factor generators; a cross has all cross-generator pairs commuting. For prime index p, a complement exists exactly when an outside element has order p. The literal reader tests every outside element's pth power. These tests pay 16,368 outside elements in the new models.

Lemma 1 (Composite-quotient lift tests)**.** *Let $N \triangleleft H$ be finite, with $|H : N| = 6$.*

1. *If H/N is cyclic, fix xN generating it. The extension splits if and only if some $a \in Nx$ satisfies $a^6 = 1$.*

2. *Suppose xN, yN generate H/N and satisfy $(xN)^2 = (yN)^3 = (xyN)^2 = N$. The extension splits if and only if some $a \in Nx, b \in Ny$ satisfy $a^2 = b^3 = (ab)^2 = 1$.*

Proof. In the cyclic case, a section supplies a lift of the fixed generator of order six. Conversely any displayed lift with sixth power one has order exactly six, since its image does, and generates a complement.

In the second case the presentation $\langle u, v \mid u^2 = v^3 = (uv)^2 = 1\rangle$ is S_3: $uv = v^{-1}u$ reduces every word to v^i or uv^i, $0 \le i < 3$, and the usual transposition and 3-cycle give all six permutations. Thus the specified quotient is S_3. A complement lifts its two fixed quotient generators and gives the stated relations. Conversely these relations define a homomorphism from S_3 into H. Its image maps onto the quotient of order six, so it has exactly six elements and meets N trivially. Hence it is a complement. □

The literal implementation checks normality first. For the cyclic case it verifies that the six cosets Nx^i cover H and that $x^6 \in N$. For the symmetric case it checks the three quotient relations and that $N, Ny, Ny^2, Nx, Nxy, Nxy^2$ are six disjoint cosets covering H. It then tests all elements of the indicated coset, or all pairs in the two indicated cosets. Each fails at least one required relation. This proves nonsplitting without assuming completeness of a complement-search routine. Eight new composite-quotient models use 39,312 literal lift tests. For instance $(2^4 : 5 : 2) \cdot S_3$ and $D_8 \cdot 6$ require different quotient presentations; the verifier does not identify quotients merely from order six.

The source atom Q_{12}. The three occurrences are `pgammal216`, row 46; `psigmal225`, row 59; and `pgammal225`, row 76. We make the meaning precise as the dicyclic group of order twelve:

$$\langle a, b \mid a^6 = 1,\ b^2 = a^3,\ b^{-1}ab = a^{-1}\rangle. \tag{1}$$

Every word reduces to a^i or a^ib, $0 \le i < 6$, so this presentation has at most twelve elements. The actual certificate gives generators a, b of a twelve-element group with $|a| = 6$ satisfying these relations; therefore it identifies the presented group exactly. The installed GAP primary documentation explicitly defines the dicyclic presentation in this way and reserves generalized quaternion for its 2-power-order case. The source article does not define Q_{12} separately. Our exact presentation fixes the nomenclature qualification rather than claiming that this is a generalized quaternion group of 2-power order.

Finite validation. A bounded witness-generation pass returned all 364 candidates in 10.42 seconds. It uses only the retained groups, not an ambient subgroup census. Normal-subgroup, complement and isomorphism routines propose witnesses; none of their failed searches proves nonsplitting. The independent permutation reader checks complete closures, all row/source joins, 42 new descriptions, 430 isomorphisms and 664,327 right-generator edges. All five top rows have literal certificates, including their nonsplit or split extension assertions.

Independent small controls build the regular dicyclic action directly from multiplication on pairs $(i, j) \in \mathbb{Z}/6 \times \{0, 1\}$:

$$(i, j)(k, l) = (i + (-1)^j k + 3jl \pmod 6,\ j + l \pmod 2).$$

Its central order-two quotient passes the nonsplit S_3 test. The explicit split product $S_3 \times C_2$ fails that test. Likewise C_{12} over its subgroup of order two passes the nonsplit cyclic-six test, while $C_6 \times C_2$ fails. Wrong quotient generators and quotient types are rejected. A dihedral group of order twelve fails the dicyclic relations. Prime-index and direct-product controls are retained as well.

Damaged full certificates alter identity images, right-generator relations, row or element coverage, actual subgroup indices and generators, source powers and dots, product association, acquired model data, factor references, complements, quotient lifts and the dicyclic witness. Every damage must fail its specific check. Ordinary and optimized Python must produce identical mathematical reports. An initial harness expected a later rejection for an invalid S_3 quotient generator; the actual quotient-relation guard rejected it earlier. That harness failure and the corrected expected guard are preserved. No scientific condition was weakened, and the successful discovery was not repeated.

Scope and attribution. This component supplies actual structure witnesses for these 364 current rows, with the explicit Q_{12} presentation qualification, using the previously accepted numerical classifications and literal source maps. The named standard models, original complete inventories and generic transport/decomposition results remain acquired premises. The new lift-test proof, exact applications and controls were authored here under root's registry-reuse task. There is no claim of unique extension labels, new ambient subgroup enumeration, historical byte identity, whole-source completion, question resolution, formalization or measurement credit.

Structural certificates for four alternating and symmetric tables

Private L054 recovery component

Theorem 1. *Every structural description in the current companion tables for A_7, S_7, A_8, S_8 has an actual decomposition certificate in its numerical row class. These four tables contain respectively $40, 96, 137, 296$ rows, for a total of 569. A description asserts the displayed decomposition; it need not determine a unique group.*

Source and acquired premises. The tables are attributed to Thomas Connor and Dimitri Leemans. All 25 native pages were directly inspected. We acquire the previously accepted complete ambient class counts and numerical row maps, together with the accepted concrete structural-model registry. These are current companion data; historical 2013 PDF-byte identity is not asserted. The 2015 naming article specifies association from the left. We preserve this convention and every native dot, colon, cross and exponent. A dot means nonsplit extension, a colon split extension, and a cross means direct product. For example, the S_8 row 13 means $(((A_4 \times 2) \times 2) : S_3) \cdot 2$. Its outer dot is not replaced by a colon. Row 200 has 4^2, meaning $C_4 \times C_4$, and row 119 has the unsigned extraspecial atom 2^{1+4}. We assert neither plus nor minus type for that atom. The inherited Q_{12} convention is the explicit dicyclic presentation from the accepted registry, not an inferred quaternion 2-power convention.

Concrete registry and transports. The input has 88 acquired concrete models and four elementary standard models, for A_7, S_7, A_8, S_8. Different concrete models may share a structural description. The new application adds 65 models. For a standard symmetric model we use a full cycle and a transposition; for an alternating model we use $(1\,2\,i)$, $3 \le i \le n$. The literal closure has order $n!$, respectively $n!/2$, and the alternating generators are even. Thus these closures are exactly the groups of all permutations, respectively all even permutations, on the indicated point set.

Every reuse has an explicit bijection fixing identity and preserving right multiplication by every model generator. Induction on generator words proves that it is a homomorphism, hence an isomorphism. This is the acquired MC.3 criterion; no group-name equality substitutes for it. The literal checker expands candidate generator maps, then checks all 692 resulting bijections and 510,303 right-generator edges.

For a split description it checks $N \lhd H$, a named subgroup K, $N \cap K = 1$ and $|N||K| = |H|$. A colon also has a noncommuting pair of factor generators; for a cross all such pairs commute. Normality and commutation on generators imply the corresponding whole-group facts. For the new prime-index nonsplit descriptions, every outside element fails $x^p = 1$: an index-p complement exists exactly when an outside element has order p. Factor references are acyclic and already verified.

Lemma 1 (Two additional atom criteria)**.** *1. Two commuting elements of order four which generate a group of order sixteen identify that group with $C_4 \times C_4$.*

2. Suppose $|H| = 32$ and z has order two. If $Z(H) = \langle z \rangle$, every square lies in $\langle z \rangle$, and generator commutators lie in $\langle z \rangle$ with at least one equal to z, then H is extraspecial.

Proof. In the first case the defining commuting generators give a surjection from $C_4 \times C_4$ onto H, and equal orders make it an isomorphism. In the second case the normal closure of the generator commutators is H', so $H' = Z(H) = \langle z \rangle$ and $H/Z(H)$ is elementary abelian. For completeness, $\Phi(H) = H'H^2$: every maximal subgroup of a finite 2-group is normal of index two, hence contains $H'H^2$; and hyperplanes in the elementary abelian quotient $H/H'H^2$ have trivial intersection. The asserted maximal-subgroup fact follows from the normalizer condition: a proper subgroup is strictly contained in its normalizer, by induction modulo the center (if the center is not contained in the subgroup it already enlarges the normalizer). A maximal subgroup is therefore normal, and its quotient has order two. Thus $\Phi(H) = \langle z \rangle$ as well, which is precisely the extraspecial condition. □

The checker computes the center by testing every group element against every generator, checks every square, and checks all generator commutators. It does not infer an extraspecial type from order alone.

Actual embedded rows despite printed point-set names. Some retained arrays print `AlternatingGroup([points])` or `SymmetricGroup([points])`. We treat their conventional point-set generators only as candidates, without claiming that GAP's display preserved the original producer's hidden embedding. We close the ambient permutations and all candidate subgroups, check membership, and enumerate the ambient conjugacy orbit of every candidate. The resulting classes are pairwise disjoint and their number equals the acquired complete class count. They therefore cover all subgroup classes.

For efficiency each actual ambient permutation is assigned its unique closure index. Conjugation by an ambient generator is an exactly computed permutation of those indices. Subgroups become subsets of this index set; this is a bijective encoding of the same literal subgroups, preserving membership, inclusion and conjugation. No class decision is approximate.

All actual subgroups now occur in the conjugacy orbits. For each retained parent H, scan proper subgroups in descending order and retain $K < H$ exactly when it lies in no already retained larger subgroup. This gives precisely the maximal subgroups of H. If a such children belong to a class with ambient length c_K, double counting conjugate incidences gives $b = ac_H/c_K$ parents over each child. Every source row order, class length, and both weighted directions for every ordered row pair are checked. This independently pays the actual source-class assignment of every candidate, including the point-set expressions.

Finite execution and independent controls. One bounded witness pass returned all 569 candidates in 12.56 seconds. It searched only the retained groups for decomposition and transport witnesses. Its failed searches supply no proof premise. The final reader requires all source occurrences and actual maps, verifies 65 new models, 19 new direct products, one C_4^2 model and one extraspecial model of order 32. Its prime-index nonsplitting tests inspect 320 outside elements. No ambient subgroup census is repeated.

The embedded audit has the following exact finite outcomes:

Table	Classes	Actual subgroups	Edges	Row pairs
A_7	40	3786	98	1600
S_7	96	11300	290	9216
A_8	137	48337	419	18769
S_8	296	151221	1080	87616
Total	569	214644	1887	117201

The new ambient-index encoding is additionally compared against the previous implementation using literal permutations on the complete S_5, A_6 and $\mathrm{P\Sigma L}_2(9)$ fixtures. Their class counts, subgroup counts, edges, every source pair and exact operation counts agree.

Independent atom controls construct regular actions on $\mathbb{F}_2^5$ using the bilinear central cocycles

$$(x, c)(y, d) = (x + y, c + d + x_2y_1 + x_4y_3 + \epsilon(x_1y_1 + x_2y_2)), \qquad \epsilon = 0, 1.$$

Both order-32 models pass. All fifteen nonzero parity assignments on the four noncentral generators yield actual order-16 kernels, with intersection exactly the center of order two. A separate coordinate action of $\mathbb{Z}/4 \times \mathbb{Z}/4$ passes. Dependent generators, $C_4 \times C_2 \times C_2$, elementary abelian order 32, and a proposed noncentral center generator are rejected.

Damaged full certificates test map identity and generator edges, missing or duplicate occurrences, wrong embedded generators and row maps, source powers, a dot changed to a colon, association, cyclic references, complements, quotient data and both new atom witnesses. Separate embedded controls reject duplicate classes, an outside ambient subgroup, wrong class lengths, row maps and incidence weights. Unsupported named group constructors are rejected. The previously acquired prime-index, composite-quotient and direct-product small controls remain executable. Ordinary and optimized Python must give identical complete reports.

Scope. These are 569 new structural row joins to current qualified data. The acquired complete class counts and earlier concrete named models remain explicit premises. The unsigned extraspecial and inherited dicyclic conventions remain visible. No assertion of canonical extension labels, historical file identity, a conjugacy-poset lattice property, whole-source completion, question resolution, formalization or measurement follows. The present proof, implementation and applications were authored under root's instruction to reuse the concrete registry. Root reviews the finished component independently.

Four linear and unitary structural tables

Private L054 finite application

Theorem 1. *All 265 structural rows of the four current companion tables below have literal decomposition certificates for their actual numerical row groups, relative to the explicitly acquired named-model interfaces. The descriptions assert existence of the displayed decompositions, not uniqueness of a group with that extension label.*

Locator	Ambient description	Rows
`psl33`	$L_3(3)$	51
`psl33d2`	$L_3(3) : 2$	78
`psu33`	$U_3(3)$	36
`pgammau33`	$U_3(3) : 2$	100

Source and numerical inputs. The current PDFs are attributed to Thomas Connor and Dimitri Leemans. All eleven native pages were directly inspected. We acquire their accepted complete numerical inventories, exact source-row bijections and named ambient-model interfaces. Every imported subgroup is an actual literal permutation group: the saved arrays contain only `Group` expressions, with degrees $13, 104, 91, 91$ respectively. No printed group constructor is reconstructed inside a proper row. The complete earlier numerical classifications are not inferred afresh from the source counts.

Native products associate from the left unless parentheses say otherwise, as prescribed in the 2015 naming article. In `pgammau33`, row 3 is $3^{1+2} : (D_8 \cdot 2)$; its right factor is parenthesized. It is not $(3^{1+2} : D_8) \cdot 2$. Source 4^2 remains a direct power $C_4 \times C_4$, and 2^{1+4} and 3^{1+2} are the unsigned extraspecial atoms paid by the acquired registry. A dot, colon and cross retain their nonsplit, split and direct meanings. The inherited explicit Q_{12} dicyclic convention remains available, with no new nomenclature claim. Current PDFs are not assumed byte-identical to a historical edition.

Two named classical inputs. We explicitly retain the documented identities of the constructors `PSL(3,3)` and `PSU(3,3)`. GAP Reference Manual 50.2-12 and 50.2-14 specify the projective determinant-one linear and unitary groups, with the default permutation action on projective lines. These are construction interfaces, not isomorphism assertions deduced from order. Their original accepted complete executions are acquired: the exact drivers, clean successful receipts, all recorded input/output hashes and the resulting literal ambient generators are checked. Those generators are identical to the retained numerical top groups. They supply the concrete atoms $L_3(3)$ and $U_3(3)$. No new constructor, automorphism-group calculation or subgroup census is run. The extension rows are separately proved to have the indicated normal subgroup and complement by actual certificates.

Registry application. The application uses 69 acquired concrete structural models and the two named classical inputs. It adds 22 models. Every factor and occurrence map is an explicit bijection preserving identity and all right-generator edges. The acquired MC.3 word-induction criterion proves each map an isomorphism. Different models with the same structural name remain distinct registry entries until an actual map is supplied. Thus model-name matching alone pays no source row.

Proof of the finite application. The literal verifier closes every listed permutation group. New model references are acyclic. In each split model it checks the actual normal subgroup N, its already verified structural description, a subgroup K with the displayed complement description, $N \cap K = 1$, and $|N||K| = |H|$. The product is all of H. A noncommuting generator pair certifies the displayed nontrivial semidirect action. Normality and commutation tests on generators extend to the whole generated groups.

There are twenty new split and two new nonsplit models. The latter use the already proved composite-quotient lift criterion: for cyclic quotient of order six, fixed generator xN lifts to a complement exactly when some $a \in Nx$ has $a^6 = 1$; for quotient S_3 with fixed generators xN, yN, a complement exists exactly when some lifts satisfy $a^2 = b^3 = (ab)^2 = 1$. The checker first verifies the actual quotient relations and six cosets, then tests every required lift or lift pair. The $Q_8 \cdot 6$ model uses eight lift checks, and the $(D_8 : 2) \cdot S_3$ model uses 256 pairs. Their failure of the splitting relations proves nonsplitting, independently of any unsuccessful complement search.

Every source occurrence is transported to its exact embedded group under the acquired numerical row map. The source text, native syntax tree, order, subgroup index and generator arrays all agree. There are 307 checked transports and 146,010 right-generator edges. These checks prove the stated finite result, using only the explicit acquired interfaces.

Execution and serialization. One bounded candidate pass completed all 265 rows in 4.65 seconds. Normal-subgroup, complement and isomorphism software only proposed witnesses; the final proof uses their literal successful certificates. The replay does not repeat candidate discovery or numerical classification.

Two GAP output strings contain physical backslash–newline continuations between a comma and the next digit. The initial reader permitted only a continuation between two digits and stopped before building a certificate. That failed reader and the exact raw outputs are retained. The final reader permits only these two numeric positions, removes only the physical continuation, and parses the resulting JSON. All permutation entries, model references and equations are then checked normally. Malformed or nonnumeric continuations are rejected. No scientific output was changed.

Controls and scope. Damaged certificates test missing/duplicate rows, incorrect identity or right-generator edges, changed subgroup generators and row maps, source exponents, dots and parentheses, wrong complements, cyclic factor references, and both new quotient witnesses. Changed named-classical generators fail the exact acquired-input check. The actual constructor-acquisition routine also rejects incomplete execution, wrong constructor, bad return code and changed-input receipts. Native-expression controls distinguish the parenthesized right factor and reject unmatched or empty parentheses. The existing independent prime-index, direct-product, dicyclic and composite-quotient small controls remain executable. Ordinary and optimized Python must produce identical complete reports.

The result is 265 current structural row joins. Numerical completeness, named constructors and earlier model proofs are retained inputs; no independent reproof of those software interfaces is claimed. There is no new ambient census, unique-name assertion, historical-byte claim, whole-original disposition, question, formalization or measurement credit. The finite application and source-sensitive implementation are authored here under root's registry-reuse task and await distinct review.

Structural certificates for M_{12} and $M_{12} : 2$

Private L054 finite application

Theorem 1. *All 147 rows of the current M_{12} companion table and all 213 rows of its $M_{12} : 2$ table have the displayed structural descriptions, relative to the explicitly acquired numerical and named-model interfaces. Each description asserts existence of its displayed decomposition, without asserting that the label determines a unique isomorphism type.*

Exact source and acquired groups. These current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All fifteen native pages were directly read. We acquire the accepted complete numerical inventories, source-row bijections and literal embedded subgroups. Every group expression in both imported arrays is a literal `Group` of permutations; no named constructor is substituted inside a proper row. Their degrees are 12 and 396. The original successful `MathieuGroup(12)` execution, its exact driver, clean receipt and recorded input/output hashes are acquired. Its top generators equal the current twelve-point top generators. GAP Reference Manual 50.1-13 supplies the named Mathieu constructor interface explicitly; it is not inferred from group order. Earlier concrete M_9, M_{10}, M_{11} and other structural models are acquired with their complete accepted proofs.

The 2015 article's left-association convention is retained. In particular, $3 : S_3 \cdot 2 \cdot 2$ in M_{12} rows 38 and 40 has two dots, whereas $2^{1+4} \cdot 2^2$ in $M_{12} : 2$ row 30 has an elementary abelian quotient of order four. We preserve $4^2 = C_4^2$, unsigned extraspecial atoms and the acquired dicyclic Q_{12} convention. There is no assertion that these current PDF bytes are a historical edition.

Lemma 2 (Literal faithful right-coset transport)**.** *Suppose a finite permutation group G, its order n, an actual subgroup H, and other actual subgroups $K_j \leq G$ are acquired. Let $|H| = h$ be verified by literal closure and set $k = n/h$. Suppose k permutations t_i, starting with identity, are verified words in the ordered generators of G, and $t_i t_j^{-1} \notin H$ for distinct i, j. For each generator x of G, let a permutation p_x of $\{1, \ldots, k\}$ satisfy*

$$t_i x t_{p_x(i)}^{-1} \in H \quad \text{for all } i.$$

These permutations induce the actual coset homomorphism. If their literal closure has order n, that homomorphism is faithful. The same equations for each generator of each K_j identify all their images exactly.

Proof. The words prove $t_i \in G$. The distinct right cosets Ht_i exhaust G by the acquired index. The displayed membership is exactly $Ht_i x = Ht_{p_x(i)}$, hence gives right multiplication on this complete coset set. With right-action permutation composition this is a homomorphism. Its image order equals $|G|$, so the kernel is trivial. Since each further generator lies in the acquired G, its equations determine its unique image under this same map. Thus the actual subgroup embeddings are preserved. Equal orders are used only after constructing the homomorphism. □

The actual compact model. For the degree-396 group we use the retained subgroup at source row 3, of order 7,920. Its M_{11} name is unnecessary for the coset argument. The acquired ambient order is 190,080, so the index is 24. Candidate breadth-first words supply representatives. The independent literal reader checks their words, 276 pairwise coset distinctions and 11,112 coset equations for the ambient generators and all 213 subgroup generator lists. The image closes to 190,080 elements. All later certificates therefore operate on exact faithful 24-point images, not unrelated groups of matching orders. The original 396-point arrays and source assignments remain bound. No ambient subgroup enumeration or new automorphism computation is used.

Lemma 3 (Nonsplitting with Klein four quotient)**.** *Let $N \lhd K$, and let $x, y \in K$ satisfy $x^2, y^2, [x, y] \in N$, with N, Nx, Ny, Nxy disjoint and covering K. Then $K/N \cong C_2^2$, and the extension splits if and only if there are $a \in Nx$, $b \in Ny$ with $a^2 = b^2 = 1$ and $ab = ba$.*

Proof. The relations and four distinct normal forms give the stated quotient. A complement supplies the indicated lifts. Conversely, such lifts define a homomorphism from C_2^2 into K; its composite with the quotient map is the prescribed isomorphism. It is therefore injective, and its image is a complement. Checking every pair in the two cosets proves nonsplitting when none satisfies the three relations. No failed complement search is a premise. □

The actual structural certificate. The registry contains 128 acquired concrete models and one new retained Mathieu constructor input. There are 23 new decompositions: eleven split, three direct and nine nonsplit. Each factor/occurrence map is an explicit identity-preserving bijection checked on all right-generator edges. The acquired MC.3 word-induction theorem proves it an isomorphism. Normality is checked on subgroup and ambient generators. A complement has trivial intersection with the normal subgroup and the required product of orders; a direct product additionally has commuting factors. A displayed nontrivial semidirect action has a noncommuting generator pair.

Six new prime-index nonsplit extensions use 152 checks of all outside elements. The three composite cases use the acquired cyclic-six and S_3 lift criteria, and the preceding Klein four lemma: respectively 32, 1,024 and 1,024 lift tests. In every case quotient relations and all quotient cosets are checked first. In particular, the new criterion pays exactly $2^{1+4} \cdot 2^2$, not a cyclic quotient substitute. Every final map lands in the exact source-row subgroup (or its proved faithful image). The 397 maps and 1,368,412 right-generator equations establish all 360 asserted structures.

Execution and independently checked interfaces. The candidate coset pass completed in 1.54 seconds. The sole bounded structural candidate pass completed all 360 rows in 3.77 seconds. GAP's normal-subgroup, complement and isomorphism operations supplied witnesses only. The replay uses literal permutations and all retained certificates; it repeats neither discovery nor subgroup classification.

An initial compact-output reader rejected a physical GAP line continuation between a digit and a comma before any mathematical check. The raw output and old reader are preserved. The corrected reader removes only physical backslash–newline at numeric array-token boundaries and then parses strict JSON; permutation domains, representative words and all equations are checked. No candidate output was edited.

The independent small action controls enumerate S_4 literally. Its nonnormal point stabilizer verifies the right-coset convention and rejects inverse-action substitution. Its normal Klein four subgroup supplies a nonfaithful action which fails the final image-order test. Other damages include repeated cosets, incorrect words, Boolean word indices, changed row images, and nonpermutation images. A changed actual 396-point row image also fails its coset equations. Independent $D_8/Z(D_8)$ and C_2^3/C_2 controls distinguish nonsplit and split Klein four extensions; dependent quotient generators fail the four-coset check.

Full-certificate controls change identity or a right-generator edge, omit an element or source row, alter source-row assignments, actual generators, source powers and dots, the named Mathieu input, complements, factor references and an actual Klein four lift. Ordinary and optimized Python must produce identical reports. The source syntax is joined to the actual retained embedded rows, not merely to their numerical orders.

Scope and attribution. The result adds 360 structural row joins for the two current tables. Their complete numerical classifications, named constructors and earlier registry proofs are explicit retained inputs. The finite source-sensitive application and the new literal coset and Klein-quotient checks are authored here; root directed compact-action and registry reuse. These elementary arguments carry no novelty claim. There is no new ambient census, historical-byte identity, uniquely determined extension name, whole-source admission, question resolution, formalization or measurement credit.

Structural certificates for $\mathrm{PSp}_4(3)$ and its extension

Private L054 finite application

Theorem 1. *Relative to the explicitly acquired numerical classifications and named constructor interface, all 116 rows of the current* `psu42` *table and all 350 rows of* `pgammau42` *have the displayed structural descriptions. These are existence assertions for the actual row groups, not assertions that a structural label determines a unique group.*

Exact source and embedded inputs. The current PDFs, attributed to Thomas Connor and Dimitri Leemans, display the ambient names $\mathrm{PSp}_4(3)$ and $\mathrm{PSp}_4(3) : 2$. Their 21 native pages, including the final caption-only page of the first table, were directly read. The source filenames are locators, not extra group-identification assumptions. All retained subgroup arrays are literal `Group` expressions in permutations, on 40 and 85 points respectively. We acquire their complete accepted numerical inventories and exact source-row bijections; no proper group is reconstructed from a printed name.

The article's left-association convention applies except where native parentheses prescribe a factor. In particular, row 4 of `psu42` and row 12 of `pgammau42` print $3^{1+2} : (Q_8 : 3)$, whereas row 20 of the latter prints $3^{1+2} : (Q_8 : 2)$. These parentheses are retained. A source 3^3 is elementary abelian of order 27, not a cyclic group of order 33. Dots, colons and crosses retain the acquired nonsplit, nontrivial split and direct meanings. The unsigned extraspecial and Q_{12} conventions are inherited with their proofs. We make no historical-byte identity claim about these current PDFs.

One explicit named construction. GAP Reference Manual 50.2-15 defines `PSp(4,3)` as the projective symplectic group preserving a nondegenerate alternating form, with default permutation action on projective lines. We retain this construction identity as an explicit software interface. The original successful numerical run has exactly this constructor in its driver, a clean terminal receipt and unchanged recorded inputs and outputs. Its actual top generators equal the 40-point generators imported here. This concrete group is the named atom; its name is not deduced from its order 25,920.

The second original numerical model was produced as a permutation image of the automorphism group of $\mathrm{PSU}_4(2)$. Its displayed symplectic extension label is paid by a new actual decomposition: the checker verifies a normal subgroup of order 25,920, an explicit isomorphism from the named PSp atom, and an actual order-two complement with nontrivial action. Separately, its source row 2 is identified with the same atom by an explicit map. Thus the source label does not follow merely by comparing orders or by assuming an exceptional unitary–symplectic isomorphism.

Registry reuse. There are 129 acquired concrete structural models and this one named input. The new certificate contains 38 models: 31 split, five direct and two nonsplit. Repeated names may select different actual models. Every factor and occurrence uses an explicit map to its embedded target. The accepted MC.3 generator-edge criterion is used unchanged: a bijection fixing identity and respecting every right multiplication by each ordered generator is a homomorphism by induction on generator words, hence an isomorphism.

Proof of all row assertions. The literal verifier closes every finite permutation group. References to new factor models are acyclic. In a split extension it verifies the actual normal subgroup N, the certified structural map to N, the actual complement K with its own certified map, $N \cap K = 1$, and $|N||K| = |H|$. Hence $H = NK$. Normality is checked on generators and extends to their generated groups. A noncommuting generator pair verifies the nontrivial action; in a direct model, all pairs of factor generators commute, which proves that the factors commute elementwise.

One new nonsplit model has normal subgroup $3 : S_3$ and quotient S_3. The acquired quotient-lift criterion first checks fixed quotient generators xN, yN, the relations $x^2, y^3, (xy)^2 \in N$, and all six distinct cosets. A complement exists exactly when some $a \in Nx$, $b \in Ny$ satisfy $a^2 = b^3 = (ab)^2 = 1$. Every one of the 324 lift pairs is tested and fails. The other new nonsplit model has normal subgroup $3^2 : S_3$ and quotient C_2; all 54 outside elements fail to have order two. These exhaustive literal checks prove nonsplitting. An unsuccessful software complement search is never used as proof.

Every final source occurrence is matched to the exact acquired subgroup under the numerical source-row map. The source literal, native syntax tree, order, actual record number and all generator arrays are checked. The certificate contains 540 explicit isomorphisms and 382,991 checked right-generator edges. Together with the new decompositions and acquired atom identities, these checks prove all 466 assertions in the theorem.

Candidate and verification evidence. One bounded GAP candidate pass completed every row in 15.69 seconds. The normal-subgroup, complement and isomorphism operations only propose witnesses. The final replay uses literal arithmetic and does not repeat that discovery, any ambient subgroup census or an automorphism calculation. Three physical line continuations in candidate JSON occur at the already accepted digit–digit or comma–digit positions. The unchanged numeric reader removes only these continuations, parses strict JSON, and validates all resulting permutation data. The sole initial development failure was an invalid loop in a preparer adaptation, rejected by Python's parser before writing any input or executing GAP; its exact bytes are retained.

The full-certificate damages test incorrect identity or generator edges, missing elements and source rows, altered subgroup assignments and generators, powers and dots, the named constructor model, wrong complements, circular references and an actual S_3 quotient lift. Separate interface controls require literal imported arrays, preserve the parenthesized factor, reject malformed syntax and nonnumeric continuation positions, and exercise the actual retained-constructor acquisition with incomplete, wrong-model, failed and changed-input receipts. Ordinary and optimized Python must yield identical complete reports.

Scope. This application adds 466 current structural row joins. The numerical classifications, original execution identities, earlier model proofs and named PSp constructor are explicit acquired inputs. No canonical uniqueness of extension notation, historical file identity, new classification algorithm, whole-original admission, question resolution, formalization or measurement credit is claimed. The application and exact source joins are authored here under root's registry-reuse task; distinct review remains separate.

Structural certificates for the A_9 and S_9 tables

Private L054 finite application

Theorem 1. *Relative to the acquired complete counts of ambient subgroup classes, exact local maximal subgroup interfaces and earlier structural registry, all 223 rows of the current A_9 table and all 554 rows of the current S_9 table have the displayed structural descriptions, with the explicit Q_{20} presentation below. These 777 descriptions assert actual decompositions, not uniqueness of an isomorphism type from a label.*

Source syntax. The current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All ten A_9 and 27 S_9 native pages were directly read. Historical identity with the 2013 source files is not asserted. The 2015 article's left-association convention is used unless the native source has parentheses. In particular, S_9 row 99 is $D_8 \times (D_{10} \cdot 2)$, not $(D_8 \times D_{10}) \cdot 2$. Row 157's $D_8 \times (D_{10})$ has redundant parentheses around a single atom. Row 22 preserves the dot in $((((A_4 \times 2) \times 2) : S_3) \cdot 2) : 2$. Powers, dots, colons and crosses are retained exactly. An unsigned extraspecial atom asserts no plus or minus type.

Recovering actual embeddings. The retained S_9 group display contains five nonliteral constructor expressions, in addition to point-set symmetric and alternating names. We do not assume that a display retained the original producer's embedding. The conventional constructor outputs, including a proposed replacement of printed $L(3,2)$ by $\mathrm{PSL}(3,2)$, are only candidate groups. Every candidate is exported as actual ordered permutation generators. The A_9 point-set names are treated in the same way.

Let the acquired complete number of subgroup conjugacy classes of G be c. If c candidate subgroups belong to G and are pairwise nonconjugate in G, their classes exhaust the subgroup classes of G. This elementary counting argument was already used in the accepted A_7, S_7, A_8, S_8 application. Here the unchanged robust local-weight kernel checks these conditions for all 223 and 554 candidates, respectively. The actual ambient models are checked against the natural alternating and symmetric groups on nine points. Thus no interpretation of a printed proper-group name is a completeness premise.

For each candidate H, the acquired local interface supplies a complete covering list of genuine maximal subgroups of a checked faithful smaller action. Literal conjugacy tests remove repetitions, and inverse transport returns actual maximal subgroups of H. Fusion into the validated ambient roster is by actual conjugacy. Each local class contributes $[H : N_H(K)]$ children; contributions with the same ambient class are added. If the ambient lengths of the parent and child classes are c_H, c_K, double counting conjugate incidences gives $b = ac_H/c_K$ parents above each child. Both integrality and the bounds $a \le c_K$, $b \le c_H$ are checked. This is the accepted robust weighting theorem and kernel, not an assumption that a software class list has no repeated classes.

The saved source bijections are tested anew against every row order, class length and both directions of every edge and nonedge. The new arrays have 722 and 2267 weighted edges, and all 356,645 ordered source pairs agree. These checks establish the source assignment of the actual candidate groups. No ambient subgroup enumeration is repeated.

Registry and structural proof. There are 189 acquired concrete registry models and two elementary natural models, A_9, S_9. A full cycle and a transposition generate the symmetric model; the alternating model uses $(1\,2\,i)$ for $3 \le i \le 9$. The literal closures have orders 9! and 9!/2, and the latter generators are even, proving the two named identifications.

The application has 82 new models: 40 split, 32 direct, nine nonsplit, and one dicyclic model. Factor references are acyclic. A structural reuse is always an explicit bijection fixing identity and preserving right multiplication by every ordered generator. Induction on generator words makes it an isomorphism; this is the acquired MC.3 criterion. All 930 such maps, comprising 4,038,442 generator-edge equations, are checked.

For a split model, the verifier checks $N \triangleleft H$, identified factors N, K, trivial intersection and $|N||K| = |H|$. Thus $H = NK$. A colon requires a noncommuting pair of factor generators. A cross requires all factor generators to commute, hence elementwise commutation. These are actual subgroup and multiplication checks.

Six new nonsplit models have prime quotient, and every outside element is checked against the requisite prime-order condition: 394 tests in all. Three models have quotient S_3. Fixed quotient generators xN, yN satisfy $x^2, y^3, (xy)^2 \in N$, and six explicit distinct cosets prove the quotient presentation. A complement exists exactly when some lifts $a \in Nx, b \in Ny$ obey $a^2 = b^3 = (ab)^2 = 1$. The relations then give a surjection from S_3 onto the generated lift group, while its quotient surjects onto S_3, forcing order six and a complement. Conversely a complement supplies such lifts. All 1940 pairs fail. These are the previously accepted exhaustive quotient-lift criteria; a failed software complement search is never evidence of nonsplitting.

The precise Q_{20} convention. The source does not separately define this notation. As with the acquired Q_{12} convention, we identify the dicyclic group by the presentation

$$Q_{20} := \langle x, y \mid x^{10} = 1,\ y^2 = x^5,\ y^{-1}xy = x^{-1} \rangle.$$

Every word reduces to x^i or $x^i y$, $0 \le i < 10$, so the presentation has at most twenty elements. The actual new model supplies generators of an exactly twenty-element group, with $|x| = 10$, satisfying these relations. Hence the presented group and the actual model are isomorphic. The installed GAP Reference Manual 50.1-8–9 defines this dicyclic presentation, and reserves generalized quaternion for the 2-power-order case. We retain the explicit nomenclature qualification rather than calling order twenty a quaternion 2-group. Source S_9 row 379 and its occurrence inside row 260's $Q_{20} : 2$ have actual maps to this model.

Validation and limits. The bounded witness pass returned all 777 rows in 58.99 seconds. It proposed witnesses only. The final replay closes and verifies every literal finite group, map, decomposition and source join, with identical ordinary and optimized reports. The embedding revalidations took 9.78 and 29.51 seconds; their exact clean receipts and all completed parent outputs are retained and acquired without an unchanged repeat.

Independent regular dicyclic actions of orders 12 and 20 test the new criterion. Wrong rotations and lifts, genuine dihedral and cyclic groups of order twenty, malformed source parentheses, wrong row generators and assignments, changed dots and orders, nonbijective source maps, duplicate edges and incorrect class lengths or weights are rejected. The accepted transport and extension controls remain acquired. No canonical uniqueness of structural notation, historical PDF-byte identity, whole-source admission, question resolution, formalization or measurement is claimed.

Structural certificates for the unitary degree-three groups over $\mathbf{F}_4$

Private L054 finite application

Theorem 1. *Relative to the acquired numerical and named-model interfaces, all 234 rows of the three current companion tables have their displayed structural descriptions: 34 rows for $U_3(4)$, 80 for $U_3(4) : 2$, and 120 for $U_3(4) : 2{\cdot}2$. Each label asserts the existence of its stated decomposition, without claiming that the label specifies a unique group.*

Exact source and actual embeddings. The current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All ten PDF pages were directly viewed, including the final footer-only page. Source powers, parentheses and punctuation are retained: 5^2 is elementary abelian, $4^2 = C_4^2$, and the top expression is $(U_3(4) : 2) \cdot 2$, with a nonsplit final extension. The 2015 article's default left association also applies to the successive dots in $2 \times 4 \cdot 2 \cdot 2 \cdot 2$. The source-row bijections and complete numerical inventories are acquired from the accepted robust reweighting and normal-restriction applications. Both robust input arrays contain only literal permutation groups; the middle table's complete state uses the accepted explicit-generator grammar. No proper subgroup is reconstructed from a displayed constructor name.

The original successful `PSU(3,4)` execution, its exact driver, clean receipt and every recorded input/output pin are acquired. Its actual 273-point generators equal the current top generators. GAP Reference Manual 50.2-14 supplies the named projective special unitary constructor interface. This is the one new named input; a matching order alone does not identify it. The actual two larger groups and all their subgroup embeddings come from the acquired numerical model certificates, with no new automorphism census.

Lemma 2 (Faithful invariant-subset restriction)**.** *Let a finite permutation group G, its order n, and actual subgroups $K_i \leq G$ be acquired. Suppose a literal subset Ω is invariant under every ordered generator of G. Restriction to Ω defines a homomorphism. If its generated image has order n, the homomorphism is injective. Restricting the actual generators of every K_i therefore gives their exact images under this same isomorphism.*

Proof. Each generator permutes the finite subset, so all words preserve it. Restriction commutes with right permutation multiplication, proving the homomorphism assertion. Its kernel has order $|G|/|\operatorname{im}| = 1$. The subgroup assertion follows by applying the same map to its generators. Thus order comparison is used after constructing a specific homomorphism. □

The three literal transports. Each acquired action has an invariant subset of 65 of its 273 points. The independent Python reader checks the complete original generator and subgroup arrays against the acquired files, all permutation domains and all 35,750 generator-point equations. Literal closures of the three image groups have orders 62,400, 124,800 and 249,600, exactly the acquired orders. Consequently all 234 images retain their original embeddings. This replaces neither an ambient subgroup classification nor a named-group identification.

Structural certificates and their proof. The registry acquires 81 earlier accepted models and the named unitary input just described. There are 38 new models: 15 split extensions, 21 nonsplit extensions and two direct products. Each factor and occurrence map is an explicit bijection fixing identity and preserving every right generator edge. The acquired MC.3 word-induction theorem proves these maps are isomorphisms. Normality is checked by conjugating subgroup generators by ambient generators. For a split extension, an actual complement has trivial intersection with the normal factor and the required product of orders. Direct products additionally have commuting factors; nontrivial semidirect actions have explicit noncommuting generator pairs.

For a prime-order quotient, a complement exists exactly when some outside element has that prime order. The reader checks all outside elements, giving 126,604 tests, including every one of the 124,800 elements outside the top index-two normal subgroup. Two other quotients are S_3: their actual normal forms and relations are checked first. The acquired S_3-lift criterion then tests all 425,984 pairs of lifts of its standard generators. Absence of a pair satisfying the presentation proves nonsplitting; failure of a candidate-search routine is not a premise. These criteria and the earlier atom definitions are acquired unchanged. In total, 289 maps and 2,477,559 generator edges prove the displayed structures and join them to the exact source rows.

Finite evidence and limits. The sole bounded witness-discovery pass completed all 234 rows in 20.84 seconds. GAP supplied candidate decompositions and maps only. Replay uses literal permutations; it repeats neither discovery nor any subgroup census. Ordinary and optimized Python must return identical reports. Twenty focused damages cover source rows, dots and powers, original embeddings, named top generators, orbit invariance, actual restriction equations and malformed source syntax. A literal C_2^2 action with an invariant 65-point subset losing one factor is rejected by the image-order guard; a faithful diagonal C_2 control passes. C_4 and C_2^2 distinguish nonsplit and split index-two extensions. Earlier complete transport and extension damage controls are acquired separately.

The first successful control harness used a temporary file in its working directory. The successor moves only that scratch file to a temporary directory, allowing immutable replay; the prior receipt and exact code remain preserved. No mathematical code or certificate changed. This application claims 234 current-table structure joins, with explicit retained classification and software inputs. It claims no historical PDF byte identity, canonical uniqueness of extension labels, whole-source admission, new question resolution, formalization or measurement. Authorship is `1044_resume`; root directed reuse and conducts the distinct review.

Literal structural certificates for M_{22} and $M_{22} : 2$

Private L054 finite application

Theorem 1. *Relative to the acquired numerical inventories and named-group interfaces, all 646 rows of the current companion tables have their displayed structural descriptions: 156 rows for M_{22} and 490 for $M_{22} : 2$. The descriptions assert existence of the displayed decompositions; they do not assert uniqueness of a group defined by an extension label.*

Source and exact models. The current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All 32 native pages were directly viewed. Powers, parentheses, colons and dots are preserved, with the published default left association. For example, row 43 of $M_{22} : 2$ is $2^{1+4} \cdot (2 \times 6)$, row 57 is $2^{1+4} \cdot 2^3$, row 179 is $D_8 \cdot (2 \times 4)$, and row 180 is $Q_8 \cdot 2^3$. A dot is verified as nonsplitting. The acquired source bijections and robust numerical inventories identify all actual subgroup occurrences. Every imported group is given by literal permutations on the same 22 points; no displayed constructor name is interpreted as a proper subgroup embedding.

The successful original `MathieuGroup(22)` execution supplies the named top model, with its exact driver, clean receipt and all recorded input/output hashes. Its ordered generators equal the current top M_{22} generators. The GAP Reference Manual, 50.1-13, is the retained constructor identification premise. The second named input, $L_3(4)$, is the accepted L100 projective construction's exact 21-point model: its retained clean export explicitly checks equality to `PSL(3,4)`. All output generators and the complete accepted construction interface are acquired. Neither name is inferred from its group order.

Registry transport. There are 175 earlier accepted structural models and these two named inputs. The application constructs 97 further models: 58 split extensions, 24 nonsplit extensions, 14 direct products and one quaternion model. Every occurrence and factor map is a literal bijection fixing identity and preserving all right-generator edges. The acquired MC.3 word-induction argument therefore proves that each map is an isomorphism. For extensions, normality is checked on generators. An actual complement has trivial intersection with the normal factor and the required product of orders. Direct factors additionally commute. A displayed nontrivial semidirect action has a literal noncommuting generator pair.

Lemma 2 (Complete lifts for a finite abelian quotient)**.** *Let $N \trianglelefteq H$ be finite. Suppose $x_1, \dots, x_k \in H$ and integers $d_i \geq 2$ satisfy $x_i^{d_i} \in N$ and $[x_i, x_j] \in N$. Suppose moreover that the cosets*

$$Nx_1^{e_1} \cdots x_k^{e_k} \qquad (0 \leq e_i < d_i)$$

are disjoint and cover H. Then $H/N \cong \prod_i C_{d_i}$, with the specified generators. The extension has a complement if and only if there exist $y_i \in Nx_i$ satisfying $y_i^{d_i} = 1$ and $[y_i, y_j] = 1$ for all i, j.

Proof. The quotient relations give a surjection from the indicated product of cyclic groups to H/N. The complete distinct normal forms give the same order, so this map is an isomorphism. If a complement exists, the quotient map restricted to it is an isomorphism; its inverse supplies the required lifts. Conversely, lifts satisfying the relations define a homomorphism from the product of cyclic groups to H. Its composition with the quotient map is the specified isomorphism. It is therefore injective, its image intersects N trivially, and its image together with N generates H. That image is a complement. □

Exact nonsplitting checks. Four new certificates use the lemma, for the actual quotients $C_2 \times C_6$, C_2^3 (twice), and $C_2 \times C_4$. The reader first checks every quotient relation, every normal-form coset, and their disjoint complete union. It then tests every tuple in $Nx_1 \times \cdots \times Nx_k$ against every order and commutation relation. No satisfying tuple exists. The largest normal subgroup has order 32, so at most 32^3 lift tuples occur in any one certificate. Failure of a candidate search is not a premise. Previously accepted prime, cyclic-six, S_3 and Klein-four criteria pay the other nonsplit descriptions.

The new Q_{16} model uses $a^8 = 1$, $b^2 = a^4$ and $b^{-1}ab = a^{-1}$, with a of order eight and a, b generating the actual group of order 16. These relations reduce every word to $a^i b^j$, $0 \leq i < 8$, $0 \leq j < 2$; the actual order proves the standard quaternion normal forms are distinct. This is the same acquired dicyclic presentation argument, here at order 16.

Evidence and limits. One bounded witness-discovery pass completed all 646 rows in 55.37 seconds, with clean output. GAP proposed decompositions and maps only. The independent reader uses literal permutation arithmetic; replay repeats neither discovery nor any ambient subgroup census. Ordinary and optimized Python must agree. Small split and nonsplit controls exercise all three new quotient types; damaged orders, dependent normal forms, missing generators and false dots are rejected. Source-row, punctuation, degree, embedding and named-model joins have separate guards, and the earlier complete transport/extension controls remain acquired.

This application supplies current-table structural joins. It does not claim historical 2013 PDF byte identity, canonical extension labels, whole-source admission, new question resolution, formalization or measurement. Authorship is `1044_resume`; root directed registry reuse and conducts the distinct review.

Structural certificates for the Suzuki and Janko small-core tables

Private L054 finite application

Theorem 1. *With the acquired numerical inventories and named-model identifications, all 247 current companion rows have their displayed structural descriptions: 40 for J_1, 146 for J_2, 22 for* Sz(8) *and 39 for* Sz(8) : 3*. An extension label asserts existence of its stated decomposition, not uniqueness of a group defined by the label.*

Sources and exact embeddings. All eleven native pages of the current Connor–Leemans companion PDFs were directly viewed. Powers and dots are preserved. The published left-association convention gives $(3 \cdot A_6) : 2$ in J_2 row 3 and $3 \cdot A_6$ in row 6. Row 13 retains the literal source $2^{1+4} : (D_{10})$; its redundant parentheses around the single atom are recorded, while the semantic tree has the same D_{10} factor. Parentheses around composite factors are not removed.

The six-table numerical component and its distinct root review are acquired. Its actual literal final states, complete row bijections and named-model checks supply the present occurrences. The states contain explicit embedded generators, on 266 points for J_1, 100 for J_2 and 65 for Suzuki. No displayed constructor name is substituted for a proper embedding.

The retained J_1 model is primitive-library entry $(266, 1)$, and J_2 is the derived subgroup of entry $(100, 2)$; their identifications are acquired from the accepted sporadic representation and six-table interfaces. The two top-row generating lists are literally equal to these already identified final-state lists. Their naming certificates are therefore exact identity joins, not fresh large Cayley-table computations. The Suzuki model is the accepted named ovoid action, with the explicit coordinate square map for its order-three field extension. These identification premises are retained explicitly; matching orders alone are not the naming argument.

Structural reuse. The registry uses 142 earlier structural models and three named inputs. There are 33 new decomposition models: 25 split extensions, six nonsplit extensions, one direct product and one extraspecial model. Every proper occurrence and factor map is a literal bijection preserving identity and every right-generator edge. The acquired MC.3 word-induction argument proves each map is an isomorphism. Normality is checked by conjugating factor generators by group generators. An actual complement has trivial intersection and the required product of orders. Direct factors commute; a displayed nontrivial semidirect action has noncommuting generators. The acquired prime, cyclic-six, S_3, Klein-four and extraspecial criteria are used without alteration. The only new nonsplitting argument follows.

Lemma 2 (Central commuting-lifts obstruction)**.** *Let $N \leq Z(H)$ and let $x, y \in H$ have nonidentity commutator $[x, y] \in N$. Then the extension of H/N by N has no complement.*

Proof. The images of x, y commute in H/N. Every pair of their lifts has the form nx, my with $n, m \in N$. Centrality gives $[nx, my] = [x, y] \neq 1$. If a complement existed, the inverse of its quotient isomorphism would give commuting lifts of these two commuting elements, a contradiction. □

The actual $3 \cdot A_6$ certificate. Inside the saved J_2 row-6 group of order 1080, an actual normal subgroup N of order three is identified with the cyclic model. The reader checks that every element of N commutes with every group generator, hence $N \leq Z(H)$. The proposed x, y belong to H, and their literal commutator is a nonidentity element of N. All nine shifted commutators $[nx, my]$ are also checked directly.

The quotient name is paid separately. The certificate gives 360 actual representatives of right cosets Nr. The reader forms every coset, verifies their disjoint complete union, and computes the right action of every ordered generator on these cosets. Since N is normal, this action has kernel exactly N. Its literal image has order 360. An explicit bijection from the acquired A_6 model to that image preserves all generator edges, proving $H/N \cong A_6$. Thus the lemma proves exactly the source dot in $3 \cdot A_6$. The row-3 group has a separate actual index-two complement, proving $(3 \cdot A_6) : 2$.

Finite evidence and development. The first bounded witness pass returned in 11.41 seconds with 245 successful rows and two explicit failures, J_2 rows 3 and 6. A proposed single-element order obstruction did not certify those rows. It is not a premise or an accepted result. Root then supplied the central commuting-lifts argument. A changed two-row pass completed in 1.77 seconds, reusing all earlier witnesses and producing the missing two certificates. Both executions and the exact earlier code are retained. No ambient subgroup census was repeated.

Ordinary and optimized replay use literal permutations and must agree. Small quaternion controls validate the central obstruction and reject a noncentral kernel, a trivial commutator, missing or duplicate quotient cosets. Separate source, named-identity and embedding damages are rejected. The reader also handles GAP's physical numeric line continuation before a comma; nonnumeric continuations remain invalid. This concerns serialization, not a mathematical change. The initial source-parser rejection of redundant atomic parentheses is preserved; raw source labels remain exact.

The first identity check rejected a GAP-proposed automorphism where literal generator equality was required. The successor writes the unchanged generators explicitly; both editions and the failed receipt are preserved. This application claims current-table structural joins only. It does not claim historical PDF byte identity, canonical extension labels, whole-source admission, new question resolution, formalization or measurement. `1044_resume` authored the application and literal certificates; root contributed the commuting-lifts argument. A different agent conducts the distinct mathematical review.

Three current structural tables over $\mathbb{F}_4$

This finite application certifies every displayed subgroup description in the current Connor–Leemans tables for $\mathrm{PSL}_3(4)$, $\mathrm{PGL}_3(4)$ and $\mathrm{P\Sigma L}_3(4)$. The complete subgroup inventories, exact source-row correspondences and corrected maximal-incidence weights are acquired from the accepted numerical components. This paper adds the structural descriptions; it repeats no ambient subgroup census.

Exact assertion. Relative to the acquired named models and earlier structural certificates, each of the 347 source rows admits its displayed decomposition. The literal certificates identify the actual embedded subgroup in that row, not merely another group of the same order. Extension descriptions need not specify a unique isomorphism type: the assertion is the existence of the displayed normal subgroup, quotient, complement or direct factors, with nonsplitting where a dot is printed.

Current companion table	Rows	Actual degree	Native pages
$\mathrm{PSL}_3(4)$	95	21	4
$\mathrm{PGL}_3(4)$	100	21	4
$\mathrm{P\Sigma L}_3(4)$	152	44	7
Total	347		15

All fifteen native pages were directly read, including the last field-extension page containing the continued-table heading and attribution. Powers, dots, colons, products and association are retained in a separate syntax tree for each row. For example, field-extension row 22 is $2^{1+4} \cdot 2^2$; rows 12–13 end with a dot, $((2^4 : 5) : 2) \cdot 2$. The final product in projective-linear row 34 is $(7 : 3) \times 3$. Flattened extracted digits are interpreted only after this native check.

Named and embedded inputs. The named $L_3(4)$ atom is the exact accepted 21-point projective construction already acquired in the M22 structural component from the L100 construction packet. Its identity is not inferred from the order $20,160$. The first two numerical files contain only literal permutation groups; every proper subgroup retains its original 21-point embedding. The field extension uses the accepted 44-point literal restriction state and its exact field-automorphism identification. No display-name constructor replaces an embedded group. The two degrees are deliberately kept separate: every transport states its actual target degree and its generator images.

There are 140 acquired registry entries, including cyclic, elementary abelian, dihedral, quaternion, extraspecial, Mathieu and projective models where they actually occur in a source factor. The present witness discovery adds only eleven models: ten semidirect products and one direct product. All nonsplit descriptions in this roster are transported from previously accepted models, including the complete quotient-lift obstructions.

Complete finite verification. We recall the acquired labelled-generator transport criterion. List every element of a finite source model H and its products by each ordered generator. A bijection f onto the actual target subgroup, fixing identity and satisfying $f(hs) = f(h)f(s)$ on every such edge, is a homomorphism: induction on generator words gives the equation for multiplication by each word. Hence it is an isomorphism. The verifier checks the entire bijection, all edges and actual generator membership. Candidate isomorphism searches merely propose these maps; their answers are never accepted without the literal checks.

For a new split model the certificate gives actual subgroups $N, K \leq H$. The finite reader checks their already certified names, normality of N on the actual generators of H, $N \cap K = 1$ and $|N||K| = |H|$. Thus $H = NK$ and the asserted split extension follows. It also checks a noncommuting generator pair for a printed nontrivial semidirect action. For the one direct product all factor generators commute instead; generation and trivial intersection give $H \cong N \times K$. The model-reference graph is checked in dependency order, so none of the new names is its own premise.

Every row is joined to its retained numerical representative before a certificate is considered. The reader checks the exact source row/order, the native syntax tree, the complete source-to-computation bijection, the literal original generators and the actual action degree. It then enumerates that finite target subgroup by permutation multiplication and checks the complete transport. In particular the top-row split assertions are certified just like the proper rows. All 347 occurrences are required; there is no implicit credit for a repeated description.

Evidence and limits. The bounded discovery returned all 347 witnesses in 6.27 seconds without any missing row. Only the resulting literal arrays are used by the independent Python reader. Ordinary and optimized executions must produce identical reports. Damage controls target the source roster, dots, powers, embedded generators, degrees, named-display substitutions and complete-state marker. A separate $Q_8 \to C_2^2$ example checks the acquired complete-coset nonsplitting guard and rejects a duplicated quotient coset.

The source attribution is Connor and Leemans, the current companion files associated with *An atlas of subgroup lattices of finite almost simple groups*; these files print November 26, 2012. Their exact current bytes are bound. No identity with a particular historical 2013 or 2015 file edition is asserted. Previously accepted numerical and named-constructor interfaces remain explicit inputs. This application makes no novelty, full-atlas, whole-original, conjecture, formalization or measurement claim.

Four further structural tables over $\mathbb{F}_4$

This finite application proves the displayed subgroup descriptions in four current Connor–Leemans companion tables. Complete numerical inventories, the exact source bijections and the actual 44-point models are acquired from the accepted nine-table restriction component. No ambient subgroup census is repeated.

Exact assertion and source convention. Relative to the acquired named atoms and earlier certified registry, each of the following 538 actual embedded subgroups admits the description in its source row. A colon asserts a split extension, a dot a nonsplit extension, and a cross a direct product. These descriptions need not determine a unique abstract group. Every occurrence is verified separately.

Current file	Displayed top group	Rows	Pages
`psl34d21`	$L_3(4) : 2$	151	6
`psl34d23`	$L_3(4) : 2$	102	4
`psl34d6`	$L_3(4) : 6$	143	6
`psl34ds3`	$L_3(4) : S_3$	142	6
Total		538	22

All twenty-two native pages were directly inspected, including the final continued-table page without data in `psl34d6`. Powers, punctuation and the default left association are recorded literally. In particular, `psl34d6` row 57 is $Q_8 \cdot 6$, whereas rows 51 and 56 are $Q_8 : 6$. The nested description in `psl34ds3` row 43 is $(D_8 \cdot 2^2) : 2$. Products do not erase an earlier dot: `psl34d21` row 19 is $(((3 : S_3) \cdot 2) \times 2) \cdot 2$.

The actual models are the accepted subgroups of the 44-point $\mathrm{Aut}(L_3(4))$ model: field times inverse-transpose, inverse-transpose, diagonal together with field times inverse-transpose, and diagonal together with inverse-transpose, respectively. Their outer groups are C_2, C_2, C_6, S_3. These are exact acquired constructions; the filename suffixes are not asserted to define a canonical numbering of outer involutions. The two printed $L_3(4) : 2$ names are therefore not silently identified. Each full and proper subgroup is parsed from literal permutation generators in its own saved state; display-name constructors cannot replace its embedding.

The named $L_3(4)$ atom is the previously certified projective model, acquired through the M22/L100 interfaces and the preceding 347-row application. There are 161 acquired registry entries and no new named simple or Mathieu atom. Candidate discovery adds 37 small structural models: 29 split, five nonsplit, two direct and one extraspecial group. The saved generator images, rather than an isomorphism-search answer or group-order comparison, supply each new identification.

Complete certificate argument. We acquire the labelled-generator transport criterion. A bijection from a finite model to an actual subgroup, fixing identity and preserving right multiplication by every ordered generator, is a homomorphism by induction on words. Hence it is an isomorphism. The reader enumerates both actual finite groups by permutation multiplication, verifies the entire bijection and checks every generator edge. References to certified factors are processed in dependency order.

For a split description the certificate supplies $N, K \leq H$ with their acquired or previously checked names. It checks normality of N on the generators of H, $N \cap K = 1$, and $|N||K| = |H|$. Thus $H = NK$ and the description follows. A noncommuting generator pair confirms the printed nontrivial action. For a direct product the factor generators commute, so the same generation and intersection checks give $H = N \times K$.

Three new nonsplit models have quotient C_2. Every element outside the specified normal subgroup is checked to have square different from identity. Any complement would have order two and contradict this check. The other two have quotient C_2^2. Four literal cosets, with the quotient relations, establish this quotient. All lifts of its two fixed basis elements are tested: no pair consists of commuting involutions. A complement would lift that basis to such a pair, so no complement exists. The printed $Q_8 \cdot 6$ description uses an earlier accepted model and its cyclic-quotient lift obstruction; it is not inferred from the absence of a found complement.

For the new group of order 32, literal multiplication shows that the center and derived subgroup are the same group of order two and that every square lies there. Thus the quotient by the center is elementary abelian and the Frattini subgroup is that same center: the group is extraspecial. The source symbol 2^{1+4} makes no further sign choice. All these arguments and the finite readers are acquired criteria applied to the new explicit arrays, not new classification premises.

Before any structural certificate is considered, the reader checks the source row, order, syntax tree, accepted source-to-computation bijection, actual state generators and degree. Each top-row extension is checked by the same literal certificates as every proper row. All 538 occurrences are required; reuse of an abstract model never skips its embedded join.

Evidence and limits. One bounded witness pass returned all 538 proposals in 11.58 seconds, with clean output and no missing row. Independent literal verification uses only the retained arrays. Ordinary and optimized Python must give identical reports. Damaged controls test source association, powers, dots, embedded generators, exact degrees, literal-state grammar and completion markers. Small quaternion examples separately test the complete lift obstructions for Klein-four and cyclic-six quotients.

The current companion PDFs credit Thomas Connor and Dimitri Leemans and print November 29, 2012. Their exact present bytes are bound; identity with a particular historical 2013 or 2015 file is not asserted. Existing numerical completeness and named-constructor interfaces remain explicit inputs. No ambient census, unique extension-name classification, novelty, whole-original, conjecture, formalization or measurement credit is claimed.

Three remaining structural tables over $\mathbb{F}_4$

This application verifies the subgroup descriptions in three current Connor–Leemans companion tables. The complete numerical inventories, source bijections and actual 44-point models are acquired from the accepted nine-table component. No ambient subgroup census is repeated.

Exact assertion. Relative to the acquired named atoms and structural registry, every one of the following 923 actual embedded subgroups admits its displayed source description. Colons assert split extensions, dots nonsplit extensions, and crosses direct products. These descriptions need not specify a unique abstract group; each occurrence has its own checked identification.

Current file	Displayed top group	Rows	Pages
`pgammal34`	$L_3(4) : S_3$	226	10
`psl34d2e2`	$L_3(4) : 2^2$	288	13
`psl34dD12`	$(L_3(4) : 2) : S_3$	409	19
Total		923	42

All forty-two native pages were directly inspected, including continued headers on pages without data. Powers, punctuation, explicit parentheses and default left association are retained. For example, the middle table has $2^{1+4} \cdot D_8$ in row 26, $2^{1+4} \cdot 4$ in row 44, and $Q_8 \cdot (2 \times 4)$ in row 85. The last table's top row is parsed as $(L_3(4) : 2) : S_3$, without replacing its displayed association by a filename shorthand. Its rows 146 and 147 both display $Q_8 \cdot D_8$ and require separate embedded joins.

The models are the acquired diagonal–field extension $P\Gamma L_3(4)$, the field–inverse-transpose extension with outer group C_2^2, and the full automorphism model, respectively. Their literal subgroup generators are parsed from their original saved states. No constructor or displayed group name can replace an embedded subgroup. The named $L_3(4)$ atom, including its convention for projective actions, is acquired from the previously accepted L100/M22 and atlas interfaces. Generic outer extension names do not assert a canonical numbering of automorphisms.

There are 272 acquired registry entries and no new named simple or Mathieu atom. Candidate generation proposes 47 new structural models: 27 split, 13 nonsplit and seven direct products. The new finite evidence consists of literal generator maps and factor/lift witnesses. Neither successful isomorphism search nor equality of group orders is accepted as a proof.

Acquired certificate arguments. A bijection fixing identity and preserving right multiplication by each ordered generator is a homomorphism, by induction on words. The reader checks all generator edges of every model-to-subgroup map. Factor names are acquired or processed earlier in an acyclic dependency order. For a split extension it verifies actual subgroups $N, K \leq H$, normality of N, $N \cap K = 1$, and $|N||K| = |H|$; hence $H = NK$. A noncommuting generator pair verifies the displayed nontrivial action. For a direct product the factor generators commute instead.

New cyclic-four and dihedral-eight lift tests. Let $N \triangleleft H$ be the actual checked normal subgroup. Suppose that $x \in H$, $x^4 \in N$, and the four cosets N, Nx, Nx^2, Nx^3 are pairwise disjoint and cover H. Then H/N is cyclic of order four. A complement exists if and only if some $a \in Nx$ satisfies $a^4 = 1$. Necessity follows by lifting the fixed quotient generator through the complement. Conversely such an a has order exactly four, since its quotient image does, and $\langle a \rangle \cap N = 1$, so it is a complement. Thus checking $a^4 \neq 1$ for every $a \in Nx$ certifies the printed dot.

For the dihedral test, take $x, y \in H$ satisfying

$$x^4,\ y^2,\ (xy)^2 \in N, \qquad H = \coprod_{j=0}^{1} \coprod_{i=0}^{3} Nx^i y^j.$$

The quotient is D_8: the presentation $\langle r, s \mid r^4 = s^2 = (rs)^2 = 1 \rangle$ reduces every word to $r^i s^j$, with $0 \leq i < 4$, $0 \leq j < 2$, while the eight disjoint cosets show these forms distinct. A complement exists if and only if there are $a \in Nx$, $b \in Ny$ with

$$a^4 = b^2 = (ab)^2 = 1.$$

A complement necessarily supplies these lifts. Conversely, such lifts define a homomorphism from the presented D_8 to H. Composing with $H \to H/N$ is the isomorphism sending r, s to Nx, Ny. Consequently the lift homomorphism is injective, its image meets N trivially, and its image maps onto H/N. It is the required complement. The certificate tests all $|N|^2$ pairs in the two specified cosets and requires each pair to fail at least one relation. It makes no inference from failure to discover a complement.

The reader first establishes group closure, the normal subgroup and its name, the quotient relations and every coset normal form. Thus neither quotient identity nor completeness of the lift choices is assumed. The three new cyclic-four and three new dihedral-eight certificates use these arguments. Four new prime-index models, one Klein-four quotient, one S_3 quotient and one $C_2 \times C_4$ quotient use the previously accepted complete lift tests, with their exact model data checked afresh.

Controls, source attribution and limits. Independent small permutation models test $C_8 \to C_4$ and $Q_{16} \to D_8$ as nonsplit examples. The split examples $C_2 \times C_4$ and $C_2 \times D_8$ must be rejected by the respective nonsplitting predicates. Damaged quotient generators, relations and trees test the boundary of each test. The existing controls also check actual source associations, powers, dots, degrees, embedded generators, literal-state grammar and terminal completion markers. Ordinary and optimized Python must produce identical complete reports.

One bounded witness pass returned all 923 proposals in 13.52 seconds, with clean output and no missing occurrence. The literal checker requires all source rows and all embedded joins. The current files credit Thomas Connor and Dimitri Leemans; `pgammal34` prints November 26, 2012, and the other two print November 29, 2012. Their present bytes are bound; identity with a particular historical 2013 or 2015 file is not asserted. No census, unique-extension classification, novelty, whole-original, question, formalization or measurement credit is claimed.

Literal structural certificates for the $U_3(5)$ family

Private L054 finite application

Theorem 1. *Relative to the acquired numerical and named-model inputs, the displayed structural assertions hold for all 565 rows of the four current companion tables: 80 for $U_3(5)$, 93 for $U_3(5) : 3$, 148 for $U_3(5) : 2$, and 244 for $U_3(5) : S_3$. Each expression asserts its indicated decomposition, without claiming that the expression determines a unique isomorphism type.*

Source and actual inputs. The current Connor–Leemans PDFs are dated November 26, 2012. All 24 native pages were directly inspected. Baseline $24 = C_{24}$ in rows 3 and 50 of `pgu35`, and rows 7 and 144 of `pgammau35`, is distinguished from the superscript 2^4. The powers $6^2 = C_6^2$, $5^2 = C_5^2$, and 5^{1+2} have their separate meanings. Explicit right parentheses in $5^{1+2} : (8 : 2)$ and $5^{1+2} : (2 \times 4)$ are preserved. Default left association applies to $5^{1+2} : 4 \cdot S_3$ and $2 \cdot A_5 \times 3 : 2$. Colons and nonsplit dots are not exchanged.

The accepted robust weights and normal/arbitrary subgroup restrictions supply the complete actual subgroup inventories and source-row bijections. All four original actions have degree 651. Their retained group files or complete states are parsed as exact permutation generators; displayed constructor names never reconstruct a proper embedded subgroup. The corrected `pgammau35` weights and its distinct review are acquired, including the retained earlier duplicate-local-class failure. No numerical census or source-pair comparison is repeated here.

The original clean `PSU(3,5)` execution and all its recorded pins are acquired. Its ordered generators equal the retained top generators. GAP Reference Manual 50.2-14 is the explicit named-constructor premise. A literal invariant 126-point subset is checked for each action, including every actual subgroup generator. The four image orders are respectively 126000, 378000, 252000, and 756000. The acquired invariant-subset lemma therefore makes each restriction an isomorphism, not merely an order match. This supplies the new named $U_3(5)$ model; the three larger named-model interfaces remain those of the accepted numerical constructions.

Lemma 2 (Exact compact encoding and generated maps)**.** *A permutation on $d \leq 256$ letters can be stored by its d image bytes. For right actions, multiplication is $ab = (b_{a_i})_i$. Let $M = \langle g_1, \ldots, g_k \rangle$ be an acquired finite model and let h_j be permutations of another set. Enumerate M by its generator edges from the identity. Assign the identity image, propagate images by right multiplication by h_j, and check every repeated edge assignment. If the resulting map is bijective onto an actual group H, it is an isomorphism $M \longrightarrow H$ taking g_j to h_j.*

Proof. Byte encoding is injective on the finite image arrays. Replacing each byte a_i by table entry b_{a_i} gives exactly the displayed multiplication; unused table entries do not affect values below d. The implementation uses this representation only for $65 \leq d \leq 256$ and ordinary tuples otherwise. Decoding preserves equality, inversion and all group operations. For the map, every element is reached because the model generators generate. Every edge is checked, including edges not used for first discovery. Induction on a generator word gives $f(xw) = f(x)w(h_1, \ldots, h_k)$. Taking $x = 1$ and then arbitrary x proves $f(xy) = f(x)f(y)$. The checked bijection proves the assertion. Thus storing the ordered generator images instead of every map value removes no edge or bijection condition from the acquired MC.3 test. □

Two explicit small atoms. For C_6^2, the reader checks two commuting elements of order 6, generation, and group order 36. The resulting surjection $C_6 \times C_6 \to H$ has equal finite orders and is an isomorphism. For 5^{1+2} it checks $|H| = 125$, $H = \langle x, y \rangle$, a central commutator $z = [x, y]$ of order 5, $Z(H) = \langle z \rangle$, and $x^5, y^5 \in \langle z \rangle$. Hence $H' = \langle z \rangle$ and $H/Z(H) \cong C_5^2$. Every index-five maximal subgroup contains H', while the inverse images of all lines of $H/Z(H)$ intersect in $Z(H)$. Therefore $\Phi(H) = H' = Z(H)$ has order 5, which is precisely the asserted extraspecial structure. The criterion prescribes no exponent and infers no unprinted extraspecial sign.

The $2 \cdot A_5$ factor. The acquired central-commutator criterion applies unchanged. The reader checks the actual central kernel $N = C_2$, all 60 quotient cosets, their literal action, and its generator-edge isomorphism with the acquired A_5 model. It checks $[x, y] \in N \setminus \{1\}$. The quotient images commute, but all four pairs of lifts in $Nx \times Ny$ have this same nonidentity commutator. A complement would supply commuting lifts, a contradiction. This criterion was previously proved with root's contribution; its present application claims no independent authorship of that argument.

All other structural maps. The registry reuses the previously accepted explicit models, including A_7, A_6, M_{10}, M_9, and $L_2(7)$. Every new extension model has an actual normal factor, and the reader checks its transported generators and normality inside the actual model. A split extension has an explicit complement with trivial intersection and the product of orders equal to $|H|$. The displayed nontrivial action has a noncommuting generator pair. A direct product additionally has commuting factors. Prime-index nonsplitting checks every outside lift. The remaining fixed small quotients use the acquired complete coset-normal-form and generator-lift criteria. Every source occurrence then receives a separately checked isomorphism to its exact retained embedded subgroup, under the same 126-point restriction map.

Finite evidence and preserved failures. The first bounded witness pass stopped at an unsupported A_5 quotient. The second completed but left four explicit missing certificates because of the erroneous baseline-24 token interpretation. No failed row was accepted. Exact prior inputs and receipts remain, with hash-bound historical aliases. The third pass, after the source-specific correction, completed all 565 candidates in 14.45 seconds. GAP proposes witnesses only; the full ordinary and optimized Python checks prove their literal properties and must return identical reports. Neither mode repeats a subgroup census.

Independent controls compare 3456 products of all S_4 permutations at six encoding-boundary degrees, and compare a complete generated S_3 map with an independently constructed explicit map. They reject incompatible generator images. Explicit C_6^2 and Heisenberg order-125 models test the two new atoms, with wrong-order, missing-generation and commutator damages. The actual orbit validator rejects a nonfaithful invariant subset, while a faithful diagonal example passes. Source punctuation, baseline 24, right parentheses, embeddings, complete states, and the named top generators have focused damages. All earlier extension and map controls are acquired.

The frozen report gives the exact registry, map, edge, lift and guard counts. This is a current-table certificate, not proof of historical PDF byte identity or a canonical naming algorithm. It grants no whole-original, question, formalization or measurement credit. Authorship of the application, compact implementation and its elementary encoding proof is `1044_resume`; prior mathematical authorship and distinct reviews remain explicit. Root conducts the new distinct review.

Literal structural certificates for $L_3(5)$ and $L_3(5) : 2$

L054 recovery component

Theorem 1. *For each of the 348 rows in the two fixed current companion tables below, the actual subgroup representative in the acquired numerical catalogue admits the displayed recursive structural description, with the source-qualified meaning of the operations used in the accepted naming components.*

Table	Group order	Rows	Native pages
$L_3(5)$	372000	140	6
$L_3(5) : 2$	744000	208	8

The tables are attributed to Thomas Connor and Dimitri Leemans. These are the currently retained companion PDF editions, not an assertion of identity with an archived historical file. All fourteen native pages were directly read. The accepted numerical row bijections identify the exact embedded groups; no group is identified by its order alone.

A colon asserts an actual normal subgroup and a complement, including a nontrivial action; a dot asserts an actual nonsplit extension. A direct-product symbol additionally requires commuting factors. Unparenthesized operations associate to the left. Explicit right parentheses are retained. The redundant parentheses around 4^2 in the indicated $5^{1+2} : (4^2)$ rows and around D_{10} in $\mathrm{Q}_8 : (D_{10})$ do not change their atomic meanings. In particular, source row 106 of the first table and row 127 of the second print the cyclic group 24, not 2^4. Powers and the extraspecial notation have their acquired literal interpretations; no unstated exponent or extraspecial sign is inferred. These labels need not determine a unique isomorphism type.

Acquired premises and actual models. The complete numerical catalogues and their source bijections, the earlier literal registry models, the generator-edge isomorphism criterion, and the previous split, direct, prime-index and small-quotient extension verifiers are acquired through the exact manifests in `dependencies.json`. The newly named atom $L_3(5)$ is the actual 31-point group of the retained clean `PSL(3,5)` execution. Its original ordered generators equal those of the acquired numerical top group. The classical constructor's named-group identification is retained explicitly, not inferred from the number 372000. The accepted 5^{1+2} and $2 \cdot A_5$ models and their elementary proofs are reused. The central commuting-lifts criterion includes the root's previously disclosed contribution; it is not a new argument of this component.

Every transport fixes the identity, is a bijection onto the actual group, and preserves right multiplication by every ordered model generator. Induction on words proves it is a homomorphism. The complete word-map format stores generator images and regenerates the entire bijection, testing every Cayley edge; no sample of words substitutes for this condition.

A smaller faithful action with the original embedding retained. The $L_3(5) : 2$ catalogue acts on 744 points. Its actual source-row-3 subgroup H has order 12000, hence index 62. The acquired right-coset method supplies 62 representatives t_i, starting at the identity, with explicit words in the original ordered generators. Literal enumeration of H verifies its order and $t_i t_j^{-1} \notin H$ for distinct representatives. For every ambient and every saved subgroup generator g, the proposed image p_g is a permutation and satisfies

$$t_i g t_{p_g(i)}^{-1} \in H \quad (1 \le i \le 62).$$

Thus it is the actual action on right cosets Ht_i. Literal closure of the image has the acquired ambient order 744000, so the action is faithful. All subgroup-image arrays are checked using the same equation. This proves the original embedded representatives are transported to the groups used by the structural reader; it is not a replacement by a convenient abstract model. The first table keeps its original 31-point action.

Lemma 1 (Complete quotient-generator lifts)**.** *Let $N \lhd X$ be finite. Let Q be a finite group with ordered generators $q_1, \dots, q_k$, and let $\alpha : Q \longrightarrow X/N$ be a verified isomorphism. Choose $x_j \in X$ lifting $\alpha(q_j)$. Then N has a complement in X if and only if at least one of the $|N|^k$ ordered choices*

$$y_j \in N x_j \quad (1 \le j \le k)$$

extends to a homomorphism $Q \to X$. Such an extension is decided by testing all edges of the complete ordered Cayley graph of Q.

Proof. A complement gives a section of the quotient map and hence one of these choices. Conversely, if $s : Q \to X$ extends a choice, its composition with the quotient map agrees with α on the generators, and therefore on all of Q. Consequently s is injective, its image meets N trivially, and the image has index $|N|$ in X; it is a complement.

For the decision procedure, list every element of Q by breadth-first multiplication in the ordered generators, with the identity first. Assign identity image, and propagate the proposed images along every generator edge. A conflicting image proves that the assignment is not a homomorphism. If there is no conflict, every group element has an image and every generator edge is preserved. The same induction on words proves a homomorphism. The list is complete, so all relations needed for the conclusion have been checked, without choosing or assuming a presentation. Every generator lift has the form nx_j, so the finite product of these lists omits no section. □

Application and literal verification. For the actual $Q_8 \cdot A_5$ and $Q_8 \cdot (A_5 : 2)$ models, the reader first verifies the normal quaternion subgroup by its acquired model transport. It partitions the actual ambient elements into disjoint complete normal cosets and computes the right quotient action on those cosets. An explicit generator-edge bijection identifies that action with an already verified model of the printed quotient. Each proposed lift has precisely the stated quotient image. The first quotient model has three ordered generators, the second two; all $8^3 = 512$ and $8^2 = 64$ choices respectively fail a complete-edge homomorphism test. The lemma proves both displayed nonsplittings. No complement-nonexistence result returned by the witness supplier is a premise of the literal checker.

All other new registry entries have the acquired elementary atom, normal-subgroup, complement or earlier nonsplitting certificates. References are checked in dependency order. For each of the 348 source rows the reader checks the exact source text, native abstract syntax, numerical row map, original generators, actual subgroup order and complete transport. The source and numerical counts are not filters in a new subgroup search.

Representation and finite controls. The reader uses byte arrays for permutation degrees from 31 through 256, and tuples otherwise. For zero-based arrays the right-action product is $(ab)[i] = b[a[i]]$. Byte translation by the array of b, padded only beyond its degree, computes exactly this formula, since every entry of a is strictly below that degree. Encoding therefore changes neither the product nor the complete closure and edge tests. Independent tuple-oracle controls check every pair of permutations of four points embedded at degrees around both representation boundaries, including the actual degrees 31 and 62. Small exact coset controls include a nonnormal subgroup, wrong orientation and a nonfaithful quotient action. The new lift criterion is checked against a separately enumerated complete subgroup lattice for five small split and nonsplit extensions, including a nonabelian quotient. Damaged actual quotient/lift and source/action inputs must be rejected. Ordinary and optimized Python executions must give the same full report.

Scope and authorship. The source descriptions belong to Connor and Leemans. The present component adds literal applications, the complete quotient-generator lift check and its controls, and the actual 62-coset transport. Prior accepted methods, models, classifications and reviews retain their original attribution. GAP is used only to propose witnesses in saved actual groups. The final replay is independent literal permutation arithmetic and does not rerun that witness search or an ambient subgroup census. This is a component for 348 current-table descriptions, not a complete L054 recovery, a canonical naming theorem, a historical byte-identity claim, or a new measurement.

Literal structural certificates for Sp(4, 4)

L054 recovery component

Theorem 1. *Each of the 496 actual subgroup representatives in the acquired numerical catalogue for the fixed current* Sp(4, 4) *companion table admits its displayed recursive structural description, with the source-qualified operation conventions below.*

The table is attributed to Thomas Connor and Dimitri Leemans, has ambient order 979200, occupies 28 native pages, and prints June 7, 2013. All 28 pages were directly inspected. This application concerns these exact retained bytes; it does not assert their identity with an earlier website edition, uniqueness of structural names, or completion of the entire article. The accepted numerical catalogue and its source-row bijection are retained inputs. No ambient subgroup census is repeated.

Actual group and named input. The accepted six-table numerical component explicitly checked that the actual complete 85-point state equals the named constructor `PSp(4,4)`. Its exact state, constructor interface, receipt, source bijection and acceptance are acquired here. Every proper representative is read from literal embedded permutations in that same state; no displayed subgroup name is substituted for its embedding. The usual named-constructor definition of the projective symplectic group is a retained software interface, not inferred from an order match.

For completeness, the source's Sp(4, 4) and that projective name agree. On a nondegenerate symplectic space over $\mathbf{F}_4$, each $T_v(x) = x + \langle x, v\rangle v$, $v \neq 0$, is a symplectic transvection. If A is central then $T_{Av} = AT_vA^{-1} = T_v$. Equality of the nonzero rank-one operators $T_{Av} - I = T_v - I$ implies $Av = \lambda v$ and $\lambda^2 = 1$. In characteristic two the latter forces $\lambda = 1$. Thus A fixes every nonzero vector and is the identity. The symplectic center is trivial, so passage to the projective quotient changes no group.

Names and source joins. The stored syntax tree preserves the displayed parentheses and punctuation. Absent parentheses, the acquired recursive naming convention is left association. A colon asserts a split extension (with nontrivial displayed action in these certificates), a dot a nonsplit extension, and a cross an actual direct product. Superscripts are recovered from native typography: in particular $2^4, 2^5, 2^6$ are elementary abelian groups, not cyclic groups of orders $24, 25, 26$. The wrapped row 22 remains $2 \times 4 : 2 \times 2 \times 2 : 2 : 2 : 3$, with its exact source text retained. The source checker requires the same row number, group order, literal name, syntax tree and acquired actual representative. Every original ambient and subgroup generator is pinned.

Complete finite certificates. The shared acquired registry supplies previously verified named atoms and extension models. New models have literal generators and finite closures. For every model-to-factor or model-to-row map, the verifier reconstructs the image of every element along the model's complete ordered-generator Cayley graph. It checks identity, all right-generator edges, injectivity and exact surjectivity onto the actual subgroup. The acquired generator-edge theorem then gives a group isomorphism: induction on words proves compatibility with multiplication by any element. No partial presentation, successful search label or comparison of orders alone is used as an isomorphism proof. The inherited byte-array representation of these permutations has the same literal composition law as the earlier accepted reader.

For an extension $H = N \star K$, literal closure and conjugation by all generators of H prove $N \lhd H$; the factor's registry map proves its name. In a split case the literal complement has trivial intersection with N and $|N||K| = |H|$, so $H = NK$. In a direct case all factor generators commute; commutation is preserved under products and inverses, so both factors commute elementwise. For a prime-index dot, the retained exhaustive test verifies that every element outside N has order different from the prime. A complement would contain such an element, giving a contradiction. Any further extension criterion used is the exact acquired literal criterion, not a negative GAP complement-search result.

For an elementary abelian factor of order 2^k, the verifier checks k commuting involutions, literal generation, and the exact closure size 2^k. The induced surjection $\mathrm{C}_2^k \to H$ therefore is an isomorphism. This applies in particular to the new $k = 5, 6$ occurrences. Cyclic, dihedral and other previously supported atoms use their acquired complete normal-form and literal relation checks.

Candidate discovery is not a premise. The first attempt stopped at an unsupported nonsolvable complement calculation; the second reached its bounded deadline. Both are preserved without success credit. The successful successor additionally proposes a centralizer as a direct factor and caches failed small-group decomposition proposals. These operations supply candidates only. The independent reader still checks every successful factor, map and row by the preceding tests, so neither a cache miss nor an incorrect proposal can validate a false label.

Validation and scope. The ordinary and optimized Python checks verify all 496 row joins and every new model and map. Damaged source trees, actual embeddings, named top inputs, action degrees, and malformed source syntax are rejected. Independent elementary abelian examples of orders $32, 64$, actual split and nonsplit index-two examples, and intersecting purported direct factors exercise the elementary boundaries. The exact earlier registry proofs and reviews retain their attribution. This component supplies only these 496 structural descriptions; it makes no new historical census, timing, question, or whole-source claim.

Structural certificates for the Suzuki and Janko small-core tables

Private L054 finite application

Theorem 1. *With the acquired numerical inventories and named-model identifications, all 247 current companion rows have their displayed structural descriptions: 40 for J_1, 146 for J_2, 22 for* Sz(8) *and 39 for* Sz(8) : 3*. An extension label asserts existence of its stated decomposition, not uniqueness of a group defined by the label.*

Sources and exact embeddings. All eleven native pages of the current Connor–Leemans companion PDFs were directly viewed. Powers and dots are preserved. The published left-association convention gives $(3 \cdot A_6) : 2$ in J_2 row 3 and $3 \cdot A_6$ in row 6. Row 13 retains the literal source $2^{1+4} : (D_{10})$; its redundant parentheses around the single atom are recorded, while the semantic tree has the same D_{10} factor. Parentheses around composite factors are not removed.

The six-table numerical component and its distinct root review are acquired. Its actual literal final states, complete row bijections and named-model checks supply the present occurrences. The states contain explicit embedded generators, on 266 points for J_1, 100 for J_2 and 65 for Suzuki. No displayed constructor name is substituted for a proper embedding.

The retained J_1 model is primitive-library entry $(266, 1)$, and J_2 is the derived subgroup of entry $(100, 2)$; their identifications are acquired from the accepted sporadic representation and six-table interfaces. The two top-row generating lists are literally equal to these already identified final-state lists. Their naming certificates are therefore exact identity joins, not fresh large Cayley-table computations. The Suzuki model is the accepted named ovoid action, with the explicit coordinate square map for its order-three field extension. These identification premises are retained explicitly; matching orders alone are not the naming argument.

Structural reuse. The registry uses 142 earlier structural models and three named inputs. There are 33 new decomposition models: 25 split extensions, six nonsplit extensions, one direct product and one extraspecial model. Every proper occurrence and factor map is a literal bijection preserving identity and every right-generator edge. The acquired MC.3 word-induction argument proves each map is an isomorphism. Normality is checked by conjugating factor generators by group generators. An actual complement has trivial intersection and the required product of orders. Direct factors commute; a displayed nontrivial semidirect action has noncommuting generators. The acquired prime, cyclic-six, S_3, Klein-four and extraspecial criteria are used without alteration. The only new nonsplitting argument follows.

Lemma 2 (Central commuting-lifts obstruction)**.** *Let $N \leq Z(H)$ and let $x, y \in H$ have nonidentity commutator $[x, y] \in N$. Then the extension of H/N by N has no complement.*

Proof. The images of x, y commute in H/N. Every pair of their lifts has the form nx, my with $n, m \in N$. Centrality gives $[nx, my] = [x, y] \neq 1$. If a complement existed, the inverse of its quotient isomorphism would give commuting lifts of these two commuting elements, a contradiction. □

The actual $3 \cdot A_6$ certificate. Inside the saved J_2 row-6 group of order 1080, an actual normal subgroup N of order three is identified with the cyclic model. The reader checks that every element of N commutes with every group generator, hence $N \leq Z(H)$. The proposed x, y belong to H, and their literal commutator is a nonidentity element of N. All nine shifted commutators $[nx, my]$ are also checked directly.

The quotient name is paid separately. The certificate gives 360 actual representatives of right cosets Nr. The reader forms every coset, verifies their disjoint complete union, and computes the right action of every ordered generator on these cosets. Since N is normal, this action has kernel exactly N. Its literal image has order 360. An explicit bijection from the acquired A_6 model to that image preserves all generator edges, proving $H/N \cong A_6$. Thus the lemma proves exactly the source dot in $3 \cdot A_6$. The row-3 group has a separate actual order-two complement, proving $(3 \cdot A_6) : 2$.

Finite evidence and development. The first bounded witness pass returned in 11.41 seconds with 245 successful rows and two explicit failures, J_2 rows 3 and 6. A proposed single-element order obstruction did not certify those rows. It is not a premise or an accepted result. Root then supplied the central commuting-lifts argument. A changed two-row pass completed in 1.77 seconds, reusing all earlier witnesses and producing the missing two certificates. Both executions and the exact earlier code are retained. No ambient subgroup census was repeated.

Ordinary and optimized replay use literal permutations and must agree. Small quaternion controls validate the central obstruction and reject a noncentral kernel, a trivial commutator, missing or duplicate quotient cosets. Separate source, named-identity and embedding damages are rejected. The reader also handles GAP's physical numeric line continuation before a comma; nonnumeric continuations remain invalid. This concerns serialization, not a mathematical change. The initial source-parser rejection of redundant atomic parentheses is preserved; raw source labels remain exact.

The first identity check rejected a GAP-proposed automorphism where literal generator equality was required. The successor writes the unchanged generators explicitly; both editions and the failed receipt are preserved. This application claims current-table structural joins only. It does not claim historical PDF byte identity, canonical extension labels, whole-source admission, new question resolution, formalization or measurement. `1044_resume` authored the application and literal certificates; root contributed the commuting-lifts argument. A different agent conducts the distinct mathematical review.

Spherical hypertopes from marked kernels and antipodal flags

Human Compression project

Abstract

We give a common argument for the spherical classification in the fixed manuscript *An exploration of locally spherical regular hypertopes*. Maxwell's normal-subgroup theorem is retained as a named input. Preservation of the marked Coxeter diagram then leaves only the universal complex and, where it exists, its antipodal quotient. A single explicit flag excludes all even D_n antipodal quotients. For E_7 and E_8, the remaining geometric condition reduces to exact orbits of fundamental weights, avoiding an enumeration of the full groups. Complete orbit certificates accompany the argument.

1 Retained inputs and the marked quotient

We use the following standard facts about a finite Coxeter system (W, S). Its geometric representation is faithful and orthogonal; a closed fundamental chamber meets each orbit in exactly one point. The stabilizer of a face is its indicated standard parabolic subgroup. The spherical Coxeter complex is a flag complex, and W acts regularly on its chambers. The irreducible finite diagrams and their group orders are the usual finite Coxeter classification, also used in Table 1 of the source. These facts are inputs, not new classifications established by this manuscript.

The additional normal-subgroup input is Maxwell's Theorem 0.1, quoted as Theorem 4.1 of the fixed source: if W is a finite irreducible Coxeter group and $N \triangleleft W$, then either $N = \{1, -1\}$ or N is the kernel of a graph homomorphism to a finite Coxeter group. Here a graph homomorphism sends each simple generator to a simple generator or to the identity, and its images contain every simple generator of the target. The full Maxwell paper has not been independently acquired; this is an explicitly retained classification input through the fixed source, with its stated meaning of "graph homomorphism."

For a regular hypertope, the distinguished chamber automorphisms are nonidentity involutions. Their pairwise product orders are exactly the labels of its Coxeter diagram, including order two for an absent edge. The resulting marked group is a quotient of the Coxeter system with that diagram. It is a C-group, and its coset geometry is the original hypertope. These are the acquired reconstruction facts of the L052 general manuscript. In particular, distinct distinguished generators cannot become equal.

Lemma 1. *Let $(W, \{s_i\}_{i\in I})$ and $(W', \{t_j\}_{j\in J})$ be Coxeter systems. Suppose a graph homomorphism $f : W \to W'$ sends every s_i to a nonidentity element, sends distinct s_i to distinct elements, and preserves the order of every product $s_i s_k$. Then f is an isomorphism of marked Coxeter systems, after a bijection of their index sets.*

Proof. Write $f(s_i) = t_{\sigma(i)}$. The assumptions on images make σ injective; the definition of graph homomorphism makes it surjective. The order assumption says that the two Coxeter matrices are equal under σ. The defining presentations therefore give a homomorphism in the opposite direction, taking $t_{\sigma(i)}$ to s_i. Both compositions fix every generator, so both are identity homomorphisms. □

Theorem 2. *Subject to the stated Maxwell input, a regular hypertope whose diagram is finite and irreducible is the universal Coxeter complex or its antipodal quotient. The latter possibility requires $-1 \in W$ and requires the quotient coset geometry to be a hypertope.*

Proof. Let G be the distinguished group of the hypertope. Its marked epimorphism $W \to G$ has kernel N. If Maxwell's alternative expresses N as the kernel of a graph homomorphism $f : W \to W'$, then $W' \cong W/N \cong G$. Thus the simple images of f are nontrivial, distinct, and preserve every original product order. Lemma 1 gives $N = 1$. The remaining alternative is precisely $N = \{1, -1\}$. Reconstruction identifies the actual hypertope with the coset geometry of the indicated quotient and marking. No assumption that its proper residues are universal has been used. □

This argument avoids a separate inspection of the normal-subgroup tables for each finite type. It also avoids using injectivity on a parabolic before knowing that the corresponding residue is universal.

2 Parabolics in the antipodal quotient

Let $\alpha_1, \dots, \alpha_n$ be simple roots in the faithful real representation, and choose fundamental vectors ω_i with $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$. Their Gram matrix is the inverse of the simple-root Gram matrix. For an irreducible finite system,

$$c_{ij} := \langle \omega_i, \omega_j \rangle > 0 \qquad (1 \le i, j \le n). \tag{1}$$

For completeness, write the positive definite simple-root Gram matrix as $D^{1/2}(1 - A)D^{1/2}$, where D is positive diagonal and A has nonnegative entries and connected support. Positive definiteness implies that the largest eigenvalue of A is less than one. For a symmetric nonnegative matrix every eigenvalue has absolute value at most its largest eigenvalue; this follows from $|x^T A x| \le |x|^T A |x|$. Hence $(1 - A)^{-1} = \sum_{m \ge 0} A^m$. Connectivity supplies a positive entry in some power for every pair of indices, proving (1).

Write $W_i = \langle s_j : j \ne i \rangle$. This subgroup fixes the nonzero vector ω_i, so it cannot contain -1. Moreover,

$$-1 \notin W_i W_j \qquad (i \ne j). \tag{2}$$

Indeed, if $-1 = ab$ with $a \in W_i$ and $b \in W_j$, then $a\omega_j = -\omega_j$ and $a\omega_i = \omega_i$. Orthogonality would give $c_{ij} = -c_{ij}$, contrary to (1).

For rank at least three, the quotient by $\{1, -1\}$ is therefore injective on every proper parabolic and satisfies the exact product-kernel criterion proved in *Kernels and intersections in quotients of hypertopes*. It is a C-group on the induced generators. This conclusion does not assert flag transitivity: the latter is precisely what fails in even type D_n. For the string diagrams, the standard string C-group reconstruction theorem does supply a regular polytope and hence a regular hypertope.

3 Incidence and signs

The vertices of type i in the spherical Coxeter complex can be represented by $W\omega_i$. Two vertices of distinct types are incident exactly when they belong to one chamber. We need the following precise numerical test.

Lemma 3. *For $v \in W\omega_j$, one has $\langle \omega_i, v \rangle \le c_{ij}$. Equality holds if and only if $v \in W_i\omega_j$, equivalently if v is incident to ω_i.*

Proof. If v is not in the closed fundamental chamber, some simple root satisfies $\langle v, \alpha_l \rangle < 0$. Reflect in that root. This increases the inner product with $\eta = \sum_k \omega_k$ strictly, and does not decrease the inner product with ω_i: the increase in the latter is zero unless $l = i$, in which case it is strictly positive. The orbit is finite, so repeating reaches its unique dominant representative ω_j. This proves the inequality. If equality held initially, no step with $l = i$ could have occurred.

All steps therefore belong to W_i, and $v \in W_i\omega_j$. Conversely W_i fixes ω_i, so every member of this orbit gives equality. The parabolic description of the Coxeter complex identifies $W_i\omega_j$ as precisely the vertices of type j incident to ω_i. □

Suppose $-1 \in W$. An antipodal vertex is $[v] = \{v, -v\}$. By Lemma 3, two such vertices of distinct types are incident exactly when their chosen representatives have inner product c_{ij} or $-c_{ij}$. Positivity makes the sign unambiguous.

Lemma 4. *Every flag in the antipodal quotient lifts to a flag in the spherical complex if and only if every incident triple has positive product of its three incidence signs. It suffices to test that there are no distinct i, j, k and $v \in W\omega_k$ such that*

$$\langle \omega_i, v \rangle = c_{ik}, \qquad \langle \omega_j, v \rangle = -c_{jk}. \tag{3}$$

Proof. Changing the sign of one representative changes the signs of its incident pairs, and leaves the product around a triangle unchanged. If all triangle products in a flag are positive, choose a first vertex and orient every other representative to have positive inner product with it. The triangle condition makes every remaining pair positive as well. These representatives are pairwise incident in the original complex; its flag property makes them a flag and hence a subflag of a chamber. Conversely any lifted flag admits all-positive representatives.

If some triangle has negative product, orient its first two representatives to make their pair positive. They are incident in the original complex, so an element of W takes them to ω_i, ω_j. Choose the sign of the third representative to make its pairing with ω_i positive. Its pairing with ω_j must then be negative, which is exactly (3). This procedure is reversible. □

When the condition holds, every quotient flag lies in the image of a chamber, and $W/\{1, -1\}$ is flag transitive. The C-group reconstruction with flag transitivity now gives the hypertope. This is stronger than merely checking that every face has the correct number of adjacent chambers.

4 A flag excluding every even D_n quotient

Use the standard realization of D_n by signed permutations with an even number of sign changes, where $n \geq 4$ is even. Its fundamental vertices include

$$e_1, \qquad w = \tfrac{1}{2}(e_1 + \cdots + e_{n-1} - e_n), \qquad v = \tfrac{1}{2}(e_1 + \cdots + e_n).$$

The central antipode belongs to this group exactly when n is even. Put $x = e_n$. The vertices v, w are incident, as they are in the displayed fundamental chamber. The vertices x, v are incident because a coordinate permutation takes e_1 to x and fixes v. Likewise $-x, w$ are incident: apply the permutation interchanging coordinates $1, n$, followed by sign changes in those two coordinates. This group element takes e_1 to $-e_n$ and fixes w.

Consequently $[v], [w], [x]$ form a flag in the antipodal incidence system. Its pairings have signs $+, +, -$, since

$$\langle v, w \rangle = (n-2)/4 > 0, \quad \langle v, x \rangle = 1/2, \quad \langle w, x \rangle = -1/2.$$

It cannot lift to a chamber. This precludes regularity with the proposed marked group: if the quotient were such a regular hypertope, its chambers would be the images of the original chambers by coset reconstruction and chamber transitivity. The negative triple contradicts that requirement. Thus no regular hypertope of type D_n can have the antipodal quotient as its distinguished group. This one flag handles every even rank, without a separate chamber lower bound or induction on rank.

5 The two exceptional incidence calculations

For E_7 and E_8, choose the simply laced Cartan matrix C with diagonal entries two and entries minus one along the tree. Number the trivalent node by 0. Its three arms have vertices (1), $(2,3)$, and $(4,5,6)$ for E_7, with $(4,5,6,7)$ as the third arm for E_8. All other entries are zero. Put $d = \det C$, so $d = 2$ or 1, respectively, and $B = dC^{-1}$. These are integral matrices with strictly positive entries.

Represent $d\omega_k$ in the simple-root basis by column k of B. A simple reflection changes an integer vector z by

$$z \longmapsto z - \left(\sum_l C_{il} z_l \right) e_i. \tag{4}$$

Moreover $\langle \omega_i, z \rangle = z_i$. Therefore the excluded pattern (3) becomes the exact pair of integer equalities

$$z_i = B_{ik}, \qquad z_j = -B_{jk}, \qquad i, j, k \text{ distinct.}$$

Starting at column k, close a set under all the reflections (4). Every stored vector belongs to the orbit, by induction on its construction. When no new vector is found, the set is invariant under every generator and contains the starting vector, so it is the full orbit. Thus no group-order estimate or randomized sampling enters the coverage argument.

The complete orbit certificates have the following sizes and contain no excluded pattern. A separate reader checks that each set contains its initial weight and is closed under every simple reflection. These two conditions already suffice for coverage of the required orbit.

	0	1	2	3	4	5	6	7
E_7	10080	576	2016	126	4032	756	56	–
E_8	483840	17280	69120	2160	241920	60480	6720	240

These fifteen orbits contain 899402 vectors in total. The reader derives B again by rational elimination from the displayed Dynkin trees. It checks strict ordering, uniqueness, the initial weight, every reflection image, and the absence of the excluded coordinate pattern. It does not import the orbit construction. The resulting finite sets are supplied in full, with deterministic compressed copies. The calculation enumerates the fundamental-weight orbits, rather than the groups of orders 2903040 and 696729600.

All arithmetic is integral. Indeed $z^T C z = dB_{kk}$ for an orbit vector; Cauchy–Schwarz gives $z_i^2 \le B_{ii} B_{kk} \le 900$. Every stored coordinate therefore has absolute value at most thirty. The producer checks this bound after each reflection; its intermediate integer expressions are bounded by the displayed matrices. A deadline or storage failure is an incomplete calculation and cannot return a complete certificate. The reader also rejects a set with a deleted vector, a changed coordinate, or an incorrect orbit header; the producer rejects a changed Cartan matrix. These controls exercise coverage and exact input identity.

6 Classification and proper residues

For clarity, the central antipode can be decided independently of the classification under discussion. In type A_n with $n \ge 2$, a coordinate permutation on the sum-zero space cannot negate every difference $e_i - e_j$: three distinct coordinates already give contradictory required images. In B_n every signed permutation occurs, whereas in D_n only an even number of sign changes is allowed. Thus -1 occurs in B_n and exactly the even D_n.

For E_6, the Cartan determinant is three. The weight lattice modulo the root lattice consequently has order three. Each simple reflection acts trivially on this quotient, since its difference from the identity is an integral multiple of a simple root. Negation acts nontrivially

on a group of order three, so it cannot belong to the Weyl group. The determinant and the usual weight/root-lattice identification are retained elementary Coxeter facts here.

For the remaining types, the accompanying exact matrix calculation verifies that the product of simple reflections, raised to the indicated power, is $-I$:

type	E_7	E_8	F_4	H_3	H_4
power	9	15	6	5	15.

The E matrices are those already displayed. For F_4 use the chain Cartan matrix with diagonal two, neighbouring entries minus one except $C_{12} = -2$ (indices start at zero). For H_3, H_4 use the symmetric matrix with diagonal two, first bond $-\phi$, and other chain bonds minus one, where $\phi^2 = \phi + 1$. Reflection i subtracts row i of this matrix from row i of the identity. Multiplication is exact in $\mathbb{Z}[\phi]$, or in $\mathbb{Z}$ in the integral cases. Faithfulness of these retained geometric representations identifies the computed matrix with the antipode in the Coxeter group. In rank one the antipodal quotient loses its generator. In rank two, quotienting $I_2(p)$ by a nontrivial central antipode reduces the order of the product of its generators, and hence changes the prescribed diagram.

The complete finite checks, Theorem 2, the string reconstruction, the even D_n flag, and Lemma 4 give the following classification. The column a records whether the antipodal quotient is also admitted. The group orders in the universal column are retained from the finite Coxeter classification, with the following correction to the fixed source. Its Table 1 prints $12 \cdot 6!$ for E_6, whereas the order is $51840 = 72 \cdot 6!$. This correction can be checked within the same orbit method. Number the E_6 tree by a central vertex 0 and arms (1), (2, 3), (4, 5). The certificate lists all twenty-seven vectors in the orbit of $3\omega_3$ and checks closure under the six reflections. The stabilizer is the standard parabolic obtained by deleting vertex 3, which has type D_5 and order $2^4 5! = 1920$. Orbit–stabilizer therefore gives $27 \cdot 1920 = 51840$. The same rational elimination verifies the Cartan determinant three used above.

diagram	restriction	$\lvert W\rvert$	a
A_n	$n \geq 1$	$(n+1)!$	0
D_n	$n \geq 4$	$2^{n-1}n!$	0
B_n	$n \geq 3$	$2^n n!$	1
$I_2(p)$	$p \geq 3$	$2p$	0
E_6		51840	0
E_7		2903040	1
E_8		696729600	1
F_4		1152	1
H_3		120	1
H_4		14400	1

There are exactly $1+a$ hypertopes in each row. When $a = 1$, the second object has distinguished group of order $|W|/2$. The two objects are distinct because they have different numbers of chambers.

Every proper residue of an admitted antipodal quotient is universal spherical. Indeed a maximal parabolic embeds, as proved above. Flag transitivity and the C-group intersection identities identify the residue of its corresponding face with the coset geometry of that embedded parabolic on its own maximal parabolics. This is its original spherical Coxeter complex. Residues of higher codimension are residues within these. Thus proper spherical residues are a consequence of the classification, not an assumption silently used to force the quotient kernel.

Attribution and limits. The classification conclusions and the standard Coxeter structure belong to the earlier literature. Maxwell's normal-subgroup theorem is a substantial retained input. The work here is the common marked rigidity argument, the explicit antipodal flag

criterion, the uniform D_n obstruction, and the exceptional orbit proof with its exact finite obligations. No historical priority or open-conjecture solution is claimed. The Euclidean and hyperbolic classifications in the original manuscript lie outside the scope of this note.

References

[1] M. E. Fernandes, D. Leemans, and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised manuscript, catalogue L033; Sections 2–4.

[2] G. Maxwell, *The normal subgroups of finite and affine Coxeter groups*, Proc. London Math. Soc. 76 (1998), 359–382. Theorem 0.1 retained through the fixed source; primary full text not acquired.

[3] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. Complete reviewed components acquired with their proofs and stated inputs.

Finite Lie-group witnesses for chiral polyhedra

Human Compression project

1 The finite source obligations

Leemans and Liebeck [1] prove that every nonabelian finite simple group except $A_7, \mathrm{PSL}_2(q), \mathrm{PSL}_3(q)$ and $\mathrm{PSU}_3(q)$ admits a chiral polyhedron. We recover the existence assertions for the thirty-five parameter entries in Table 1. These are existing results, not new existence theorems.

There are twenty-one simple entries in their list (4.1), physical page 7:

$$L_4^\epsilon(2), L_4^\epsilon(3), L_4^\epsilon(4), L_5^\epsilon(2), L_6^\epsilon(2),$$
$$\mathrm{PSp}_4(3), \mathrm{Sp}_4(4), \mathrm{Sp}_6(2), \mathrm{PSp}_6(3), \mathrm{Sp}_8(2),$$
$$\Omega_7(3), \Omega_8^\epsilon(2), \mathrm{P}\,\Omega_8^+(3), \Omega_{10}^\epsilon(2).$$

Both signs occur, with $L^+ = \mathrm{PSL}$ and $L^- = \mathrm{PSU}$. The additional printed $\mathrm{Sp}_4(2) \cong S_6$ is not simple; it is not a positive simple-case obligation. The eleven remaining finite exceptions in Section 4(2), physical page 8, are

$$L_5^\epsilon(3), L_6^\epsilon(3), L_8^\epsilon(2), \mathrm{Sp}_6(4), \mathrm{PSp}_8(3), \mathrm{Sp}_8(4), \Omega_{12}^\epsilon(2).$$

The last three entries are $G_2(3), G_2(4), {}^2F_4(2)'$, treated at the end of Section 3, physical page 6. We do not include the nonsimple $G_2(2)$, nor turn its derived group $\mathrm{PSU}_3(3)$ or ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ into positive exceptions.

Lemma 1. *Let $G = \langle x, t\rangle$, where $|t| = 2$. If $|w(x,t)| \neq |w(x^{-1},t)|$ for a group word w, no automorphism of G inverts x and fixes t.*

Proof. Such an automorphism would carry the first word value to the second. It preserves element order, which contradicts the inequality. □

This is the elementary word criterion already proved in the accepted twelve-sporadic component [4]; it is restated to make the present argument self-contained. In every row we take $w(x,t) = xtx^2tx^3t$.

Theorem 1. *Each parameter entry in Table 1 has generators x, t satisfying conditions (i)–(iii) of [1, Theorem 1.1]. It therefore admits a chiral polyhedron of the displayed type $\{|x|, |x^{-1}t|\}$, with that named simple group as full automorphism group.*

Proof. The exact checks described below establish membership, $|t| = 2$, full generation, and the unequal word orders. Lemma 1 rules out even noninvolutory reversing automorphisms. Put $y = x^{-1}t$. Then $xy = t$. An element of $\langle x\rangle \cap \langle y\rangle$ commutes with both x, y, so is central. The named groups are nonabelian simple, hence this intersection is trivial. The first two orders exceed two. The retained Schulte–Weiss reconstruction theorem [2], in the precise form used in [1, Section 1], now gives the stated geometric conclusion. □

2 Named models and exact verification

Every `pair.g` is a literal permutation record containing the two witnesses, the full ambient generating list, the parameters, the degree and the expected orders. A fresh GAP process resets the same recorded constructor seed and calls `SimpleGroup` with those parameters. It requires exact equality of the saved and newly constructed generating lists. There is no recognition of a group from its order.

The named-model identifications are retained GAP 4.15.1 construction and catalogue premises [3]. The `L`, `U` and `S` branches construct `PSL`, `PSU` and `PSp`. Their projective matrix constructions and subsequent faithful smaller actions are the documented named constructors. The odd and signed even orthogonal branches take the derived subgroup of the appropriate special orthogonal action on lines, followed when applicable by the constructor's smaller action. Their identification and simplicity are retained; neither the stored `Size` nor a simplicity flag proves the identification here. The three exceptional calls are explicitly

$G_2(3)$	`PrimitiveGroup(351,7)`
$G_2(4)$	`PrimitiveGroup(416,7)`
${}^2F_4(2)'$	`PrimitiveGroup(1600,20).`

The exact primitive catalogue records name these groups. The constructor, classical projective interface, primitive loader and actual catalogue files are pinned in the dependency ledger. Correctness of GAP's exact permutation algorithms and the named library inputs is explicit computational infrastructure, not a new classification proof.

For every row the checker reconstructs H from the saved ambient generators without supplying its order. It verifies $H = G$, computes $|H|$, checks $x, t \in H$ and $|t| = 2$, and forms $K = \langle x, t \rangle$ without supplying its order. It verifies $|K| = |H|$ and $K = H$. Membership and equal finite order already imply generation; the latter equality is a redundant explicit check. The four table entries are then computed by exact permutation operations. The new groups are checked to have no preassigned size before the relevant calculations.

A separate standard-library Python parser reads the original cycle notation directly, checks disjoint cycles, and composes the corresponding right-action permutations. Cycle lengths and integer least common multiples reproduce all four orders and $|t| = 2$, independently of GAP's word calculations. It also parses every ambient generator, but it does not purport to duplicate GAP's large subgroup computations.

The source roster is constructed independently as the explicit sets of parameters above. Missing, duplicate, nonsimple, misrouted and displaced entries are rejected. Seven GAP controls damage parameters, ambient generators, the involution, membership, generation, an order, and the word inequality. All must reject before producing a positive certificate. Both Python entry points reject optimized execution before output work. Every replay is finite; a timeout or nonclean terminal result grants no certificate. The recorded searches merely supplied candidates and are not run by verification.

3 Attribution and limits

The root agent supplied the bounded candidate searches and the acquired word argument; this component's author supplies the deterministic certificate, independent literal oracle, source routing and explanation. All original candidate and execution bytes are retained. An initial author setup used a directory ordinal instead of the actual recorded seed for some constructors; that failed preliminary setup is preserved outside the scientific replay. The final replay uses the recorded seed and changes no candidate.

These are thirty-five parameter entries, not thirty-five newly classified groups. In particular $L_4(2) \cong A_8$ overlaps the alternating family, and $U_4(2) \cong \mathrm{PSp}_4(3)$ occurs twice under the source's parameters. The group identities, geometric reconstruction and original existence theorem retain their attribution. This component does not settle the infinite Lie families, ancillary enumerations, or the whole original source, and adds no measurement or conjecture claim.

4 The complete finite table

G	$\lvert x\rvert$	$\lvert x^{-1}t\rvert$	$\lvert w\rvert$	$\lvert w^{-}\rvert$
$L_4(3)$	8	5	9	12
$U_4(3)$	8	9	7	5
$L_4(4)$	30	85	30	15
$U_4(4)$	65	30	65	15
$L_5(2)$	15	15	3	7
$U_5(2)$	12	18	12	11
$L_4(2)$	15	6	5	6
$U_4(2)$	9	12	6	5
$L_6(2)$	12	31	28	31
$U_6(2)$	6	18	10	11
$S_4(3)$	4	9	4	5
$S_4(4)$	15	5	17	10
$S_6(2)$	6	9	12	10
$S_6(3)$	13	9	24	9
$S_8(2)$	30	30	8	6
$O_7(3)$	13	13	7	8
$O_8^+(2)$	12	10	10	12
$O_8^-(2)$	30	6	7	6
$O_8^+(3)$	15	12	15	20
$O_{10}^+(2)$	12	31	12	15
$O_{10}^-(2)$	11	30	6	15
$L_5(3)$	121	9	18	121
$U_5(3)$	24	28	21	40
$L_6(3)$	52	121	10	39
$U_6(3)$	91	80	13	56
$L_8(2)$	30	105	62	124
$U_8(2)$	85	66	12	85
$S_6(4)$	51	63	20	63
$S_8(3)$	13	84	24	40
$S_8(4)$	195	126	255	65
$O_{12}^+(2)$	63	18	24	18
$O_{12}^-(2)$	24	102	34	60
$G_2(3)$	13	12	7	6
$G_2(4)$	7	13	6	15
${}^2F_4(2)'$	12	8	13	12

Table 1: Exact orders in the saved witnesses. $L_n(q), U_n(q), S_n(q)$ mean $\mathrm{PSL}_n(q), \mathrm{PSU}_n(q), \mathrm{PSp}_n(q)$; $O_n^{\pm}(q)$ means $\mathrm{P}\,\Omega_n^{\pm}(q)$ and $O_7(3) = \Omega_7(3)$. Let $w = xtx^2tx^3t$ and $w^- = w(x^{-1}, t)$.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, fixed arXiv:1603.07713v3, Sections 1, 3 and 4.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

[3] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1, reference manual Sections 39.15-14 and 50.2; PrimGrp 4.0.1. Exact local implementation and data pins accompany this proof.

[4] Human Compression project, *Chirality detected by the order of one word*, accepted twelve-sporadic component, final edition v2; its distinct review and root adoption are retained.

A complete small-order domain for the subgroup-atlas reconstruction

Private L054 recovery component

Boundary and retained inputs. The fixed 2013 article, Section 4, physical pages 5–6, restricts its small-order assertion to almost simple groups appearing in its cited 2012 online ATLAS. The final 2015 version, page 264, has the same qualification. We instead construct a complete mathematical superset. This does not assert historical file identity. Nor does it establish the correctness of any unverified, named larger companion table.

We explicitly retain the classified nonabelian-simple-group roster below 10^6 documented in GAP 4.15.1, Reference Manual 39.15–15 and 39.15–17, and implemented in the pinned `grp/simple.gi` and `simple.gd`. Its non-PSL_2 data give nineteen entries, of which A_5 and A_6 are respectively $\mathrm{PSL}_2(5)$ and $\mathrm{PSL}_2(9)$. We retain simplicity, the indicated exceptional isomorphisms, and the full automorphism/outer-group classifications below. For classical groups the explicit presentations in `OuterAutoSimplePres` cite Bray–Holt–Roney-Dougal, pages 36–37; the same pinned implementation specifies the alternating and sporadic exceptions. These are mathematical classification inputs, not conclusions inferred from finite order tests. The ATLAS entries for $L_3(4)$ and $U_3(5)$ independently record the nonabelian outer groups D_{12} and S_3. No new proof of CFSG or of automorphism classification is claimed.

Lemma 1 (Preimages and isomorphism classes)**.** *Let S be a nonabelian finite simple group. The almost simple groups with socle S are exactly the preimages $\pi^{-1}(K)$, where $\pi : \mathrm{Aut}(S) \to \mathrm{Out}(S)$ and $K \leq \mathrm{Out}(S)$. Two such preimages are isomorphic if and only if their outer subgroups are conjugate in* $\mathrm{Out}(S)$.

Proof. Identify S with $\mathrm{Inn}(S)$. A group between S and $\mathrm{Aut}(S)$ containing one representative of an outer coset contains the whole coset, proving the preimage assertion. Its order is $|S||K|$. The centralizer of S in $\mathrm{Aut}(S)$ is trivial: if α commutes with every inner automorphism, then $c_{\alpha(s)} = \alpha c_s \alpha^{-1} = c_s$, so $\alpha(s) = s$ because $Z(S) = 1$. Any nontrivial normal subgroup N of a preimage meets S nontrivially, for otherwise $[N, S] \leq N \cap S = 1$. Simplicity then implies $S \leq N$. Thus S is the unique minimal normal subgroup and is characteristic. An isomorphism of two preimages restricts to $\beta \in \mathrm{Aut}(S)$; conjugation on S shows that its value on any element h is $\beta h \beta^{-1}$. Therefore the preimages are conjugate in $\mathrm{Aut}(S)$, exactly when their images are conjugate in $\mathrm{Out}(S)$. The converse is immediate. □

Lemma 2 (The linear rank-one part)**.** *There are precisely* 86 *almost simple isomorphism classes of order at most 10^6 whose socle is $\mathrm{PSL}_2(q)$. They have* 39 *distinct simple socles, taking $q = 5$ in place of the duplicate $q = 4$.*

Proof. For $q = p^f \geq 4$ put $d = (2, q-1)$. Counting ordered bases gives $|\mathrm{GL}_2(q)| = (q^2-1)(q^2-q)$. Determinant is onto $\mathbb{F}_q^\times$, and the center of $\mathrm{SL}_2(q)$ has order d. Hence $|S| = q(q^2-1)/d$. The projective action has precisely the scalar kernel: a linear map fixing every one-dimensional subspace fixes the two coordinate lines and their sum, forcing equal diagonal entries. The retained automorphism theorem is $\mathrm{Aut}(S) = \mathrm{P\Gamma L}_2(q)$, including $q = 9$. The determinant modulo squares and the field automorphism give $\mathrm{Out}(S) = C_d \times C_f$. The field action on the order-two diagonal quotient is trivial (odd p acts trivially modulo two).

For $q \geq 127$, the lower bound $q(q^2-1)/2$ exceeds 10^6 and is increasing. Thus it suffices to test prime powers $5 \leq q \leq 126$. The literal program `roster.py` tests primality by trial division, lists these prime powers, and forms every subgroup of the small explicit additive group $\mathbb{Z}/d \times \mathbb{Z}/f$. Its generic procedure starts with the identity subgroup and adjoins every possible element to each known subgroup, taking closure; every finite generating sequence is therefore represented. A separate check generates every pair of elements. This also suffices: projection of any subgroup onto C_f is cyclic, and the kernel is contained in the cyclic group C_d; a lift of a projection generator and a kernel generator generate the subgroup. The two inventories agree. Keeping precisely those K with $q(q^2-1)|K|/d \leq 10^6$ gives 86 entries. Since the outer group is abelian, no further fusion occurs. The only duplicate simple name in this range is $\mathrm{PSL}_2(4) = \mathrm{PSL}_2(5)$, removed explicitly; the other listed linear socle orders are distinct. □

Theorem 1 (Conditional classified roster)**.** *With the explicit classified inputs above, there are exactly* 127 *almost simple isomorphism classes of order at most* 10^6*. The accepted eighty current small tables cover eighty of them; the forty-seven additional targets are all in the* PSL_2 *family.*

Proof. The remaining seventeen simple socles and their retained outer groups are shown below. D_{12} denotes the dihedral group of order twelve. The last column counts conjugacy classes of outer subgroups whose preimages satisfy the bound.

S	$\lvert S\rvert$	$\mathrm{Out}(S)$	count
A_7, A_8, A_9	$2520, 20160, 181440$	C_2	$2+2+2$
$L_3(3)$	5616	C_2	2
$L_3(4)$	20160	D_{12}	10
$L_3(5)$	372000	C_2	2
$U_3(3)$	6048	C_2	2
$U_3(4)$	62400	C_4	3
$U_3(5)$	126000	S_3	4
$U_4(2) = \mathrm{PSp}_4(3)$	25920	C_2	2
$\mathrm{Sp}_4(4)$	979200	C_4	1
$\mathrm{Sz}(8)$	29120	C_3	2
M_{11}	7920	1	1
M_{12}	95040	C_2	2
M_{22}	443520	C_2	2
J_1	175560	1	1
J_2	604800	C_2	1

The program explicitly multiplies the relevant cyclic and dihedral groups, closes all subgroups and fuses by every inner conjugation. Their 41 surviving classes, together with the 86 linear classes, give 127. The checked roster assigns all eighty accepted targets once, with exact outer-index multiplicities. Their identities as the stated almost simple constructions are acquired from their accepted named-model transports, not deduced from orders. Where several outer classes have the same order, distinctness is additionally forced by the accepted numbers of subgroup conjugacy classes: $25, 26, 56$ for the three $A_6.2$ groups; $43, 103, 39$ for the three $L_2(25).2$ groups; $151, 152, 102$ for the three $L_3(4).2$ groups; and $143, 226, 142$ for the three index-six $L_3(4)$ extensions. These are intrinsic isomorphism invariants. Each relevant multiplicity is therefore exhausted, with no assumption based on a displayed name. The difference is $127 - 80 = 47$. □

Actual finite applications and limits. The forty-seven additions comprise twenty prime-field parent descents, eight nonprime parent descents and nineteen index-two restrictions. Each uses the previously acquired complete-local-cover descent or normal-restriction theorem; source counts are never inputs. The projective models of orders 235200, 531360, 885720, 976500, 524160 and 786240 are literal defining actions, with full generator and receipt records. Final

states use the accepted explicit generator serializer. The separate two-page applications proof and exact literal certificates now cover every new row. This roster argument supplies the domain classification; distinct acceptance of the combined component remains a separate review step.

Section 4 does not name an individual larger group: it refers to some larger groups, including large sporadic groups, and then reports an atlas of more than a hundred outputs. The actual companion index supplies individual names. A fully verified 127-table reconstruction would prove the complete small-order superset and a stronger cardinality statement. It would not by itself verify every larger PDF named in that index or prove the exact historical output set. Those claims must retain their own source-qualified dispositions.

Forty-seven additional exact subgroup tables

Private L054 recovery component

Claim and acquired interfaces. The companion roster proof supplies a classified superset of the fixed article's small-order domain. Here we construct its forty-seven missing members. We acquire the previously reviewed maximal-descent theorem, its duplicate-local-cover correction and explicit checkpoint writer, the index-two normal-restriction theorem, and the complete-generator-edge isomorphism and extension-certificate criteria. Their full proofs and exact acceptance records remain dependencies. GAP's complete local maximal subgroup cover and standard named finite-group constructors are retained software/classification inputs. No new proof of these library interfaces is inferred from a successful run.

Theorem 1. *Under those acquired interfaces, the forty-seven fixed new targets have complete subgroup-class inventories with* $2,468$ *total rows and* $6,248$ *weighted maximal-incidence edges. Every row has an independently checked structural description, interpreted as the displayed existence of a named group or extension, without asserting that an extension expression uniquely determines an isomorphism class.*

Numerical coverage. Twenty prime-field parents and eight nonprime parents were processed by the acquired complete maximal descent. Every subgroup lies in a maximal proper subgroup unless it is the full group, so descent through a complete local maximal cover reaches every subgroup. Actual local conjugacy removes duplicate cover entries; ambient conjugacy merges only actual conjugates. Local weights are normalizer indices, not the potentially unreliable size attribute of a supplied conjugacy-class object. If a local class of $K < H$ contributes weight a, the reverse weight is

$$b = a\,|G : N_G(H)|/|G : N_G(K)|.$$

All edges satisfy integral balance and both ambient-class bounds.

The other nineteen targets are normal index-two restrictions. For each actual $K \lhd G$ of index two, the acquired parent catalogue is complete. Retain its subgroups inside K. Every ambient conjugacy class splits into at most two K-classes, represented by H and H^t for a fixed $t \notin K$; the accepted routine tests actual K-conjugacy. Local maximal inclusions are inherited and fused with their actual local normalizer weights. This uses the reviewed index-two implementation only. No claim that its specialized weight routine handles arbitrary normal index is made.

The prime parents use the literal executed standard PSL_2 and PGL_2 constructors. In the nonprime cases the stored projective-line model starts from $\mathrm{SL}_2(q)$, adjoins the indicated diagonal map δ and coordinate Frobenius F, and checks normalization and the exact resulting orders. The retained classical automorphism theorem identifies their outer coordinates. For $q = 49$, the three index-two targets are the preimages of $\langle\delta\rangle, \langle F\rangle, \langle\delta F\rangle$ in C_2^2, together with the full outer preimage and the socle. For $q = 81$, the three permitted outer subgroups are $\langle\delta\rangle, \langle F^2\rangle, \langle\delta F^2\rangle$. For $q = 64$, the additions use F^3 and F^2, of orders two and three. The other new nonprime socles are $q = 121, 125$. The prime-field derived subgroups in the restrictions are the respective $\mathrm{PSL}_2(q)$: they are perfect, normal in the index-two $\mathrm{PGL}_2(q)$, and the quotient is abelian.

Every actual execution returned completely with empty stderr and unchanged input pins. The audit rechecks all $2,456$ recorded runtime bindings, actual executed drivers, every parent-begin/return marker, every saved complete-parent state, and all final arrays. Explicit literal generators are used throughout; GAP display names are not reconstruction instructions. The

nineteen restriction records also bind the actual named-model identification and complete origin records. This acquires the completed runs; no ambient enumeration is repeated by the checker. □

Structural coverage. There are 47 ambient rows and 2,421 proper rows. Each saved row is an actual subgroup of its named ambient group, by the acquired numerical method. A full-order row is therefore that group, even when restriction produced a different ordered generating set.

For 2,394 proper rows, literal permutation closure and generator-edge maps transport 223 acquired registry models and 371 new certified models. A new cyclic or elementary abelian model has the required actual generators and full order. A dihedral model has actual generators r, s with $r^n = s^2 = 1$, $srs = r^{-1}$, and order $2n$. For an extension, the verifier checks a literal normal subgroup N, transports its earlier certified name, and either checks a complement K with $N \cap K = 1$, $|N||K| = |H|$, or verifies nonsplitting. Direct products additionally check commutation; displayed nontrivial semidirect products check a nontrivial conjugation action. All new nonsplitting certificates have quotient prime order two or three and check that no element outside N has that order. Such an element would generate a complement, and every complement would supply one. References are acyclic. The checked data contain 65,036 actual outside checks and sixteen direct-product models.

A transport lists the images of the ordered source generators. The reader constructs the candidate image of every source element by words, checks independence of the chosen words, a bijection onto the actual target subgroup, identity, and every right-generator edge. Induction on words proves multiplicativity; bijectivity proves isomorphism. This is the previously acquired elementary criterion. The present application checks 2,822 transports and 2,389,759 right-generator edges.

The other 27 proper rows have large named defining models. Each of their 55 actual generators is evaluated as a stored word in the exact named model's generators. Thus the actual row is contained in that named model; the independently acquired equal orders imply equality. This is a membership-and-order argument, not identification from order alone. The $q = 64$ socle uses the exact sixty-five-point defining embedding, and the $q = 29$ row uses the actually executed canonical constructor record. No canonical small-degree subgroup is substituted for an embedded row.

The first three bounded witness-discovery attempts stopped at their caps. Only complete individual model and row certificates are used. Later passes visit unpaid rows and reuse earlier complete models; the last seven affine cases use direct elementary abelian normal-core/complement witnesses. Their discovery algorithm is not a proof premise: every positive certificate is checked literally. Run-local registry IDs are explicitly renamed into disjoint namespaces, with the actual inherited prefixes preserved. GAP's physical numeric line continuations are decoded by the acquired strict reader. No incomplete record supplies a premise. □

Roster join and source disposition. The exact machine-readable join binds every new target to its parameter, outer subgroup and complete numerical/structural record. Together with the accepted eighty tables and 8,226 structural rows, this establishes **127 small-order tables with 10,694 structural rows**. The classified roster shows that every almost simple group of order at most one million occurs exactly once up to isomorphism. Consequently the fixed article's qualified small-order domain is contained, regardless of the unavailable exact 2012 website bytes, and this is also a mathematical reconstruction of more than a hundred tables.

The new descriptions are generated exact existence assertions; they are not transcriptions of unavailable historical PDFs. We do not identify this roster with the precise historic output set. The article's statement about some larger groups, including large sporadic groups, and the correctness of individually named larger companion files retain separate source-qualified dispositions. The already acquired six larger numerical tables are not promoted to complete structural tables here. Whole-original acceptance is not self-issued by this component.

Structural certificates for the A_{10} companion table

Private L054 finite application

Theorem 1. *Relative to the acquired complete ambient subgroup-class count, the complete local maximal-subgroup interfaces and the accepted structural registry, all 430 rows of the current A_{10} companion table have the displayed structural descriptions. Each description asserts an actual decomposition, with the previously specified atomic conventions; it does not assert that an extension label determines a unique abstract group.*

Source and scope. The current table is attributed to Thomas Connor and Dimitri Leemans. All 21 native pages were directly inspected. Products and extensions are associated to the left unless the source supplies parentheses, as in the 2015 article's naming convention. For example, row 23 is $(A_6 \times 2) \cdot 2$; the dot is not replaced by a colon. Row 263 has 3^3, whereas row 299 has the baseline integer 21. The dicyclic Q_{12} and unsigned extraspecial 2^{1+4} conventions are acquired explicitly from the earlier structural proofs. Historical identity of these PDF bytes with the 2013 website is not asserted.

Actual subgroup embeddings. The previously saved group file has five proper representatives printed as alternating groups on explicit point supports. Those expressions are used only to propose candidate subgroups. We do not assume that a displayed name preserved the original producer's embedding.

The acquired complete subgroup-class count is 430. A new application of the accepted robust local-weight kernel verifies that the 430 candidates are subgroups of the actual natural A_{10} and are pairwise nonconjugate in it. Consequently their classes exhaust all subgroup classes. The actual ambient model is checked against AlternatingGroup(10), and all candidate generators are then exported as literal ten-point permutations. This is the same counting and reconstruction argument accepted for A_9, S_9; no ambient subgroup enumeration is repeated.

For each candidate H, an acquired complete local maximal-subgroup interface is applied to a verified faithful smaller action. Duplicate local classes are removed by actual conjugacy, and inverse transport returns maximal subgroups of H. A local class of $K < H$ contributes $[H : N_H(K)]$ children. Contributions with the same ambient class are added. Writing c_H, c_K for the ambient class lengths, the reverse weight is ac_H/c_K by double counting; integrality and both class-length bounds are checked. The resulting 1592 weighted edges and all 184900 ordered source row pairs agree with the saved source bijection. In particular, the actual reconstructed subgroups have the correct source assignments.

Literal structural verification. The application acquires 286 concrete registry seeds and proposes 21 new models, with acyclic references to factors. A reused structure is transported by an explicit identity-preserving bijection which respects right multiplication by each ordered generator. Induction on generator words proves it is an isomorphism, the previously accepted generator-edge criterion. Every occurrence is joined to its exact embedded subgroup, not merely to another group of the same order.

For a split description the verifier checks $N \triangleleft H$, identifies N and a subgroup K, verifies $N \cap K = 1$ and $|N||K| = |H|$, and hence proves $H = NK$. A colon includes a noncommuting pair of factor generators. For a direct product, all factor generators commute; then the factors commute elementwise. All subgroup, normality, product and commutation conditions are checked by literal finite permutation arithmetic.

There are twelve new split models, five direct models and four nonsplit models. Each new nonsplit model has quotient of order two. For such an extension $N \triangleleft H$, a complement exists if and only if some $x \in H \setminus N$ is an involution: the forward implication takes its nonidentity complement element; the reverse uses $\langle x \rangle$ and $[H : N] = 2$. The verifier checks all 8208 elements outside the four normal factors and finds none of order two. This is an exhaustive positive obstruction certificate, not a failed software complement search. The inherited normal and factor identifications are checked by the same literal maps.

Row 1 acquires the exact completed numerical construction $G :=$ AlternatingGroup(10) and the new ambient equality check. Its assigned subgroup has full order in G, so it is G itself. The named constructor convention is recorded explicitly; the full ambient group is not expanded again for a redundant structural map.

Validation and boundaries. The embedding and source checks completed in 32.68 seconds. The single bounded witness pass returned all 429 proper-row witnesses in 29.01 seconds, with empty error output. These are proposal receipts; the independent literal verifier checks all model closures, factor maps and source joins. The verification checks 467 maps and 1948088 generator edges. Ordinary and optimized modes must return identical results. Controls damage actual generators, row assignments, source syntax, dots, degrees, input hashes, class lengths, weights, edges and bijections, and test a genuine split extension against the nonsplitting criterion.

All earlier mathematical criteria and their authorship remain acquired. No new classification algorithm, unique interpretation of extension labels, historical byte-identity claim, whole-source admission, question result, formalization or measurement is asserted. This is one of the larger source-associated tables, all of which remain in the recovery contract.

Structural certificates and a correction for the A_{11} atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and structural registry,* 787 *of the* 788 *structural labels in the retained companion table headed* A_{11} *hold for their corresponding embedded subgroups. Row* 186 *prints* $2 \cdot A_5 : 2$*. Under the article's convention of left association, this expression is impossible in* A_{11}*; the corresponding subgroup instead has structure* $(A_5 \times 2) \cdot 2$*, equivalently* $2 \cdot (A_5 : 2)$ *with the displayed parentheses. An extension expression asserts a normal subgroup, quotient and the stated splitting behaviour; it need not specify a unique isomorphism class.*

Source and acquired results. The companion table by Thomas Connor and Dimitri Leemans is dated 27 November 2012 and accompanies [1]. Its numerical catalogue, source correspondence, embedded generators and orders are acquired from the separately accepted computation in degree 11. There are 788 conjugacy classes and 3304 weighted maximal-subgroup edges. The subgroup census and incidence counts are not repeated here. The registry contains 638 finite models reconstructed from previously accepted applications; the files used to reconstruct them are bound to those reviews. The acquired hypotheses and qualifications remain part of the present statement.

Section 3.2 of the article explicitly gives left association for unparenthesized extensions and distinguishes split from nonsplit extensions. We acquire that convention from its separately reviewed native reading. A new source audit joins the native powers, subscripts, parentheses and operators of all 788 labels to the retained expressions. The table has 41 pages and 17498 retained text states. Native pages $1, 2, 11, 13, 18, 40, 41$ were inspected directly; the remaining pages have position checks, not a claim of visual inspection. The position audit uses the same pypdf decoder as the flat extraction. Row 357 places redundant parentheses around the single atom D_{10}; only that exact glyph sequence is normalized. The final caption is checked explicitly and excluded from the last cell. Every printed expression, including row 186, is retained unchanged. The claim concerns this pinned edition of the table.

Lemma 1. *Let* H *be generated by specified permutations and let* S *be the union of their supports, with* $n = |S|$*. If the order of this same embedded group is* $n!$*, then* $H = \mathrm{Sym}(S)$*. If all its specified generators are even and its order is* $n!/2$*, then* $H = \mathrm{Alt}(S)$*.*

Proof. Every generator fixes the complement of S, so $H \leq \mathrm{Sym}(S)$. Equality of finite orders proves the first assertion. For the second, the sign homomorphism gives $H \leq \mathrm{Alt}(S)$, and equality of orders again gives equality of groups. The application first joins each source row to the accepted numerical record and exact generator tuple. The order used here is consequently the order of that group, rather than an inference from its printed name. □

This identifies rows $1, 2, 4, 12, 39, 138, 414, 706$. Symmetric subgroups embedded in an alternating group can have extra moved points, so a second criterion is useful for eight further rows.

Lemma 2. *Let* H *act on a finite set* X*, let* S *be invariant, and put* $n = |S| \geq 3$*. Suppose that* t *is an involution on* $X \setminus S$*, possibly the identity, and that each specified generator* h *acts on the complement as* $t^{\epsilon(h|_S)}$*, where* ϵ *is the additive sign in* $\mathbb{Z}/2$*. If the acquired order of* H *is* $n!$*, restriction identifies* H *with* $\mathrm{Sym}(S)$*.*

Proof. The sign map is a homomorphism, so the displayed relation on generators holds on every word and its inverse. Any element acting trivially on S has sign zero and therefore fixes the complement as well. Thus restriction is injective. Its image in $\mathrm{Sym}(S)$ has order $n!$, so it is the whole symmetric group. The implementation finds actual generator orbits, computes each restriction sign by inversion parity, and checks the same complement involution on every generator. It does not enumerate H or infer its order from this test. □

The second lemma applies to rows $3, 9, 26, 82, 277, 287, 762, 766$. It is the usual argument for the graph of a homomorphism. The adaptation here recognizes the sign action on the complementary points.

Lemma 3. *Let M and H be finite groups given by complete closures of specified permutation generators. A bijection $f : M \to H$ with $f(1) = 1$ and $f(xg) = f(x)f(g)$ for every $x \in M$ and every specified generator g is an isomorphism.*

Proof. Every element is a positive word in the generators: in a finite group the inverse of a generator is a nonnegative power of it. Induction on such a word proves $f(xy) = f(x)f(y)$. The reader constructs the proposed images throughout the complete closure, checks consistency on repeated elements and checks every right-generator edge. It verifies that the image is exactly the target subgroup. Thus the checked map is a bijective homomorphism, rather than only a correspondence between generating sets. □

Products and extensions. The new registry has 155 models: 96 split extensions, 30 direct products, 28 nonsplit extensions and one dicyclic group. Each model is proved before any later model or source occurrence can use it. For an extension model H, the reader closes the displayed subgroup N, checks $N \triangleleft H$, and transports an earlier proved model onto N. A split certificate supplies a subgroup K of the indicated type, checks $N \cap K = 1$ and $|N||K| = |H|$, and hence proves $H = NK$. For the colon notation it also witnesses nontrivial conjugation of K on N. A direct product instead checks commutation for all pairs of factor generators; the factors then commute elementwise.

Fifteen nonsplit models have quotient C_2. A complement exists exactly when there is an involution outside N. Indeed, such an element generates a subgroup disjoint from N whose product with N has order $|H|$. The reader checks every outside element, so absence of an involution proves nonsplitting.

The other thirteen nonsplit models have quotient S_3. The quotient certificate identifies generators with defining relations $a^2 = b^3 = (ab)^2 = 1$. A complement provides lifts satisfying the same relations. Conversely, such lifts define a homomorphism from S_3 whose composition with the quotient map is the identity, and hence a section. All possible lifts are considered. Those failing the first two relations may be rejected before the third is checked; the recorded pair count still includes the full original Cartesian product. Every pair fails at least one relation, which excludes a section.

For the source symbol Q_{28}, the new literal model has generators x, y of a group of order 28, with

$$|x| = 14, \qquad y^2 = x^7, \qquad y^{-1}xy = x^{-1}.$$

These relations reduce every word to x^i or $x^i y$, with $0 \le i < 14$. Thus the presented dicyclic group has at most 28 elements. The actual generated permutation group has exactly 28, so its canonical quotient map is an isomorphism. The inherited predicate already checked these relations, complete generation and cardinality. Its finite allowed list was extended from orders $8, 12, 16, 20$ to include 28; no relation was weakened. The earlier prepared requests were not executed.

The separate correction. The accepted companion note [2] proves that a nonsplit extension $2 \cdot A_5$ cannot embed in A_{11}. Such an extension is perfect and has a central involution. Its restriction to the support of that involution would lie in $C_2 \wr S_m$ for $m = 2$ or 4, a solvable group. A perfect group has trivial image there, contradicting the involution's nontrivial restriction. The note also identifies the actual row-186 subgroup as the parity pullback

$$\{(\sigma, k) \in S_5 \times C_4 : \operatorname{sgn}(\sigma) = (-1)^k\}.$$

Its subgroup of index two is $A_5 \times C_2$, and every element outside it has nontrivial square in the C_4 coordinate. The extension is therefore nonsplit. The full proof and its literal 240-element certificate are acquired from the distinct review of that note. The present reader replays the certificate on the exact saved generators. The correction remains separate from the 787 recovered printed labels; it is not an open-conjecture resolution.

Verification and attribution. The complete literal check passed in 23.697 seconds, with 788 source joins, 155 new models, 1051 transports and 3892103 generator edges. It performed 19706 outside-element tests and considered 87108 quotient lift pairs. The support and restriction lemmas each account for eight rows; 771 source occurrences use literal transports, and row 186 uses its separate corrected certificate. Inputs remained unchanged and stderr was empty. The native source wrapper independently replayed all 17498 retained text states and all 788 glyph joins.

Fourteen damaged source inputs are rejected, including changes to the printed row-186 assertion and the final caption. Five narrow parenthesis controls permit only the single D_{10} normalization. The two short recognition lemmas have their separate literal controls. Eleven damaged Q_{28} certificates are rejected in ordinary and optimized Python, including a cyclic group of order 28 satisfying the order and square conditions but failing the inversion relation.

The numerical catalogue, source expressions and notation belong to Connor and Leemans. The finite presentation, restriction, transport and extension arguments are standard group theory or acquired common results. The work here provides the explicit joins and finite certificates for this table, together with the separately proved correction. Candidate searches supply witnesses only; the literal reader establishes their assertions. Distinct final source admission remains to be recorded. No Lean formalization, whole-article completion or compression measurement is claimed here. Other large companion tables remain in scope for L054.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed A_{11} and dated 27 November 2012.

[2] human_compression project, *A structural label in the subgroup table of* A_{11}, companion proof and literal certificate, acquired in the present packet.

The 204 structural rows of the companion M_{23} table

Private L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named-model interfaces, each of the 204 subgroup-class representatives in the current Connor–Leemans M_{23} table has the structural description printed in its row. A compound label asserts the existence of the displayed normal factors, complements or nonsplit extensions; it need not characterize a unique isomorphism type.*

Source and numerical interface. The ten-page companion PDF is dated 25 November 2012. All ten native pages were directly inspected. The separately accepted robust numerical audit supplies the complete 204-class catalogue, 679 weighted maximal inclusions, and the exact source-row bijection. Those results and their computational classification premises are acquired unchanged; no ambient subgroup census is repeated here. The original and reweighted group files contain only literal permutation groups, so every proper subgroup keeps its actual 23-point embedding. This proves a current companion result, without asserting historical PDF byte identity or discharging the other larger tables.

The source's default left association is retained. In particular row 42 is $2\cdot L_2(7)$, not $2 : L_2(7)$. Baseline integers and superscripts are recorded separately: row 49 is $23 : 11$ and row 155 is the cyclic group of order 23, whereas row 188 has the elementary abelian label 2^3. The flat text string 23 is insufficient to distinguish these cases; the explicit native assertion ledger makes the distinction.

Named ambient group. The acquired driver constructs `MathieuGroup(23)`. Its clean complete receipt, all recorded input/output hashes, and actual generator equality with the numerical catalogue are checked. GAP Reference Manual 50.1-13 supplies the named constructor interface. The numerical top representative is an actual subgroup of this group with the full ambient order 10 200 960, and hence is the group itself. This acquisition pays row 1 without enumerating ten million elements anew. Its name is not inferred from an abstract order match.

Actual structural transports. The registry acquires 224 earlier accepted named and elementary models. For each proper row, the certificate specifies images of all ordered model generators. The independent reader closes the source model and verifies that the induced word map is a bijection onto the actual embedded subgroup, fixes identity and preserves every right-generator edge. Induction on words, as in the acquired MC.3 transport theorem, makes this map a homomorphism; it is therefore an isomorphism. All previously proved atom identifications retain their original authorship and exact acceptance routes.

The five new models. Four new models are split extensions:

$$2^4 : A_7, \qquad (2^3 : L_2(7)) : 2, \qquad 2^4 : L_2(7), \qquad (((2 \times 4) : 2) : 2) : 2 : S_3.$$

For each extension the reader closes actual normal and complement subgroups, checks their generator conjugates, trivial intersection and product of orders, and verifies the factor isomorphisms. The nontrivial displayed actions have noncommuting factor generators. These are explicit existence certificates, not conclusions from a structure-description heuristic.

The fifth model is $2 \cdot L_2(7)$. Its actual normal subgroup N has order two and is central. Literal cosets give a faithful quotient action, which an explicit generator-edge bijection identifies with the acquired $L_2(7)$ model. Two actual elements x, y have nonidentity commutator in N. Their images in the quotient commute. Centrality gives $[nx, my] = [x, y] \neq 1$ for all $n, m \in N$. A complement would provide commuting lifts of those commuting quotient elements, a contradiction. Thus the extension is nonsplit. This is an application of the already accepted central commuting-lifts criterion, whose original contribution by root is retained; no new general criterion is claimed here.

Finite certificate closure. The literal reader proves every new model from previously verified factors, then all 203 proper occurrence maps. Together with the named ambient join this covers all 204 rows through 213 maps and 1,848,806 generator edges. Every source order, exact name tree, embedded generator array and source-row bijection is checked against the acquired inputs. The five new models, the completed witness records and their raw execution receipts are retained. GAP discovery only proposes witnesses; correctness is established by literal permutation arithmetic.

The first bounded discovery stopped at the unsupported nonabelian quotient case in row 42. Its forty completed witnesses are retained individually, without calling that execution a complete success. A successor adds only candidate generation for the already proved central criterion and preserves those forty rows. The two baseline-23 ambiguities then produced explicit `NO_CERTIFICATE` results; after native correction only those two rows were retried. Their new witnesses pass the same independent reader. No failed candidate search is evidence of nonsplitting or of a missing subgroup.

Ordinary and optimized Python must return identical complete reports. Focused damages cover missing rows, wrong embeddings and degrees, altered source pins and names, dots changed to colons, baseline 23 changed to 2^3, malformed syntax and an unlicensed displayed constructor. The acquired extension and generator-edge controls remain explicit dependencies. This application is authored by `1044_resume`; its independent review is separate. It claims no new original, question, measurement, formalization, canonical label uniqueness or whole-L054 completion.

Structural certificates for the $\mathrm{Sp}(4,5)$ atlas table

Private L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and structural registry, all 307 rows of the current companion table headed* $\mathrm{Sp}(4,5)$ *have the displayed structural descriptions. The ambient group is* $\mathrm{PSp}_4(5)$*, of order 4680000, as constructed in the acquired numerical execution. Each extension description asserts an actual normal subgroup and quotient, with splitting or nonsplitting as displayed; it is not a unique abstract classification by an extension label.*

Source and actual groups. The table is attributed to Thomas Connor and Dimitri Leemans and is dated 7 June 2013. All fourteen native pages were directly inspected. The source's $\mathrm{Sp}(4,5)$ notation is used for the projective group in this table; the group of all form-preserving matrices has twice the displayed order. The precise named input is the original completed GAP construction $G := \mathrm{PSp}(4,5)$. Its saved ordered ambient generators agree literally with the accepted robust numerical catalogue. Every proper subgroup is saved by literal 156-point generators, with no constructor or other printed group name substituted for an embedding.

The numerical source bijection and weighted inclusions are acquired, and each structural occurrence is checked against its exact source row, order and embedded generators. The source punctuation is retained. In particular row 10 is $2 \cdot (A_5 \times A_5)$ and row 30 is $Q_8 \cdot (A_5 \times 3)$. Parentheses around the single atoms Q_{20}, D_{20}, D_{10} in rows 18,19,31,33,133 are redundant; their original text is preserved. The 5^3 and 5^{1+2} atoms are distinct. The earlier dicyclic conventions for Q_{12}, Q_{20} and unsigned extraspecial conventions are acquired unchanged.

Finite proof interface. There are 196 acquired concrete seeds and 55 new models: 38 split, five direct, eleven nonsplit, and one elementary abelian model. Every factor reference is acyclic. All finite groups are closed from their literal generators. An identity-preserving bijection which respects right multiplication by each ordered generator is a homomorphism by induction on words, hence an isomorphism. This acquired generator-edge criterion verifies each factor and occurrence map; equal orders alone never supply an identification.

For a split model, literal checks establish $N \triangleleft H$, a subgroup K, $N \cap K = 1$, and $|N||K| = |H|$. Thus $H = NK$. A colon includes an actual noncommuting generator pair; a direct product has elementwise commuting factors. The new elementary abelian model has commuting prime-order basis generators and the exact required order. Row 1 acquires the named ambient construction and its full-order subgroup identity.

Nonsplitting certificates. The literal reader uses only previously accepted exhaustive criteria. For a prime-order quotient, it checks every outside element: a complement would supply an element of that prime order. For a quotient S_3, six verified cosets give quotient generators xN, yN and the relations $x^2 = y^3 = (xy)^2 = 1$. Every pair of lifts is tested; a pair satisfying these relations would generate a complement, and a complement supplies such a pair. Both directions follow from the presentation and quotient order.

For the central-kernel model, the reader verifies $N \leq Z(H)$ and $x, y \in H$ with $1 \neq [x, y] \in N$. Their quotient images commute, whereas all pairs of lifts have the same nonidentity commutator. A complement is therefore impossible. The actual quotient is separately identified by its literal coset action and a certified registry map. This is the previously reviewed commuting-lifts criterion, retaining its original contribution.

Two models use the acquired general Cayley-lift criterion, which permits a nonabelian kernel or quotient. Let $Q \cong H/N$ be an identified concrete model with ordered generators $q_1, \ldots, q_r$, and let x_i be actual lifts. Enumerate every tuple in

$$(Nx_1) \times \cdots \times (Nx_r).$$

For each tuple, follow the complete right-generator graph of Q, assigning the identity first and multiplying by the chosen lifts. A consistent assignment on every edge gives a homomorphism $s : Q \to H$. Its composition with the quotient map is the identified isomorphism, so s is injective and its image is a complement. Conversely any complement supplies one of these tuples and a consistent assignment. Thus one failed edge for every tuple proves nonsplitting. The reader verifies complete disjoint cosets, the ordered generator lifts, the quotient identification, and all $|N|^r$ choices. A failed software complement search is used only to propose a certificate, never as its proof.

Validation and boundaries. The first bounded proposal pass stopped at an omitted candidate quotient case after 28 positive rows. Its failed receipt and completed witnesses are retained. A successor changed only that candidate case and visited the 278 unfinished rows; all returned positive with clean error output. The source trees and accepted mathematical criteria were unchanged.

Independent literal replay verifies all model and occurrence certificates in ordinary and optimized modes. The reader also checks GAP's four physical numeric-array continuations at bracket boundaries without permitting continuations inside JSON strings. Focused damages test source assignments, actual embeddings, punctuation, hashes, quotient lifts and parsing. The acquired Cayley criterion's small complete-subgroup controls are included.

No new ambient census, unique interpretation of extension notation, historical PDF-byte identity, whole-source admission, question resolution, formalization or measurement is asserted. The larger source-associated tables remain in scope until individually discharged.

Structural certificates for the Sp(4, 7) atlas table

Private L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named structural registry, all 603 rows of the current companion table headed* $\mathrm{Sp}(4,7)$ *have the displayed structural descriptions. Here the ambient group is* $\mathrm{PSp}_4(7)$*, of order* 138297600*. An extension description asserts the indicated actual normal subgroup, quotient and splitting behaviour; it does not classify an abstract group uniquely by its label.*

Exact source and embeddings. The Connor–Leemans table is dated 7 June 2013. All 28 native pages were read. Its numerical catalogue, source-row bijection and weighted maximal inclusions are acquired from the separately reviewed projective seed and completed descent. That input retains the explicit classical-maximal-cover premise; the present application supplies no replacement classification. The ordered 400-point ambient generators are identical throughout the initial projective seed, completed state and retained subgroup file. Every proper subgroup is given by literal embedded permutation generators. No printed constructor name substitutes for a proper embedding.

The native source fixes powers and punctuation before any group proposal. In particular rows 25 and 410 contain cyclic 42, rows 40,460,473 cyclic 24, and row 458 cyclic 25; elsewhere the flattened strings may denote 4^2 or 2^4. The atoms 7^3 and 7^{1+2} are distinct. Parentheses around the single atoms Q_{16}, D_{16}, D_{14} are redundant and their original text is retained. Every parsed tree is joined to the exact native assertion, row, order and acquired embedded generators.

Reusable finite certificate. The registry contains 332 acquired models and 242 new models: 174 split, 41 nonsplit, 25 direct, one elementary abelian and one extraspecial model. The additional $L_2(49)$ atom is the exact named top from the accepted small-order-domain application, not an inference from its order. All factor references are acyclic. Complete literal generator closures and identity-preserving maps respecting every right-generator edge prove all model and occurrence isomorphisms by induction on words.

For a displayed split extension the reader checks $N \triangleleft H$, $K \leq H$, $N \cap K = 1$ and $|N||K| = |H|$. A colon also has a nontrivial actual action; direct factors commute. Elementary abelian factors have a commuting prime-order generating basis and the exact order. The new extraspecial factor has order 7^3, generators x, y, and $z = [x, y]$ of order seven with $Z(H) = \langle z \rangle$ and $x^7, y^7 \in \langle z \rangle$. Hence $H/Z(H) \cong C_7^2$ and $H' = Z(H)$; maximal-subgroup quotients give $\Phi(H) = Z(H)$, proving the unsigned extraspecial assertion used by the source.

The two central-product certificates. The nonsplit groups underlying rows 8 and 11 have a central kernel $N \cong C_2$ and quotient $L_2(7) \times L_2(7)$. Their certificates exhibit actual subgroups A, B of order 336 with

$$[A, B] = 1, \qquad A \cap B = N, \qquad |A||B| = |H||N|.$$

Thus $H = AB$. Complete right N-cosets in each factor give two regular quotient actions, each independently transported to the acquired $L_2(7)$ model. The map $(A/N) \times (B/N) \to H/N$ induced by multiplication is well defined and surjective. Its kernel is trivial by $A \cap B = N$, so it is the asserted direct-product isomorphism.

Actual witnesses $x, y \in A$ have $1 \neq [x, y] \in N$, and $N \leq Z(H)$. Their images commute in H/N, while every pair of lifts has the same nonidentity commutator. A complement would supply commuting lifts, which is impossible. This last argument is the previously accepted commuting-lifts criterion, originally suggested by the root collaborator; its authorship is retained. The new work here is the explicit central product and its factor-quotient identifications. The reader also checks all four shifted pairs of lifts literally.

The other dots use the acquired complete prime, cyclic-six, S_3, abelian-quotient or general Cayley-lift criteria. For the latter, every choice of lifts of the identified quotient's ordered generators is followed along its full Cayley graph. A consistent assignment is a section of the quotient map; conversely a complement supplies such an assignment. One inconsistent edge for every lift choice proves nonsplitting. For S_3, first retain only lifts with $a^2 = 1$ and $b^3 = 1$, then test $(ab)^2$ on their Cartesian product. Every discarded pair already fails a necessary relation; the result and total original-pair count are unchanged. Exact old/new small truth tables test this factoring. Failed complement or isomorphism searches are merely proposal outcomes, never certificates of nonexistence.

Exact larger-degree encoding. For $257 \leq d \leq 512$, each image a_i is stored in two byte planes $a_i = \ell_i + 256h_i$, where $h_i \in \{0, 1\}$. For right-action composition $(ab)_i = b_{a_i}$, each plane of b is split into its first 256 and remaining entries. Two byte lookups indexed by ℓ_i, selected by the all-zero or all-one byte mask determined by h_i, give exactly both planes of b_{a_i}. The selection is bitwise and introduces no carries. Canonical encoding preserves equality and hashing; iteration, inversion and JSON serialization recover the original point images. Other degree ranges retain their acquired representation. Independent tuple multiplication, sparse high-label permutations and actual source-generator pairs check this implementation boundary, including an explicit regular Heisenberg group over $\mathbb{F}_7$.

Validation and scope. Bounded proposal attempts and their failures are preserved individually. An early driver returned before its intended calls; other attempts reached their caps. Later passes used exact earlier positive witnesses and changed only the remaining witness proposal strategy. All 602 proper-row certificates now exist, with the top acquired by its named construction. The final literal reader and damage controls are replayed in ordinary and optimized Python. All source joins, embeddings, quotient maps and lift choices are checked independently of proposal success messages.

No new ambient subgroup census, historical-byte identity, unique interpretation of extension labels, whole-source admission, question resolution, formalization or measurement is claimed. The remaining larger source-associated tables stay in scope.

Structural certificates for the $\mathrm{Sp}(6,2)$ atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named structural registry, all* 1369 *rows of the current companion table headed* $\mathrm{Sp}(6,2)$ *have the displayed structural descriptions. The ambient group is* $\mathrm{PSp}_6(2)$*, of order* 1451520*, in the acquired action on* 63 *points. An extension description asserts the indicated normal subgroup, quotient and splitting behaviour. It does not determine an abstract isomorphism class uniquely from the printed expression.*

Source and acquired results. The source is the Connor–Leemans companion table dated 7 June 2013, associated with [1]. Its numerical catalogue, source-row bijection, actual embedded subgroup generators and named top group are acquired from the separately accepted numerical computation. No new ambient subgroup census is used here. The 515 acquired structural models come from earlier accepted finite applications. Each of the 52 files read to reconstruct this registry agrees with its entry in the accepted $\mathrm{Sp}(4,7)$ review's dependency record. Acquiring these inputs retains their existing hypotheses and qualifications.

The paper's Section 3.2 adopts left association for an unparenthesized sequence of extensions. We have read that convention on native printed page 263. A position audit joins the powers, subscripts, operators, parentheses and wrapped structural cells of all 1369 rows to their parsed expressions and orders. Direct inspection of the companion covers pages $1, 2, 49, 86, 87$. It is not an inspection of every page. The position audit uses the same pypdf library as the flat extraction, so these are not two independent extraction engines. The exact retained files, rather than an assertion of identity with every historical edition, specify the source.

What is supplied by this application. There are 199 new finite model certificates: 143 split extensions, 35 direct products and 21 nonsplit extensions. Every proper source occurrence has a transport from a proved model to its acquired subgroup of the 63-point action. The source row, its literal expression, its order and its actual generators are joined before the transport is checked. Candidate models used to help later searches are still proved individually; they are never promoted to acquired facts merely because a search returned them. All references to new factor models precede the model using them.

Lemma 1. *Let* M *and* H *be finite groups given by complete closures of specified permutation generators. A bijection* $f : M \to H$ *satisfying* $f(1) = 1$ *and* $f(xg) = f(x)f(g)$ *for every* $x \in M$ *and every specified generator* g *is an isomorphism.*

Proof. Every element $y \in M$ is a positive word in the generators: in a finite group, an inverse of a generator is a nonnegative power of that generator. Induction on such a word gives $f(xy) = f(x)f(y)$ for every $x, y \in M$. Thus f is a homomorphism, and the assumed bijection makes it an isomorphism. The reader constructs the proposed images along the complete closure, checks their consistency whenever a word reaches an old element, and then checks every right-generator edge. Bijectivity is checked against the whole target subgroup, not merely against its generating set. □

Extension certificates. For each new model H, the reader constructs the subgroup N generated by the displayed permutations, verifies $N \triangleleft H$, and transports an earlier model onto N. For

a split extension it similarly verifies a subgroup K, its structure, $N \cap K = 1$ and $|N||K| = |H|$. Normality makes NK a subgroup, and the order equation gives $H = NK$. This proves the semidirect product assertion. For a colon the retained convention also requires nontrivial actual conjugation of K on N; a pair of noncommuting generators supplies the witness. For a direct product all generator pairs commute, hence all elements of the two factors commute. These are the previously accepted finite extension predicates.

The nonsplitting certificates. Thirteen models have prime quotient. If that quotient has order p, a complement would be generated by an element of order p outside N. Conversely such an element generates a complement by the order formula. The reader checks every outside element and excludes this possibility. One model has an abelian quotient. Its certificate gives complete, disjoint normal forms for the quotient and checks every choice of lifts of its stated cyclic generators. Every choice fails an order relation or a commutation relation, so none defines a section.

Five models have quotient S_3. The certificate identifies quotient generators satisfying $a^2 = b^3 = (ab)^2 = 1$ and gives all their lifts. A complement would provide lifts satisfying these same relations. First discarding lifts which fail $a^2 = 1$ or $b^3 = 1$ preserves every possible section; the remaining pairs are checked against $(ab)^2 = 1$. The source count records the full original Cartesian product, including discarded pairs. This filtering is inherited from the earlier accepted reader.

The remaining two dots use the previously accepted Cayley-graph criterion. The quotient is identified with an acquired finite model. For each choice of lifts of its ordered generators, the reader propagates a proposed section along the complete quotient Cayley graph. Consistency on every edge would give a homomorphism by the lemma, and the quotient images would make it a section. Conversely a section supplies such a consistent choice. One inconsistent edge for every lift choice therefore proves that no complement exists. The present instances require 128 lift choices and 2176 edge tests. A failed GAP complement search is never a certificate of nonsplitting.

The last source expression. Row 28 has order 1536 and expression

$$2^2 \times 4 \cdot 2 : 2 : 2 : 2 : S_3.$$

Left association gives successive orders $16, 32, 64, 128, 256, 1536$. In particular, the direct product at the start is formed before the nonsplit extension. The last four newly generated models complete the required chain from order 64 onwards and its final S_3 complement. The literal reader proves these models by the same predicates as every other row; no special interpretation is introduced to obtain this result.

The proposal search used two elementary improvements. For an abelian requested expression it rejects a subgroup with different cyclic prime-power invariants, after checking abelianness. These invariants are obtained by multiset union for direct products. It also requires the requested action before selecting the first prime-order complement generator. An earlier selection could choose a centralizing involution and overlook a later involution with nontrivial action. Both changes concern proposal efficiency and completeness, and leave the proof predicates unchanged. They do not constitute new structure theorems.

Verification and attribution. The complete literal check passed in 33.710 seconds. It checked all 1369 source joins, the 199 new model proofs, 1747 transports and 1682351 generator edges. The nonsplitting checks include 1376 outside elements and 9472 composite-quotient lift pairs, in addition to the Cayley-graph counts above. Input hashes remained unchanged and stderr was empty. Twenty damaged source or certificate inputs are rejected in both ordinary and optimized Python. The complete optimized replay reproduces the same proof counts. Source

admission is recorded separately by the distinct review, with the extraction qualifications stated above.

The original table, its notation and association convention are due to the source authors. The numerical catalogue and named models are acquired results. The finite transport, extension and lift criteria are reused from the project's accepted common method. The contribution here is their explicit application to every occurrence of this table, with literal certificates and an exact source join. No new conjecture resolution, Lean formalization, historical-byte identity, whole-article completion or compression measurement is claimed. The other large source tables remain in scope.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed $\mathrm{Sp}(6,2)$ and dated 7 June 2013.

Structural certificates for the S_{10} atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and structural registry, the* 1593 *rows of the current companion table headed S_{10} have the displayed structural descriptions. An extension expression asserts a normal subgroup, quotient and the stated splitting behaviour. It need not specify a unique abstract isomorphism class.*

Source and acquired results. The companion table by Thomas Connor and Dimitri Leemans is dated 27 November 2012 and accompanies [1]. Its numerical catalogue, source-row bijection, actual embedded generators and subgroup orders are acquired from the separately accepted computation in degree 10. The acquired source contains 1593 conjugacy classes and 8023 weighted maximal-subgroup edges. The subgroup census and incidence counts are not recomputed here; proper-group orders are checked again when constructing literal transports. The registry contains 676 finite structural models reconstructed from accepted earlier applications; the exact files read to reconstruct it are bound to those reviews. Existing hypotheses and qualifications remain part of the acquired statements.

The source article's Section 3.2 gives the convention of left association for an unparenthesized sequence of extensions. This convention is acquired from its separately reviewed native reading. A new audit joins the native powers, subscripts, parentheses and operators of every structural cell to its expression and order. The table has 103 pages and 38011 retained text states. Native pages $1, 6, 33, 52, 102, 103$ were directly inspected. This does not amount to visual inspection of all pages. The position audit uses the same pypdf decoder as the flat extraction, rather than an independent extraction engine. Rows 77 and 576 put redundant parentheses around the single atom D_{10}; only that exact glyph sequence is removed when comparing their expressions. The raw positions are retained. The pinned source edition, rather than every historical edition, specifies the claim.

Lemma 1. *Let H be generated by specified permutations, let S be the union of their supports, and put $n = |S|$. Suppose that the order of this same embedded group is known. If $|H| = n!$, then $H = \mathrm{Sym}(S)$. If every specified generator is even and $|H| = n!/2$, then $H = \mathrm{Alt}(S)$.*

Proof. Every generator fixes the complement of S, so H is a subgroup of $\mathrm{Sym}(S)$. Equality of finite orders proves the first claim. The sign map is a homomorphism. If all generators are even, then $H \leq \mathrm{Alt}(S)$, and equality of orders again gives equality of groups. No transitivity assumption is needed. In the application the source row is first joined to its accepted numerical record and exact generator tuple. The acquired order is then the order of the group in this lemma; it is not inferred from the printed name. □

This identifies rows $1, 2, 3, 4, 7, 11, 29, 45, 109, 191, 451, 738, 1397, 1547$ directly. The other 1579 source occurrences are handled using the acquired models, 172 new finite model certificates and complete permutation transports. The model roster consists of 87 split extensions, 69 direct products and 16 nonsplit extensions. In particular, an abstractly symmetric or alternating subgroup with another action is not accepted by the support lemma unless its actual support and order meet the stated conditions.

Lemma 2. *Let M and H be finite groups given by complete closures of specified permutation generators. A bijection $f : M \to H$ satisfying $f(1) = 1$ and $f(xg) = f(x)f(g)$ for every $x \in M$ and every specified generator g is an isomorphism.*

Proof. Every element is a positive word in the generators, because the inverse of a generator in a finite group is a nonnegative power of it. Induction on such a word gives $f(xy) = f(x)f(y)$. The bijection is therefore an isomorphism. The reader constructs proposed images along the complete closure, checks consistency whenever a word reaches an old element, and checks every right-generator edge. It compares the entire image with the actual target subgroup. Merely mapping a generating set is insufficient. □

Products and extensions. For a new model H, the reader closes the displayed generators of N, checks $N \triangleleft H$, and transports an earlier proved model onto N. A split certificate also supplies K, its structure, $N \cap K = 1$ and $|N||K| = |H|$. Normality makes NK a subgroup; its order then gives $H = NK$. For a colon, the retained convention requires nontrivial conjugation of K on N, witnessed by a pair of noncommuting generators. A direct product requires every pair of factor generators to commute, which implies that the factors commute elementwise. The finite transport and extension predicates are reused from the accepted common method. Every newly proposed model is proved before it is used, including models obtained in an earlier candidate search.

Twelve nonsplit models have a prime quotient of order p. A complement exists exactly when there is an element of order p outside N: its cyclic subgroup intersects N trivially, and the order formula gives all of H. The nonsplitting reader checks every outside element, so the absence of such an element proves the assertion.

The other four nonsplit models have quotient S_3. Their specified quotient generators satisfy $a^2 = b^3 = (ab)^2 = 1$ and generate the quotient. These are the standard defining relations of S_3. A complement would lift them to elements satisfying the same relations. Conversely such lifts give a homomorphism from S_3 whose composite with the quotient map is the identity, hence a section. All lifts are considered; those failing the first two relations may be discarded before testing the third. The recorded count includes the full original Cartesian product. Each possible pair fails at least one relation, so no section exists.

The proposal search. The first pass stopped at row 34, with expression $A_5 \times A_5$: GAP's general complement routine does not handle the case where both the normal factor and the quotient are nonsolvable. A process exit code of zero after the resulting break-loop error is recorded as an incomplete search. The retained witnesses are not discarded or accepted on that basis.

The following elementary criterion supplies a suitable direct-product search. Let $N \triangleleft H$ and $n = [H : N]$. If $C_H(N)$ has order n and intersects N trivially, it is the unique direct complement of N. Indeed, a direct complement lies in that centralizer and has the same order. Otherwise one can inspect the normal subgroups K of order n with $N \cap K = 1$. Normality puts every commutator in the intersection, so these factors commute, and the order formula gives $H = NK$. Conversely every internal direct complement is normal, since conjugation by N fixes it pointwise and conjugation by itself preserves it. This standard direct-product criterion changes proposal generation only. The literal proof predicates remain unchanged.

Verification and attribution. The numerical catalogue, source expressions and association convention belong to the source and acquired results. The support criterion and the internal direct-product criterion above are elementary group theory. The contribution of this application is an explicit source join and finite certificates for its remaining occurrences using the common method. The complete literal check passed in 22.115 seconds, with 1593 source joins, 14 support identifications, 172 model proofs, 1907 transports and 3209070 generator edges. Nonsplitting requires 10236 outside-element tests and 5764 quotient-lift pairs. Inputs remained unchanged and stderr was empty. The initial incomplete candidate run is retained; the 27 remaining rows

were subsequently supplied in 2.531 seconds using the direct-factor improvement. Candidate generation itself is never proof.

Ten damaged source inputs are rejected, and eight narrow glyph checks confirm that only the two parenthesized D_{10} atoms are normalized. Support recognition has its separate eight small literal model controls and nine damaged cases. Ordinary and optimized modes are checked separately. The final independent review records source admission and retains the extraction limitations stated above. No conjecture resolution, Lean formalization, historical-byte identity, whole-article completion or compression measurement is claimed. Other large companion tables remain in scope for L054.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed S_{10} and dated 27 November 2012.

A structural label in the subgroup table of A_{11}

human_compression project

Connor and Leemans' companion table for A_{11}, dated 27 November 2012, prints $2 \cdot A_5 : 2$ in row 186, of order 240 and class length 20790 [2]. The native page is page 11. Section 3.2 of their article distinguishes the nonsplit middle dot from the general extension symbol and explicitly adopts left association [1, p. 263]. Thus the printed expression denotes $(2 \cdot A_5) : 2$ under the stated convention. We prove that this interpretation is impossible in A_{11}. We then identify the actual subgroup supplied by the independently checked numerical table. The original expression remains in the source ledger; the corrected structure is recorded separately.

Lemma 1. *A nonsplit extension of A_5 by a normal subgroup of order two is perfect.*

Proof. Write $1 \to Z \to N \xrightarrow{q} A_5 \to 1$, with $|Z| = 2$. The subgroup Z is central, since its automorphism group is trivial. Recall that A_5 is perfect. For completeness, the commutator of $(12)(34)$ and $(12)(35)$ is a 3-cycle. The commutator subgroup therefore contains every 3-cycle, since these form one conjugacy class in A_5; they generate A_5. Consequently $q(N') = A_5$. If $N' \cap Z = 1$, then N' is a complement to Z, contradicting the nonsplitting assumption. Hence $Z \subseteq N'$ and $N' = N$. □

Proposition 2. *No nonsplit extension $2 \cdot A_5$ embeds in A_{11}. In particular, no subgroup of A_{11} has the structure $(2 \cdot A_5) : 2$.*

Proof. Suppose that such a subgroup N exists, and let z be its central involution. Since z is an even permutation on eleven points, it is a product of $m = 2$ or $m = 4$ disjoint transpositions. Put $\Omega = \mathrm{supp}(z)$. Centrality makes Ω invariant under N, and restriction defines a homomorphism

$$N \longrightarrow C_{\mathrm{Sym}(\Omega)}(z) \cong C_2^m \rtimes S_m.$$

Indeed, a permutation commuting with z permutes its two-point orbits and independently interchanges the two points in each orbit. The group on the right is solvable: its base group is abelian, and S_2 and S_4 are solvable. One may use the normal series $1 \triangleleft V_4 \triangleleft A_4 \triangleleft S_4$ to see the latter. The image of the perfect group N is perfect, while every perfect solvable group is trivial. But z has nontrivial restriction to Ω, a contradiction. The final assertion follows because the indicated semidirect product contains $2 \cdot A_5$ as a subgroup. □

This obstruction concerns every subgroup of A_{11}; it does not depend on the choice of a correspondence between the numerical source rows and the computed representatives. It does not assert that the authors intended the impossible interpretation. A missing pair of parentheses would change the expression to a valid one.

Proposition 3. *The saved representative corresponding to row 186 is*

$$H = \langle (12345),\ (12)(8\,10\,9\,11),\ (8\,9)(10\,11) \rangle.$$

It is isomorphic to

$$\{(\sigma, k) \in S_5 \times C_4 : \mathrm{sgn}(\sigma) = (-1)^k\}.$$

Its structure can be written $(A_5 \times 2) \cdot 2$, or equivalently $2 \cdot (A_5 : 2)$, with the displayed parentheses.

Proof. Put $r = (8\,10\,9\,11)$, acting on a set disjoint from the first five points. Each displayed generator belongs to the indicated pullback. The first two generators project onto S_5: a 5-cycle and an adjacent transposition generate S_5. Moreover, the square of the second generator is $r^2 \neq 1$. The projection kernel therefore has order at least two. The pullback has order $2|S_5| = 240$, so containment and projection show that H is the entire pullback.

The elements with even k form $N = A_5 \times \langle r^2 \rangle$, of index two. Every element outside N has odd k, so its square has second coordinate two and cannot be the identity. Thus the extension of C_2 by N has no complement and is nonsplit. This gives the first expression. Projection onto S_5 has central kernel $\langle r^2 \rangle$. Every lift of a transposition has square r^2; a section of this projection would preserve the order of a transposition, which is impossible. Hence this central extension is nonsplit as well. Finally, $S_5 = A_5 : C_2$, with the usual nontrivial conjugation action, giving the second expression. □

The accompanying literal certificate uses only permutations on eleven points. It closes the saved generators and checks equality with all 240 pullback elements. It verifies the direct product of order 120, the absence of involutions outside it, and all twenty lifts of the ten transpositions. As an additional check, it propagates all eight possible assignments of the three generators to C_2 across every Cayley edge. Exactly one assignment is surjective, and its kernel is the displayed $A_5 \times C_2$. The centre of that kernel has order two. Thus the index-two subgroup cannot itself be a nonsplit extension $2 \cdot A_5$.

The table and its naming convention are due to Connor and Leemans. The arguments above use elementary facts about central extensions, perfect groups, and permutation centralizers. The contribution here is the explicit correction and its connection to the checked representative; no general priority claim is made for those facts. This note corrects one structural entry. It neither changes the numerical subgroup lattice nor proves an open conjecture.

References

[1] T. Connor and D. Leemans, *An atlas of subgroup lattices of finite almost simple groups*, Ars Mathematica Contemporanea **8** (2015), 259–266.

[2] T. Connor and D. Leemans, *The subgroup lattice of* A_{11}, companion table, 27 November 2012, 41 pages; row 186 on page 11.